Exam A

Jeremy's IT Lab Ordered Edition

All 74 items reorganized to follow the CCNA course sequence.

Original question numbers, answer choices, answers, explanations, diagrams, and lab

Searchable / selectable text layer is added during final OCR processing.

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Day 1 - Network Devices & Fundamentals

Original Question 54/74

54/74

Your company maintains a large, geographically dispersed WAN. Recently, the WAN has been experiencing intermittent broadcast storms.

Which of the following should you add to the network in order to reduce the broadcast storms? *(Select the best answer.)*

- A. more Layer 2 switches
- B. more Layer 3 devices
- C. more firewalls
- D. more IPSs

Incorrect

Correct Answer(s): B

Explanation:

You should add more Layer 3 devices to the network. Layer 3 devices, such as routers and multilayer switches, do not forward broadcast traffic, thereby segmenting the network into multiple broadcast domains. Segmenting a network into multiple smaller broadcast domains reduces the impact of broadcast storms, which can occur on large Layer 2 wide area networks (WANs) during periods of network instability.

You should not add more Layer 2 devices to the network. Adding Layer 2 devices, such as Layer 2 switches and bridges, will further increase the size of the broadcast domain, thereby increasing the possibility of broadcast storms.

You need not add more firewalls to the network. A firewall is a device that separates a network into zones. The firewall will allow or deny traffic to these zones based on the zone's security policy. For example, a firewall might be configured to deny traffic to or from a network port that is typically used by peer-to-peer network sharing software. However, firewalls are not as responsive as routers and multilayer switches. Firewalls should be used to protect mission-critical networks and devices, not to reduce broadcast storms.

You need not add more intrusion prevention systems (IPSs). An IPS is used to prevent malicious or unauthorized traffic from entering a network. Like firewalls, most network-based IPS devices sit inline

unauthorized traffic from entering a network. Like firewalls, most network-based IPS devices sit inline, which means that they sit in the path of traffic and intercept it. When an IPS detects malicious traffic, it will drop the malicious traffic to prevent it from infiltrating the network.

Reference:

Boson CCNA Curriculum 200-301, Module 1: Network Fundamentals, Switches

Boson CCNA Curriculum 200-301, Module 1: Network Fundamentals, Routers

Auvik: How to Troubleshoot a Broadcast Storm

Category:

Network Fundamentals

Day 2 - Interfaces & Cables

Original Question 39/74

38/74

Which of the following connectors are used to terminate fiber connections? *(Select the best answer.)*

- A. RJ-45
- **B. ST**
- C. RJ-11
- D. BNC
- E. DB-9

Incorrect

Correct Answer(s): B

Explanation:

ST connectors are used to terminate fiber connections. Fiber-optic cables consist of a glass core, reflective cladding, and a buffer to provide rigidity. The outside of the cable is covered with a protective jacket, which is typically made of plastic. Data is transmitted through fiber-optic cable as pulses of light. Other common fiber-optic connectors include SC and LC.

RJ-45 connectors are used to terminate four-pair twisted copper cables, typically in a wired Ethernet local area network (LAN). Each of the eight connector pins found in the RJ-45 connector corresponds to a copper wire found within a twisted-pair cable. Copper cable contains a conductive core surrounded by bedding. The bedding is surrounded by an outer sheathing layer and an optional metallic shield to protect against electromagnetic interference (EMI).

DB-9 connectors are often used to terminate point-to-point serial connections between network devices, such as data terminal equipment (DTE) and data communications equipment (DCE) devices. Serial connections send and receive data one bit at a time, which greatly limits speed. DB-9 ports are commonly found only in legacy peripherals.

Day 2 - Interfaces & Cables

Original Question 39/74 - continued

BNC connectors are used to terminate legacy coaxial cable connections. Coaxial cables are flexible plastic cables that consist of an inner copper core surrounded by a dielectric insulator. The insulator is covered with a woven copper shield. Although coaxial cable was used in older computer networks, it is now most commonly used for carrying cable television signals.

RJ-11 connectors are most often used to terminate legacy plain old telephone system (POTS) connections. The RJ-11 connector includes six connector pins that are used to terminate pairs of copper wires. RJ-11 connectors typically terminate one pair of wires for a single telephone line. However, two additional pairs of wires can be used to accommodate additional phone lines.

Reference:
Boson CCNA Curriculum 200-301, Module 1: Network Fundamentals, Fiber-Optic Cable Connectors

Cisco: SC, ST, LC fibre connections

Category:
Network Fundamentals

Days 5-6 - Ethernet LAN Switching

Original Question 44/74

44/74

Which of the following Ethernet fields contains a 4-byte CRC value? *(Select the best answer.)*

- A. preamble
- B. data
- C. FCS
- D. type

Incorrect

Correct Answer(s): C

Explanation:

The Ethernet frame check sequence (FCS) field contains a 4-byte cyclic redundancy check (CRC) value. Ethernet, which operates at the Data Link layer of the Open Systems Interconnection (OSI) networking model, is calculated by the frame's sender. When the frame arrives at its destination, the receiver recalculates the CRC and compares it to the value that is stored in the FCS field. If the values match, the frame is sent to the next-layer protocol. If the values do not match, the frame is typically discarded and the interface FCS-err value is incremented. A large number of FCS errors could indicate a faulty interface or damaged cable.

Methods similar to CRCs are used at several other layers of the OSI networking model. The CRCs used by Ethernet and the Institute of Electrical and Electronics Engineers (IEEE) 802.3 standard are a form of checksum. Network interface cards (NICs), which operate at the Physical layer of the OSI networking model, can also use checksums as a form of digital error detection. Internet Protocol (IP), which operates at the Network layer of the OSI networking model, uses a header checksum to protect the header. Checksums used in protocols that operate at layers higher than the Network layer typically protect both the transmission's header and its data.

The Ethernet type field contains the protocol that receives the frame after Ethernet processing. This field is 2 bytes in length. The type field is not present in the IEEE 802.3 standard on which Ethernet is based. Instead, the 802.3 standard contains a length field at the location of the type field. This length field is a 2-byte representation of the number of bytes that follow the length field as well as the number of bytes that precede this field.

The Ethernet preamble field contains a 7-byte field of alternating bits. This pattern enables bit-level synchronization between the frame sender and the frame receiver.

The Ethernet data contains the frame payload. In the 802.3 standard, this field also contains an 802.2 payload header.

Reference:
Boson CCNA Curriculum 200-301, Module 2: Network Addressing and Transport, Ethernet Overview

Cisco: Troubleshooting Ethernet: Frame Formats

Category:
Network Fundamentals

Days 5-6 - Ethernet LAN Switching

Original Question 50/74

50/74

Which of the following addresses will a switch use to populate the CAM table? *(Select the best answer.)*

- A. destination IP addresses
- B. destination MAC addresses
- C. source IP addresses
- D. source MAC addresses

Incorrect

Correct Answer(s): D

Explanation:

A switch will use source Media Access Control (MAC) addresses to populate the Content Addressable Memory (CAM) table. The CAM table, which is also called the switching table or the MAC address table, is used by a switch to discover the relationship between the Open Systems Interconnection (OSI) Layer 2 address of a device and the physical port used to reach the device. The switch populates the CAM table by recording the source MAC address of an inbound Layer 2 frame and the corresponding switch port that the frame arrived on.

Switches make forwarding decisions based on the destination MAC address contained in a frame's header. The switch first searches the CAM table for an entry that matches the frame's destination MAC address. If the frame's destination MAC address is not found in the table, the switch forwards the frame to all its ports, except the port from which it received the frame. This is referred to as flooding. If the destination MAC address is found in the table, the switch forwards the frame to the appropriate port. The source MAC address is also recorded if it did not previously exist in the CAM table.

A switch will not use destination MAC addresses to populate the CAM table. However, if a switch already has entries in the CAM table for a frame's source and destination MAC addresses, the switch can forward frames between them.

A switch will not use source or destination Internet Protocol (IP) addresses to populate the CAM table. IP addresses are Layer 3 addresses that are typically used to populate the routing table, which is stored in the Ternary CAM (TCAM) table.

Reference:
Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Using the CAM Table

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 1, Chapter 5: Analyzing Ethernet LAN Switching, Learning MAC Addresses

Cisco: Catalyst 6500/6000 Switches ARP or CAM Table Issues Troubleshooting: Background Information

Category:
Network Fundamentals

Day 9 - Switch Interfaces

Original Question 28/74

You issue the **show interfaces status** command while troubleshooting intermittent connectivity issues.

Which of the following causes are you most likely attempting to isolate? *(Select 2 choices.)*

- ☐ A. late collisions
- ☐ B. speed mismatches
- ☐ C. switching loops
- ☐ D. broadcast storms
- ☐ E. duplex mismatches

Incorrect

Correct Answer(s): B, E

Explanation:

Most likely, you are attempting to troubleshoot duplex mismatches or speed mismatches if you issue the **show interfaces status** command while troubleshooting intermittent connectivity issues. Duplex mismatch errors can cause a number of problems on switches, including intermittent connectivity, performance problems, a high number of collisions, and late collisions. A duplex mismatch error occurs when the ends of a network link are configured with different duplex settings. Cisco switches support half-duplex mode and full-duplex mode, but both ends of the link should be configured to use the same duplex setting. If they are configured with different duplex settings, network performance could be adversely affected due to collisions being detected. One symptom of a duplex mismatch is that the half-duplex side of the connection will report late collisions. The full-duplex side, on the other hand, will report runts, Frame Check Sequence (FCS) errors, and alignment errors, You can configure the duplex settings explicitly, or you can configure the link to autonegotiate the duplex settings. Duplex mismatches can sometimes be difficult to diagnose because the problems might be intermittent and the interface will report an up/up state. If you suspect that a duplex mismatch error is causing network problems, you can use the **status** parameter of the **show interfaces** command to verify the duplex settings for all the interfaces on a switch.

Speed mismatch errors can prevent an interface from sending or receiving traffic. A speed mismatch error occurs when one end of a network link is configured to use a different speed than the other end of the link, such as when one end of a link is configured to transmit at 10 megabits per second (Mbps) while the other end of the link is configured to transmit at 100 Mbps. In such a scenario, a link between the two interfaces would not be able to be established and the links would remain in the down state. If you suspect that a speed mismatch error is causing network problems, you can use the **status** parameter of the **show interfaces** command to verify the speed settings for all the interfaces on a switch.

It is not likely that you are troubleshooting broadcast storms if you issue the **show interfaces status** command. A broadcast storm is the result of an extreme amount of broadcast traffic on a network, thereby causing network congestion. Broadcast traffic is traffic, such as Internet Control Message Protocol (ICMP) packets, that is sent to all hosts on a network. A broadcast storm occurs when a device repeatedly sends out broadcast packets, resulting in continual broadcast responses from the other hosts on the network. This continuous flow of broadcast traffic can be the result of connecting multiple switches together in such a way as to create a loop or the result of connecting a patch cable to two different ports on the same switch.

A sluggish or unresponsive network is characteristic of a network suffering from a broadcast storm. In addition, you might witness unusually high CPU utilization by connected hosts as well as any affected switches. High CPU usage results from the devices processing the large volume of packets flowing around the network during the storm. You might also see a high collision count after issuing the **show interfaces** command. To view the CPU usage of a switch, you can issue the **show processes** command.

It is not likely that you are troubleshooting late collisions if you issue the **show interfaces status** command. A late collision is a collision that occurs after the 512th bit (64th byte) of a frame has been transmitted by a device. A switch detects such a collision if it begins sending a frame and a collision occurs after 512 bits of the outgoing frame have been sent. The amount of time it takes to send the first 512 bits of a frame is dependent on the network technology in use. For example, it takes 51.2 microseconds to send 512 bits over a 10-Mbps Ethernet segment, whereas it only takes 5.12 microseconds to send 512 bits over a 100-Mbps FastEthernet segment. Similar to other collision errors, late collisions can occur as a result of duplex mismatch errors or a network segment that extends farther than the cable length supports. If you notice late collisions occurring on an interface, you should ensure that the duplex settings are configured correctly and that the network segments are not too long. You can issue the **show interfaces** *interface* command to troubleshoot late collisions.

It is not likely that you are troubleshooting switching loops if you issue the **show interfaces status** command. There can be only one active path at any given time between any two endpoints on an Ethernet network. If multiple paths between the same two endpoints exist at the same time, switching loops can occur. You can issue the **show spanning-tree** command to examine the Spanning Tree Protocol (STP) configuration on a switch. STP is designed to prevent switching loops in topologies with redundant connections between switches.

Reference:
Boson CCNA Curriculum 200-301, Module 1: Network Fundamentals, Duplex Mismatch

Boson CCNA Curriculum 200-301, Module 1: Network Fundamentals, Speed Mismatch

Cisco: Cisco IOS Interface and Hardware Component Command Reference: show interfaces status

Category:
Network Fundamentals

Day 11 - Routing Fundamentals & Static Routing

Original Question 24/74

24/74 OVO

You issue the **show ip route** command on RouterA and receive the following partial output:

```
S 10.20.0.0/16 [1/0] via 192.168.10.2
D 10.20.0.0/20 [90/2809856] via 192.168.10.4, 00:02:14, Serial0/4
R 10.20.0.0/24 [120/3] via 192.168.10.3, 00:33:38, Serial0/3
O 10.20.0.0/28 [110/64] via 192.168.10.1, 00:02:38, Serial0/1
```

RouterA receives a packet that is destined for 10.20.0.17.

To which next-hop IP address will RouterA send the packet? *(Select the best answer.)*

- ○ A. 192.168.10.4
- ○ B. 192.168.10.2
- ○ C. 192.168.10.1
- ○ D. 192.168.10.3

Incorrect

Correct Answer(s): D

Explanation:

RouterA will send the packet to the next-hop address 192.168.10.3. RouterA will use the Routing Information Protocol (RIP) route, because it is the route with the longest prefix match. When a packet is sent to a router, the router checks the routing table to see if the next-hop address for the destination network is known. If multiple routes to a destination are known, the most specific route is used. Therefore, the following rules apply on RouterA:

- Packets sent to the 10.20.0.0/28 network use the Open Shortest Path First (OSPF) route. This includes destination addresses from 10.20.0.0 through 10.20.0.15.
- Packets sent to the 10.20.0.0/24 network, *except those sent to the 10.20.0.0/28 network*, use the RIP route. This includes destination addresses from 10.20.0.16 through 10.20.0.255.
- Packets sent to the 10.20.0.0/20 network, *except those sent to the 10.20.0.0/24 network*, use the Enhanced Interior Gateway Routing Protocol (EIGRP) route. This includes destination addresses from 10.20.1.0 through 10.20.15.255.
- Packets sent to the 10.20.0.0/16 network, *except those sent to the 10.20.0.0/20 network*, use the static route. This includes destination addresses from 10.20.16.0 through 10.20.255.255.
- Packets sent to any destination not listed in the routing table are forwarded to the default gateway, if one is configured.

Because the most specific route to 10.20.0.17 is the route toward the 10.20.0.0/24 network, RouterA will forward a packet destined for 10.20.0.17 to the next-hop address 192.168.10.3 through the Serial0/3 interface.

RouterA will not use the OSPF route to send a packet destined for 10.20.0.17 to the next-hop address 192.168.10.1, because 10.20.0.17 is outside the 10.20.0.0/28 address range. Packets destined to addresses within the 10.20.0.0/28 subnet will be sent by using the OSPF route to the next-hop address 192.168.10.1.

RouterA will not use the static route to send a packet destined for 10.20.0.17 to the next-hop address 192.168.10.2. Although the static route has the lowest administrative distance (AD), AD values are used only to determine which route is placed in the routing table when multiple routes to a destination are known. A router considers routes with different prefix lengths as separate routes. If the static route were configured so that the destination network were 10.20.0.0/24, the static route would be preferred over the RIP route.

RouterA will not use the EIGRP route to send a packet destined for 10.20.0.17 to the next-hop address 192.168.10.4. If OSPF, EIGRP, and RIP had all advertised routes to 10.20.0.0/24, the EIGRP route would have been selected because EIGRP has the lowest AD of the three dynamic routing protocols. The following list contains the most commonly used ADs:

Route Source	AD
Directly connected route	0
Static route	1
EIGRP summary route	5
eBGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
iBGP	200
Unknown	255

Reference:
Boson CCNA Curriculum 200-301, Module 6: IP Routing, Understanding AD

Cisco: Configure Route Selection for Routers

Category:
IP Connectivity

Day 11 - Routing Fundamentals & Static Routing

Original Question 72/74

72/74

You issue the **show ip route** command on a router named KeyWest and receive the following output:

```
KeyWest#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route

Gateway of last resort is not set

     200.120.45.0/24 is variably subnetted, 10 subnets
O       200.120.45.0/26 [110/586] via 200.120.45.253, 00:04:11, FastEthernet0/1
O       200.120.45.64/26 [110/391] via 200.120.45.253, 00:04:11, FastEthernet0/1
O       200.120.45.128/27 [110/196] via 200.120.45.253, 00:04:11, FastEthernet0/1
O       200.120.45.160/27 [110/391] via 200.120.45.253, 00:04:11, FastEthernet0/1
C       200.120.45.192/28 is directly connected, FastEthernet0/0
O       200.120.45.236/30 [110/585] via 200.120.45.253, 00:04:11, FastEthernet0/1
O       200.120.45.240/30 [110/585] via 200.120.45.253, 00:04:11, FastEthernet0/1
O       200.120.45.244/30 [110/390] via 200.120.45.253, 00:04:11, FastEthernet0/1
O       200.120.45.248/30 [110/390] via 200.120.45.253, 00:04:11, FastEthernet0/1
C       200.120.45.252/30 is directly connected, FastEthernet0/1
```

Which of the following best describes the `200.120.45.192/28` output in the routing table? *(Select the best answer.)*

- ○ A. It is the broadcast address for the 200.120.45.192/28 network.
- ○ B. It is the CIDR representation for the 200.120.45.192 255.255.255.248 subnet.
- ○ C. It is the 28-bit prefix for a directly connected network.
- ○ D. It is the network address for the 200.120.45.190/28 network.

Day 11 - Routing Fundamentals & Static Routing

Original Question 73/74

73/74

Which of the following codes from the output of the **show ip route** command indicates that the route is a default route? *(Select the best answer.)*

- A. D
- B. *
- C. EX
- D. S

Incorrect

Correct Answer(s): B

Explanation:

The * in the output of the **show ip route** command indicates that the route is a default route. For example, the following route entry indicates a default route that is redistributed into Enhanced Interior Gateway Routing Protocol (EIGRP):

```
D*EX 0.0.0.0/0 [170/2195456] via 10.10.10.1, 00:00:15, FastEthernet0/1
```

The `D` indicates that the route is an EIGRP route. The `EX` indicates that the route is an external route, which is a route that was redistributed into EIGRP. The address `0.0.0.0/0` specifies that any packet that does not have a specific route in the routing table should be sent over this default route. The first number in the square brackets, `170`, is the administrative distance (AD), which is used to determine the routing protocol that should be preferred when multiple routes to a destination network exist; external EIGRP routes have an AD of 170 by default. The second number is the feasible distance (FD), which is the total cost to a destination network. The entry `via 10.10.10.1` indicates that packets destined for the default route should be sent to the next-hop router at 10.10.10.1. The entry `00:00:15` specifies the number of hours, minutes, and seconds since the route was updated. Finally, `FastEthernet0/1` indicates that the default route can be reached through the FastEthernet0/1 interface.

A `D EX` indicates an external EIGRP route, which is a route that has been redistributed into EIGRP. However, the absence of the `*` indicates that the route is not a default route.

A `D` without an `EX` indicates that the route is an internal EIGRP route. Internal EIGRP routes are routes that are learned by EIGRP. By default, internal EIGRP routes have an AD of 90.

An `S` indicates a static route, which is created manually with the **ip route** command. To configure EIGRP to propagate a static route, you must redistribute the static route into EIGRP and you must specify the network by issuing the **ip default-network** command. An `S*` indicates a static default route.

Reference:

Boson CCNA Curriculum 200-301, Module 6: IP Routing, Verifying a Default Route

Cisco: Cisco IOS IP Routing: Protocol-Independent Command Reference: show ip route

Cisco: Configure a Gateway of Last Resort that Uses IP Commands

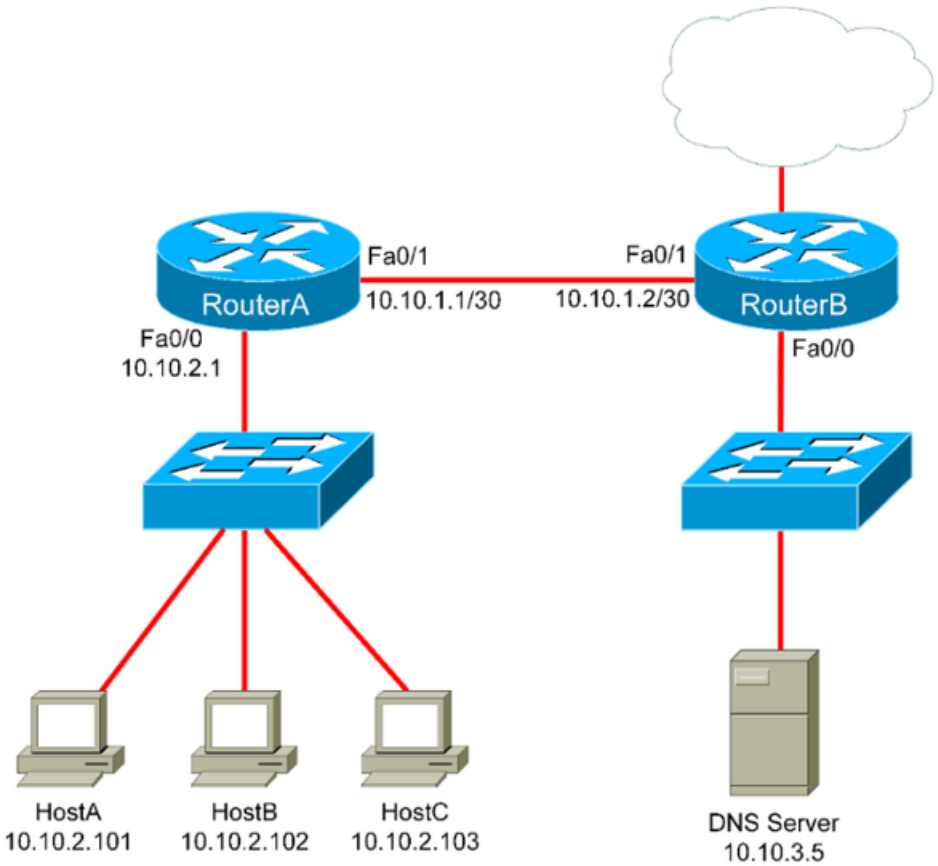
Category:

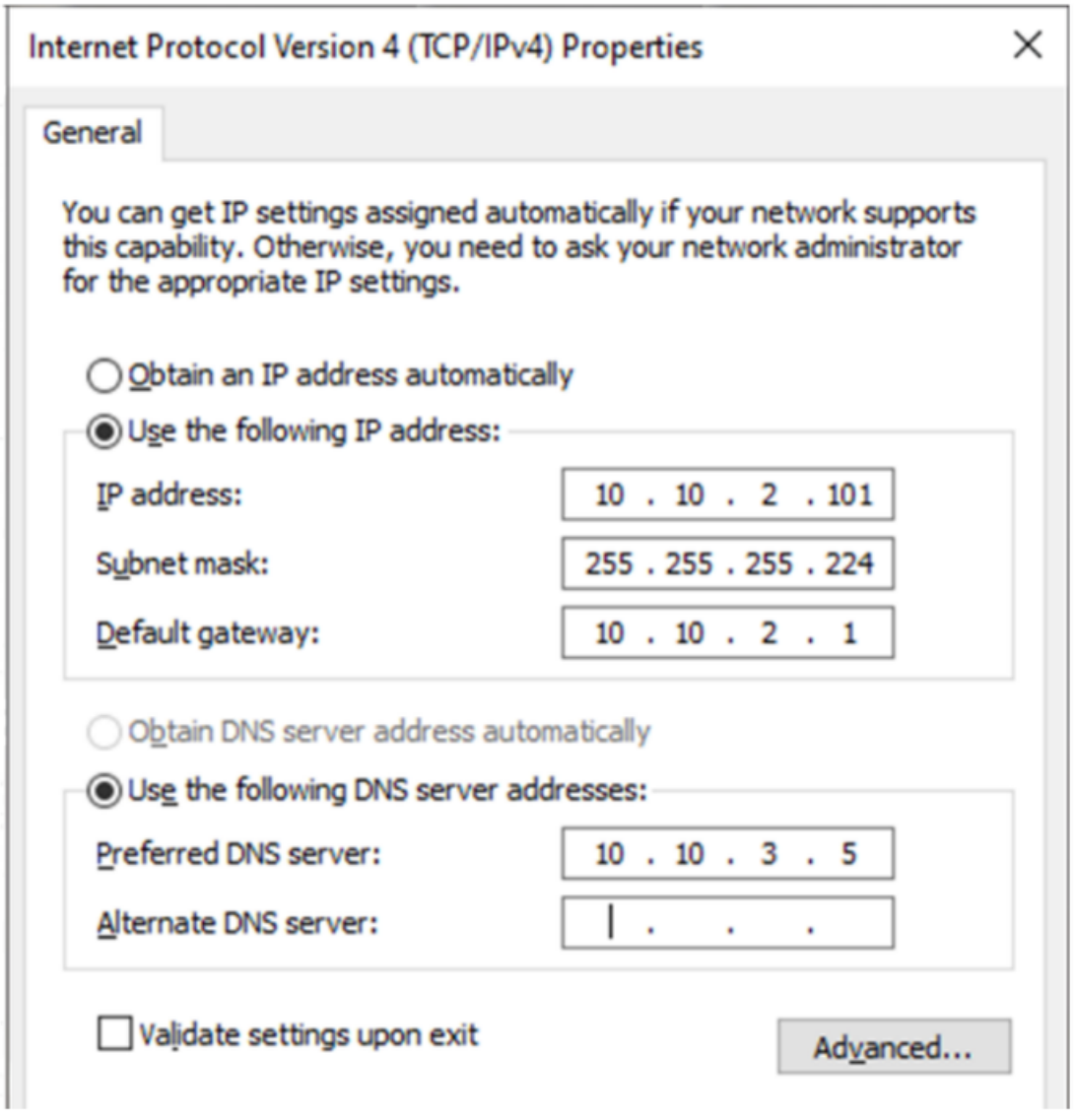
IP Connectivity

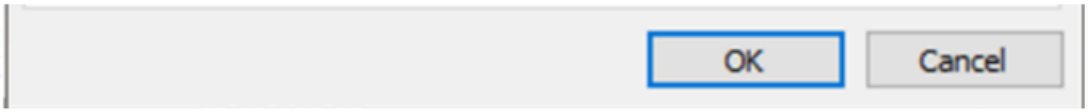
Days 13-15 - IPv4 Subnetting

Original Question 25/74

25/74 OVO




Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 10 . 10 . 2 . 101
Subnet mask: 255 . 255 . 255 . 224
Default gateway: 10 . 10 . 2 . 1
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server: 10 . 10 . 3 . 5
Alternate DNS server: . . .
Validate settings upon exit
Advanced...



Refer to the exhibit above. HostA cannot connect to the Internet. HostB and HostC are able to connect to the Internet. The IPv4 configuration for HostA is displayed in the exhibit.

Which of the following parameters should you change to enable HostA to connect to the Internet? *(Select the best answer.)*

- ○ A. the default gateway
- ○ B. the DNS server
- ○ C. the subnet mask
- ○ D. the IP address

Incorrect

Correct Answer(s): C

Explanation:

You should change the subnet mask to enable HostA to connect to the Internet. The subnet mask for HostA is currently 255.255.255.224, which is equivalent to /27 in Classless Inter-Domain Routing (CIDR) notation. The /27 notation means that 27 bits are used for the network portion of the 32-bit Internet Protocol version 4 (IPv4) address. The remaining bits belong to the host portion of the Internet Protocol (IP) address. To determine how many addresses are defined by a subnet mask, use the formula 2^n, where n is the number of bits in the host portion of the address. A /27 subnet mask uses 5 bits for host addresses, so 2^5 equals 32 addresses for the subnet. Networks that are subnetted by using /27 masks are separated into groups of 32 addresses each. For example, the 10.10.2.0 network can be divided into the following eight subnets:

10.10.2.0/27

10.10.2.0/27
10.10.2.32/27
10.10.2.64/27
10.10.2.96/27
10.10.2.128/27
10.10.2.160/27
10.10.2.192/27
10.10.2.224/27

The IP address of HostA is 10.10.2.101, which exists within the 10.10.2.96/27 subnet. However, the IP address of HostA's default gateway is 10.10.2.1, which does not exist within the 10.10.2.96/27 subnet. Changing the subnet mask of HostA to a subnet mask of 255.255.255.0 or 255.255.255.128 would allow HostA to communicate with devices beyond RouterA, including those on the Internet. A subnet mask of 255.255.255.0 corresponds to /24 in CIDR notation. The 10.10.2.0/24 network includes all addresses from 10.10.2.0 through 10.10.2.255. A subnet mask of 255.255.255.128 corresponds to /25 in CIDR notation. The 10.10.2.0/25 network includes all addresses from 10.10.2.0 through 10.10.2.127.

You should not change the IP address to enable HostA to connect to the Internet. The IP address of HostA in the network diagram is the same as the IP address shown in the configuration window of HostA. Furthermore, HostB and HostC are able to connect to the Internet; therefore, those devices must be configured to use a different subnet mask than the one used by HostA.

You should not change the default gateway of HostA. The FastEthernet 0/0 interface of RouterA is the default gateway for HostA, HostB, and HostC. The IP address of RouterA's FastEthernet 0/0 interface is displayed in the network diagram as 10.10.2.1, which is the same as the IP address of the default gateway in the configuration window of HostA; therefore, HostA is already configured with the correct default gateway.

You should not change the Domain Name Server (DNS) address of HostA. The IP address of the DNS server address is displayed in the network diagram as 10.10.3.5, which is the same as the preferred DNS server address in the configuration window of HostA; therefore, HostA is already configured with the correct DNS server. If HostA had been able to connect to devices on the Internet by their IP addresses but not by their fully qualified domain names (FQDNs), HostA might need to use a different DNS server. However, HostA is unable to connect to the Internet at all.

Reference:
Boson CCNA Curriculum 200-301, Module 2: Network Addressing and Transport, Subnetting

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 1, Chapter 14: Analyzing Existing Subnets, Analyzing Existing Subnets: Decimal

Cisco: Configure IP Addresses and Unique Subnets for New Users

Category:
Network Fundamentals

Days 13-15 - IPv4 Subnetting

Original Question 31/74

31/74 OVO

You have been asked to summarize your network addresses into a single summarized subnet. The addresses on your network consist of the following addresses:

- 10.1.1.64/27
- 10.1.1.96/27
- 10.1.1.128/27
- 10.1.1.160/27

In order to preserve address space, you should summarize these networks by using the most efficient subnet mask.

Which of the following subnet masks should you use? *(Select the best answer.)*

- A. /26
- B. /25
- C. /23
- D. /24

Incorrect

Correct Answer(s): D

Explanation:

You should use a /24 subnet mask to summarize the four /27 subnets in this scenario. Summarization involves combining subnets to create a larger subnet. The following list demonstrates the relationship between subnets and Internet Protocol (IP) addresses:

- /20 contains 2^{12} = 4,096 addresses
- /21 contains 2^{11} = 2,048 addresses
- /22 contains 2^{10} = 1,024 addresses
- /23 contains 2^{9} = 512 addresses
- /24 contains 2^{8} = 256 addresses
- /25 contains 2^{7} = 128 addresses
- /26 contains 2^{6} = 64 addresses
- /27 contains 2^{5} = 32 addresses
- /28 contains 2^{4} = 16 addresses
- /29 contains 2^{3} = 8 addresses
- /30 contains 2^{2} = 4 addresses

You can use a "rule of two" to determine how many subnets can be summarized within a larger subnet. Each step up to a larger subnet multiplies the number of available subnets by two. The following list demonstrates how many /27 subnets can be summarized within each larger subnet:

- Two /27 subnets can be summarized within a /26 subnet
- Four /27 subnets can be summarized within a /25 subnet
- Eight /27 subnets can be summarized within a /24 subnet
- Sixteen /27 subnets can be summarized within a /23 subnet...
- ...and so on.

Because there are four /27 subnets on your network, you must use a /25 or larger subnet to summarize these /27 subnets. Therefore, a /26 subnet cannot summarize the networks in this scenario.

Each of the /27 subnets in this scenario contains 32 addresses. The following address ranges exist on each of the subnets in this scenario:

- 10.1.1.64/27 = 10.1.1.64 through 10.1.1.95
- 10.1.1.96/27 = 10.1.1.96 through 10.1.1.127
- 10.1.1.128/27 = 10.1.1.128 through 10.1.1.159
- 10.1.1.160/27 = 10.1.1.160 through 10.1.1.191

A /25 subnet contains 128 addresses. For example, the 10.1.1.0/25 subnet contains addresses from 10.1.1.0 through 10.1.1.127, and the 10.1.1.128/25 subnet contains addresses from 10.1.1.128 through 10.1.1.255. Both of these subnets together would encompass the /27 subnet ranges in this scenario, but you cannot summarize all of these /27 subnets by using a single /25 subnet. Therefore, you must use a larger subnet.

A /24 subnet contains 256 addresses. For example, the 10.1.1.0/24 network contains addresses from 10.1.1.0 through 10.1.1.255. This address range contains all of the /27 subnets in this scenario. Therefore, you can use a /24 subnet to summarize the addresses in this scenario.

A /23 subnet contains 512 addresses. For example, the 10.1.1.0/23 network contains addresses from 10.1.1.0 through 10.1.2.255. This address range is enough to contain two /24 subnets, four /25 subnets, eight /26 subnets, and so on. Although a /23 subnet can summarize the /27 subnets in this scenario, a /23 subnet is not as efficient as a /24 subnet.

Reference:
Boson CCNA Curriculum 200-301, Module 2: Network Addressing and Transport, Subnetting

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 1, Chapter 14: Analyzing Existing Subnets, Defining a Subnet

Cisco: Configure IP Addresses and Unique Subnets for New Users

Category:
Network Fundamentals

Days 13-15 - IPv4 Subnetting

Original Question 70/74

70/74

Select the addresses, and place each address into the appropriate category. Use all addresses. Some boxes underneath the categories will not be filled.

Addresses	Subnet Address	Host Address	Broadcast Address
192.168.2.32/27			
192.168.2.32/26			
192.168.4.127/25			
192.168.4.255/23			
192.168.12.0/22			
192.168.12.0/21			

Reset

Incorrect

Answer:

You should place the addresses as shown in the following graphic:

Subnet Address	Host Address	Broadcast Address
192.168.2.32/27	192.168.2.32/26	192.168.4.127/25
192.168.12.0/22	192.168.4.255/23	
	192.168.12.0/21	

Explanation:

The first address in a subnet is the subnet address, the last address in a subnet is the broadcast address, and all the addresses in between are host addresses. Therefore, to determine whether an address is a subnet address, broadcast address, or host address, you must determine what address range is used for each subnet.

The /27 in the 192.168.2.32/27 address indicates that 27 bits belong to the network portion of a 32-bit Internet Protocol version 4 (IPv4) address. The remaining bits belong to the host portion of the Internet Protocol (IP) address. To determine how many addresses are defined by a subnet mask, use the formula 2^n, where n is the number of bits in the host portion of the address. A /27 subnet mask uses 5 bits for host addresses, so 2^5 equals 32 addresses for the subnet. Networks that are subnetted by using /27 masks are separated into groups of 32 addresses each. For example, the 192.168.2.0 network can be divided into the following eight subnets:

192.168.2.0/27
192.168.2.32/27
192.168.2.64/27
192.168.2.96/27
192.168.2.128/27
192.168.2.160/27
192.168.2.192/27
192.168.2.224/27

These addresses are the subnet addresses for each subnet defined by the subnet mask. Therefore, the 192.168.2.32/27 address is a subnet address. The last address in this subnet, 192.168.2.63/27, is the broadcast address, and the 30 addresses from 192.168.2.33/27 through 192.168.2.62/27 are host addresses.

The /26 in the 192.168.2.32/26 address indicates that 26 bits belong to the network portion of a 32-bit IP address and 6 bits belong to the host portion. Because 2^6 equals 64, networks that are subnetted by using /26 masks are separated into groups of 64 addresses each. For example, the 192.168.2.0 network can be divided into the following four subnets:

192.168.2.0/26
192.168.2.64/26
192.168.2.128/26
192.168.2.192/26

The 192.168.2.32/26 address is not a subnet address, because it does not match any of these subnet addresses. Nor is 192.168.2.32/26 a broadcast address, because it is not the last address in any of these subnets. Therefore, the 192.168.2.32/26 address is a host address within the 192.168.2.0/26 subnet.

The /25 in the 192.168.4.127/25 address indicates that 25 bits belong to the network portion and 7 bits belong to the host portion. Because 2^7 equals 128, networks that are subnetted by using /25 masks are separated into groups of 128 addresses each. For example, the 192.168.4.0 network can be

masks are separated into groups of 128 addresses each. For example, the 192.168.4.0 network can be divided into the following two subnets:

192.168.4.0/25
192.168.4.128/25

The 192.168.4.127/25 address is not a subnet address, because it does not match either of these subnet addresses. However, 192.168.4.127/25 is a broadcast address, because it is the last address in the 192.168.4.0/25 subnet.

The /23 in the 192.168.4.255/23 address indicates that 23 bits belong to the network portion and 9 bits belong to the host portion. Because 2^9 equals 512, networks that are subnetted by using /23 masks are separated into groups of 512 addresses each. A single Class C network address includes 256 addresses, so you must combine two networks to create a single subnet with 512 addresses. For example, you can create the following subnets:

192.168.0.0/23
192.168.2.0/23
192.168.4.0/23
192.168.6.0/23
192.168.8.0/23
192.168.10.0/23
...and so on

All of the addresses from 192.168.4.0/23 to 192.168.5.255/23 are on the same subnet; the next subnet starts with 192.168.6.0/23. The 192.168.4.255/23 address is not a subnet address, because the subnet address is 192.168.4.0/23. Nor is 192.168.4.255/23 a broadcast address, because the broadcast address for this subnet is 192.168.5.255/23. Because the 510 addresses from 192.168.4.1/23 to 192.168.5.254/23 are host addresses, the 192.168.4.255/23 address is a host address within the 192.168.4.0/23 subnet.

The /22 in the 192.168.12.0/22 address indicates that 22 bits belong to the network portion and 10

bits belong to the host portion. Because 2^{10} equals 1,024, networks that are subnetted by using /22 masks are separated into groups of 1,024 addresses each. You must combine four Class C networks to create a single subnet with 1,024 addresses. For example, you can create the following subnets:

192.168.0.0/22
192.168.4.0/22
192.168.8.0/22
192.168.12.0/22
192.168.16.0/22
192.168.20.0/22
...and so on

The 192.168.12.0/22 address is a subnet address, because it is the first address in the 192.168.12.0/22 subnet. The 192.168.15.255/22 address is the broadcast address for the subnet, and the 1,022 addresses from 192.168.12.1/22 to 192.168.15.254/22 are host addresses.

The /21 in the 192.168.12.0/21 address indicates that 21 bits belong to the network portion and 11 bits belong to the host portion. Because 2^{11} equals 2,048, networks that are subnetted by using /21 masks are separated into groups of 2,048 addresses each. You must combine eight Class C networks to create a single subnet with 2,048 addresses. For example, you can create the following subnets:

192.168.0.0/21
192.168.8.0/21
192.168.16.0/21
192.168.24.0/21
192.168.32.0/21
192.168.40.0/21
...and so on

All of the addresses from 192.168.8.0/21 through 192.168.15.255/21 are on the same subnet. The 192.168.12.0/21 address is not a subnet address, because the subnet address for this subnet is 192.168.8.0/21. Nor is 192.168.12.0/21 a broadcast address, because the broadcast address for this

subnet is 192.168.15.255/21. Because the 2,046 addresses from 192.168.8.1/21 through 192.168.15.254/21 are host addresses, the 192.168.12.0/21 address is a host address within the 192.168.8.0/21 subnet.

Although it is important to learn the formula for calculating how many addresses exist on a subnet, the following list demonstrates the relationship between subnet masks and valid host addresses:

- /16 contains 2^{16} = 65,536 addresses (2^{16} – 2 = 65,534 hosts)
- /17 contains 2^{15} = 32,768 addresses (2^{15} – 2 = 32,766 hosts)
- /18 contains 2^{14} = 16,384 addresses (2^{14} – 2 = 16,382 hosts)
- /19 contains 2^{13} = 8,192 addresses (2^{13} – 2 = 8,190 hosts)
- /20 contains 2^{12} = 4,096 addresses (2^{12} – 2 = 4,094 hosts)
- /21 contains 2^{11} = 2,048 addresses (2^{11} – 2 = 2,046 hosts)
- /22 contains 2^{10} = 1,024 addresses (2^{10} – 2 = 1,022 hosts)
- /23 contains 2^{9} = 512 addresses (2^{9} – 2 = 510 hosts)
- /24 contains 2^{8} = 256 addresses (2^{8} – 2 = 254 hosts)
- /25 contains 2^{7} = 128 addresses (2^{7} – 2 = 126 hosts)
- /26 contains 2^{6} = 64 addresses (2^{6} – 2 = 62 hosts)
- /27 contains 2^{5} = 32 addresses (2^{5} – 2 = 30 hosts)
- /28 contains 2^{4} = 16 addresses (2^{4} – 2 = 14 hosts)
- /29 contains 2^{3} = 8 addresses (2^{3} – 2 = 6 hosts)
- /30 contains 2^{2} = 4 addresses (2^{2} – 2 = 2 hosts)

Reference:

Boson CCNA Curriculum 200-301, Module 2: Network Addressing and Transport, Subnetting

Boson CCNA Curriculum 200-301, Module 2: Network Addressing and Transport, Subnetting

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 1, Chapter 14: Analyzing Existing Subnets, Defining a Subnet

Cisco: Configure IP Addresses and Unique Subnets for New Users

Category:
Network Fundamentals

Days 16-18 - VLANs

Original Question 17/74

17/74 OVO

Which of the following commands should you issue to configure a switch to send traffic from VLAN 17 as untagged? *(Select the best answer.)*

- A. **switchport trunk allowed vlan remove 17**
- B. **switchport trunk native vlan 17**
- C. **switchport voice vlan 17**
- D. **switchport trunk allowed vlan add 17**

Incorrect

Correct Answer(s): B

Explanation:

You should issue the **switchport trunk native vlan 17** command to configure a switch to send traffic from virtual local area network (VLAN) 17 as untagged. The **switchport trunk native vlan 17** command is used to configure a trunk port to use VLAN 17 as the native VLAN. Traffic that is passed over the native VLAN is sent untagged, which means that the packet is sent without 802.1Q encapsulation. Changing the native VLAN can have unexpected consequences. For example, when Dynamic Trunking Protocol (DTP) is used to negotiate the formation of a trunk link between switches, DTP uses the native VLAN for its packets. If the native VLAN is not the same on both ends of the link, a trunk will not dynamically form. In addition, if a trunk is manually formed and the native VLAN is not the same on both ends of the link, any untagged traffic received on the trunk link is considered to belong to the locally configured native VLAN, not the native VLAN configured on the sending side of

the link. This could potentially cause connectivity issues such as failed pings between devices that reside in the same VLAN on both sides of the trunk.

You should not issue the **switchport trunk allowed vlan add 17** command or the **switchport trunk allowed vlan remove 17** command. The **switchport trunk allowed vlan** {**add** | **all** | **except** | **remove**} *vlan-list* command is issued from interface configuration mode to manually prune VLANs. Manual pruning enables an administrator to strictly specify which VLANs are allowed or denied on a trunk port. The **switchport trunk allowed vlan add 17** command configures a trunk port to allow VLAN 17 traffic, and the **switchport trunk allowed vlan remove 17** command configures a trunk port to disallow VLAN 17 traffic. In either case, VLAN 17 traffic is tagged unless VLAN 17 is configured as the native VLAN.

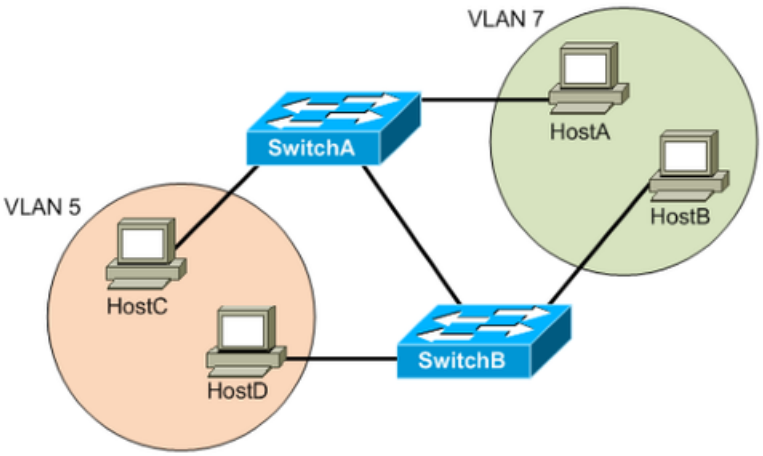
You should not issue the **switchport voice vlan 17** command. The **switchport voice vlan** *vlan-id* command configures voice traffic to be sent over a user-specified voice VLAN. Voice traffic will be carried in 802.1Q frames and will be carried on a different VLAN than data traffic. If you were to issue the **switchport voice vlan 17** command, voice traffic would be tagged and sent over VLAN 17.

Reference:
Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Configuring Trunk Ports

Days 16-18 - VLANs

Original Question 74/74

74/74



You connect SwitchA and SwitchB as shown above. SwitchA is a Cisco switch, and SwitchB is a non-Cisco switch. You want to be able to trunk VLAN traffic between the two switches.

What commands should you issue on SwitchA? *(Select 2 choices.)*

- ☐ A. `SwitchA(config-if)#switchport trunk encapsulation dot1q`
- ☐ B. `SwitchA(config-if)#switchport mode access`
- ☐ C. `SwitchA(config)#switchport trunk encapsulation isl`
- ☐ D. `SwitchA(config)#switchport trunk encapsulation dot1q`
- ☐ E. `SwitchA(config-if)#switchport mode trunk`
- ☐ F. `SwitchA(config-if)#switchport trunk encapsulation isl`

Incorrect

Correct Answer(s): A, E

Explanation:

You should issue the **switchport mode trunk** command and the **switchport trunk encapsulation dot1q** command in interface configuration mode. A trunk is used to carry traffic from multiple virtual local area networks (VLANs) over a single link. Traffic from each VLAN is encapsulated in a frame with a special header when it is transmitted onto the trunk link and then de-encapsulated when it is received on the other side of the trunk. This encapsulation of frames, which is also referred to as *VLAN tagging*, ensures that traffic does not lose its VLAN membership properties as it passes from one switch to another. To enable trunking on the port, you must enter interface configuration mode and issue the **switchport mode trunk** command.

There are two trunking encapsulation methods: Inter-Switch Link (ISL) and 802.1Q. ISL is a Cisco-proprietary trunking encapsulation method and works only when you are configuring trunking between Cisco devices. The Institute of Electrical and Electronics Engineers (IEEE) 802.1Q standard can be used to trunk between any two industry-standard switches. In this scenario, you must use 802.1Q encapsulation because SwitchB is a non-Cisco switch. Therefore, you should issue the **switchport trunk encapsulation dot1q** command in interface configuration mode on SwitchA.

The following command sequence would configure Gigabit Ethernet port 1/0/1 for trunking with 802.1Q encapsulation and save the configuration:

```
SwitchA#configure terminal
SwitchA(config)#interface gigabitethernet 1/0/1
SwitchA(config-if)#switchport mode trunk
SwitchA(config-if)#switchport trunk encapsulation dot1q
SwitchA(config-if)#exit
SwitchA(config)#exit
SwitchA#copy running-config startup-config
```

You should not issue the **switchport trunk encapsulation isl** command, because SwitchB is not a Cisco device. If SwitchB were a Cisco device, you could issue the **switchport trunk encapsulation isl** command in interface configuration mode on SwitchA. The encapsulation methods on each side of the trunk link must match; therefore, you would also be required to issue the **switchport trunk encapsulation isl** command in interface configuration mode on SwitchB.

You cannot issue the **switchport mode trunk** command in global configuration mode, because it must be issued in interface configuration mode. Similarly, you cannot issue the **switchport trunk encapsulation** command in global configuration mode, because it must be issued in interface configuration mode.

A port that is not configured for trunking is called an access port. Access ports carry traffic for a single VLAN and are typically used to connect a workstation to a switch. To configure a port for access mode, you must enter interface configuration mode and issue the **switchport mode access** command.

Reference:
Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Configuring Trunk Ports

Cisco: Configuring VLANs

Category:
Network Access

Days 20-21 - Spanning Tree Protocol (STP)

Original Question 57/74

57/74

When BPDU guard detects a BPDU on a port, into which state will it place the port? *(Select the best answer.)*

- A. root-inconsistent
- B. shutdown
- C. loop-inconsistent
- D. err-disabled

Incorrect

Correct Answer(s): D

Explanation:

When BPDU guard detects a bridge protocol data unit (BPDU) on a port, it will place the port into the err-disabled state. BPDU guard is used to disable ports that erroneously receive BPDUs. BPDU guard is a Spanning Tree Protocol (STP) enhancement that is typically applied to edge ports that have PortFast enabled. Because PortFast automatically places ports into a forwarding state, a switch that has been connected to a PortFast-enabled port could cause switching loops. However, when BPDU guard is applied, the receipt of a BPDU on a port will result in the port being placed into the err-disabled state, which prevents loops from occurring. BPDU guard should be enabled on ports that have been enabled with PortFast so that BPDU guard can prevent a rogue switch from modifying the STP topology. When such a port receives a BPDU, BPDU guard immediately puts that port into the err-disabled state and shuts down the port. The port must then be manually re-enabled, or it can be recovered automatically by configuring the **errdisable recovery cause bpduguard** command and the **errdisable recovery interval** *interval* command.

The loop guard feature, not BPDU guard, will place a port into the loop-inconsistent state. The loop guard feature prevents nondesignated ports from inadvertently forming bridging loops if the steady flow of BPDUs is interrupted. When the port stops receiving BPDUs, loop guard puts the port into the loop-inconsistent state, which keeps the port in a blocking state. After the port starts receiving BPDUs again, loop guard enables the port to transition through the normal STP states

again, loop guard enables the port to transition through the normal STP states.

The root guard feature, not BPDU guard, will place a port into the root-inconsistent state. Root guard will prevent a switch port from forwarding traffic until it no longer receives superior BPDUs from an adjacent switch. When enabled on a port, the root guard feature prevents the port from becoming a root port if it receives superior BPDUs. Receiving superior BPDUs would cause a root guard–enabled port to enter the root-inconsistent state and to block all data flowing through the port until the port stops receiving superior BPDUs.

Reference:
Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, BPDU Guard

Category:
Network Access

Day 22 - Rapid Spanning Tree Protocol (RSTP)

Original Question 27/74

27/74 OVO

Which of the following statements is accurate regarding alternate ports? *(Select the best answer.)*

- A. An alternate port is the port on a segment that has the best path to the root bridge.
- B. An alternate port is a blocked port that receives more useful BPDUs from a port on another switch.
- C. An alternate port is the port on a switch that has the best path to the root bridge.
- D. An alternate port is a blocked port that receives more useful BPDUs from another port on the local switch.
- E. An alternate port is always in the forwarding state.

Incorrect

Correct Answer(s): B

Explanation:

An alternate port is a blocked port that receives more useful bridge protocol data units (BPDUs) from a port on another switch. Rapid Spanning Tree Protocol (RSTP) defines the following four port roles:

- Root port
- Designated port
- Alternate port
- Backup port

The root port on a switch is the port that receives the best BPDU, which indicates the best path to the root bridge based on the best root port cost. All switches except the root bridge contain exactly one root port. Because there is only one best path to the root bridge, a switch can have only one root port; only the root bridge does not have a root port. Root ports are always in the forwarding state.

The best path to the root bridge is the one that has the lowest cost to the root bridge. The cost is based on the bandwidth of a link. The higher the bandwidth, the lower the cost. RSTP uses the following link costs by default:

Bandwidth	Cost
4 Mbps	250
10 Mbps	100
16 Mbps	62
45 Mbps	39
100 Mbps	19
155 Mbps	14
622 Mbps	6
1 Gbps	4
10 Gbps	2

In the event of a tie, the port connected to the device with the lowest bridge ID (BID) becomes the root port. If both ports are connected to the same switch, the port that receives the BPDU with the lowest sending port ID becomes the root port. The port ID consists of the port priority and the port number. The port configured with the lowest port priority will also have the lowest port ID; port numbers are considered only when port priorities are equal.

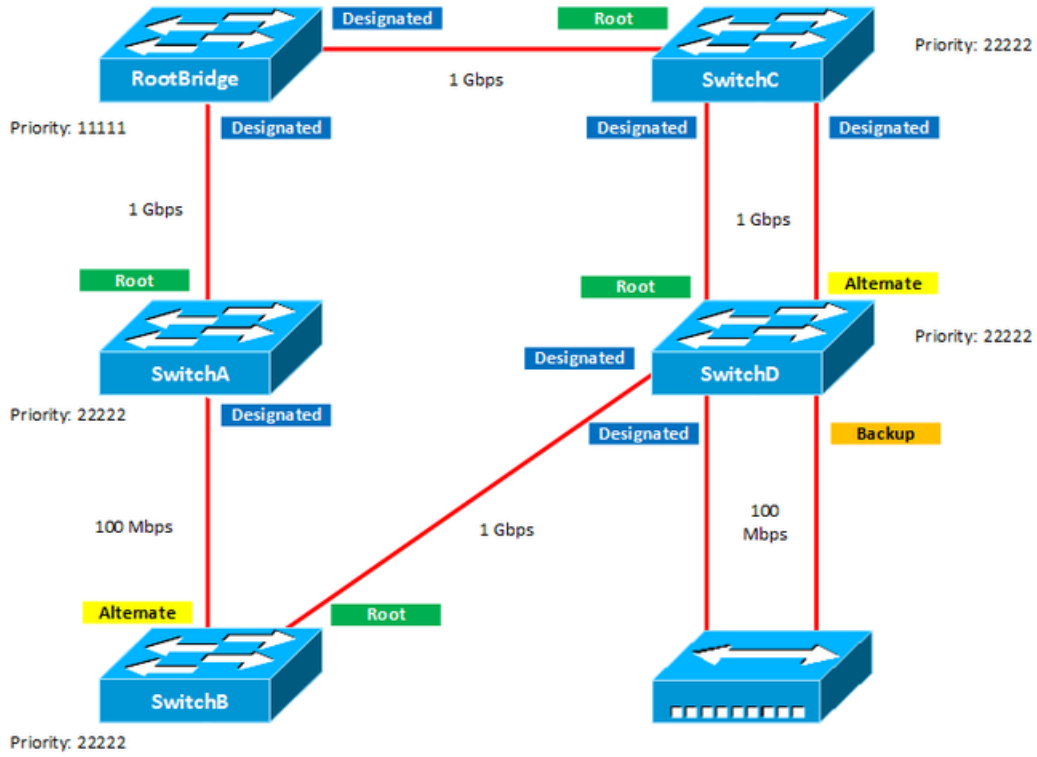
A designated port is the port on a segment that has the best path to the root bridge. One designated port is selected for each segment. If multiple ports on a segment have the same root port cost, the port on the switch with the lowest BID becomes the designated port. Switches may have one or more designated ports, and some switches may not have any designated ports. All the ports on a root bridge are designated ports. Designated ports are normally in the forwarding state.

The blocking port role in Spanning Tree Protocol (STP) is split into two RSTP port roles: the alternate port role and the backup port role. An alternate port receives more useful BPDUs from a designated port located on another switch, and a backup port receives more useful BPDUs from a designated port on the switch itself. For a port to receive more useful BPDUs from the same switch, it must be connected to the same collision domain as another port on the same switch or it must be connected to itself by a loopback device, such as a loopback adapter. An alternate port guarantees a path to the root bridge should the current root port become unavailable; however, a backup port only guarantees redundant access to a particular network segment and not necessarily an alternate path to the root bridge. Alternate ports and backup ports are always in the blocking state.

The following graphic displays the correlation between the various RSTP port roles:

Day 22 - Rapid Spanning Tree Protocol (RSTP)

Original Question 27/74 - continued



Reference:

Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, RSTP Alternate and Backup Port Roles

Cisco: Understand Rapid Spanning Tree Protocol (802.1w)

Category:

Network Access

OVO

Day 22 - Rapid Spanning Tree Protocol (RSTP)

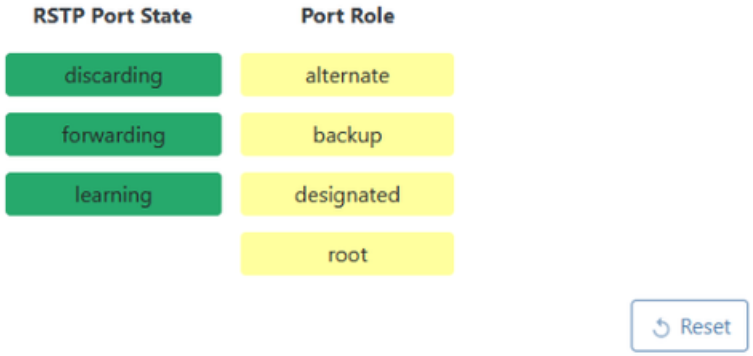
Original Question 27/74 - continued

Day 22 - Rapid Spanning Tree Protocol (RSTP)

Original Question 37/74

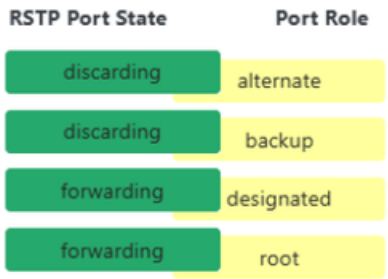
37/74

RSTP is configured on your company's network, and RSTP convergence is completed. Drag the RSTP port states, and place them on the appropriate port role. Choices can be used more than once. Some choices may not be used.



Incorrect

Answer:



Explanation:

Rapid Spanning Tree Protocol (RSTP) is used to prevent bridging loops in a network. Defined by the Institute of Electrical and Electronics Engineers (IEEE) 802.1w standard, RSTP was created to increase the convergence speed of Spanning Tree Protocol (STP). Unlike STP, which has five port states, RSTP has only three: discarding, learning, and forwarding. The disabled, blocking, and listening states of STP are combined into the discarding state in RSTP. RSTP designates how individual ports should behave by assigning one of four port roles to each port in the topology. These port roles are alternate, backup, designated, and root. Only the designated and root port roles are used to forward

packets. Alternate and backup ports remain in a discarding state.

After RSTP convergence, the alternate and backup ports are placed into the discarding state. Alternate ports reside on switches that provide an alternate path to the root bridge in the event of a failure of the designated device. Backup ports reside on designated switches and can be used for failover if a designated port fails. After the failure of a designated port, a backup port assumes the role of the designated port and moves from the discarding state to the forwarding state.

After RSTP convergence, the root and designated ports are placed into the forwarding state. Root ports provide the lowest path cost to the root bridge. Root ports receive the highest-priority bridge protocol data units (BPDUs) from the root bridge. Only one root port can be configured per network segment. Designated ports receive frames sent to the root bridge. Designated ports are used to send the highest-priority BPDUs on a network segment. Devices that house designated ports are known as designated devices.

After RSTP convergence, the root and designated ports are placed into the forwarding state. Root ports provide the lowest path cost to the root bridge. Root ports receive the highest-priority bridge protocol data units (BPDUs) from the root bridge. Only one root port can be configured per network segment. Designated ports receive frames sent to the root bridge. Designated ports are used to send the highest-priority BPDUs on a network segment. Devices that house designated ports are known as designated devices.

The learning state is a transition state that occurs while STP and RSTP convergence is occurring. During the learning state, ports that will later be used for forwarding learn device Media Access Control (MAC) addresses in the network topology. The MAC address entries are then stored in a filtering database on the device. These MAC addresses are used to forward packets once the port transitions to the forwarding state.

Reference:
Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Understanding RSTP Port States

Cisco: Understand Rapid Spanning Tree Protocol (802.1w): Port Roles

Day 22 - Rapid Spanning Tree Protocol (RSTP)

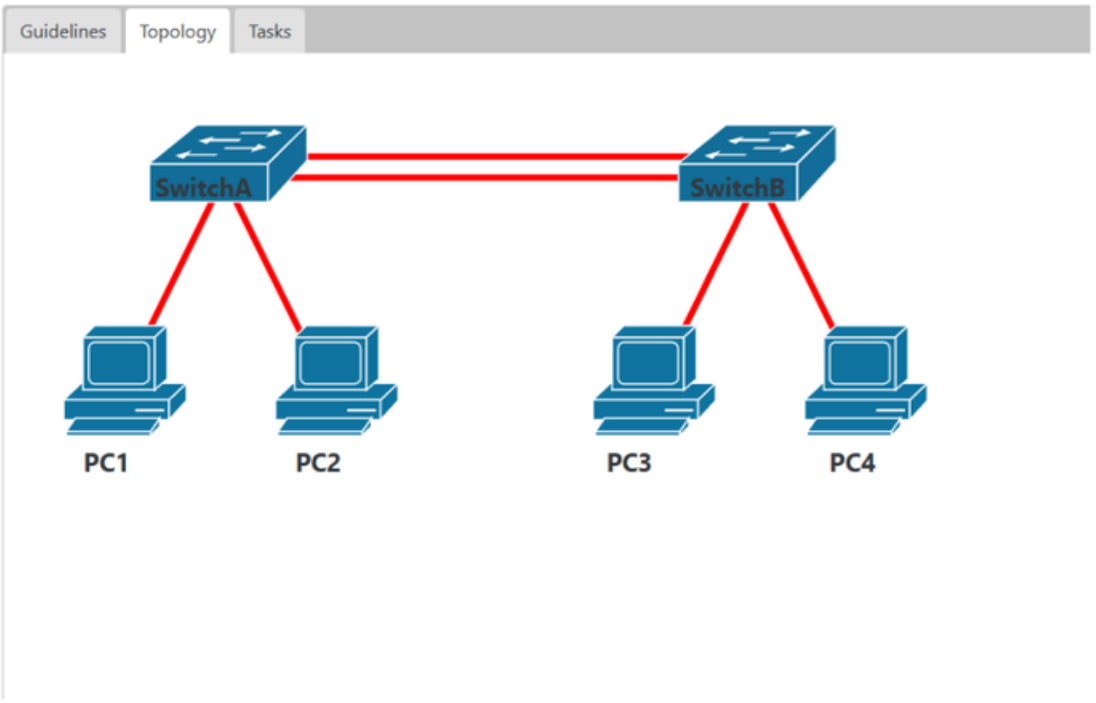
Original Question 37/74 - continued

Category:
Network Access

Day 23 - EtherChannel

Original Question 1/74

1/74



Day 23 - EtherChannel

Original Question 1/74 - continued

Tasks

Your company's network includes two Cisco switches, SwitchA and SwitchB. SwitchB has been configured with the correct VLAN, trunk, and EtherChannel settings.

Configure SwitchA with the following parameters:

- Switch ports FastEthernet 0/3, FastEthernet 0/4, FastEthernet 0/8, and FastEthernet 0/9 should be explicitly configured as access ports.
- VLAN 10 should contain switch ports FastEthernet 0/3 and FastEthernet 0/4.
- VLAN 20 should contain switch ports FastEthernet 0/8 and FastEthernet 0/9.
- Switch ports FastEthernet 0/1 and FastEthernet 0/2 should be configured to use a standards-based EtherChannel negotiation protocol.
- Switch ports FastEthernet 0/1 and FastEthernet 0/2 should be configured as members of EtherChannel port-group 1 and should actively negotiate EtherChannel links.
- The EtherChannel virtual interface should be configured to use a standards-based encapsulation protocol and to always function as a trunk.

Day 23 - EtherChannel

Original Question 14/74

14/74 OVO

You want to establish an EtherChannel between SwitchA and SwitchB by using a Cisco-proprietary protocol. After you configure the ports on SwitchA with the correct channel protocol, you issue the channel-group 1 mode auto command on those ports.

Which of the following commands should you issue on SwitchB? *(Select 2 choices.)*

- ☐ A. **channel-protocol lacp**
- ☐ B. **channel-group 1 mode desirable**
- ☐ C. **channel-group 1 mode passive**
- ☐ D. **channel-group 1 mode auto**
- ☐ E. **channel-protocol pagp**
- ☐ F. **channel-group 1 mode active**

Incorrect

Correct Answer(s): B, E

Explanation:

You should issue the **channel-protocol pagp** command and the **channel-group 1 mode desirable** command on SwitchB. EtherChannel is used to bundle two or more identical, physical interfaces into a single logical link between switches. An EtherChannel can be permanently established between switches, or it can be negotiated by using one of two aggregation protocols: the Cisco-proprietary Port Aggregation Protocol (PAgP) or the open-standard Institute of Electrical and Electronics Engineers (IEEE) 802.3ad protocol, which is also known as Link Aggregation Control Protocol (LACP). To configure a switch port to use a particular aggregation protocol, you should use the **channel-protocol** {**lacp** | **pagp**} command.

You can issue the **show etherchannel summary** command to verify the status of an EtherChannel link and to determine which aggregation protocol, if any, was used to establish the link. The following sample output indicates that the PAgP protocol was used to successfully establish an EtherChannel link:

```
SwitchA#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-along s - suspended
        H - Hot-standby (LACP only)
```

Day 23 - EtherChannel

Original Question 14/74 - continued

```
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
Number of channel-groups in use: 1
Number of aggregators:           1
```

```
Group  Port-channel  Protocol    Ports
------------------------------------------------------------------
1      Po1(US)         PAgP        Fa0/5(P)   Fa0/6(P)
```

An EtherChannel can have up to eight active switch ports in the bundle that forms the logical link between switches. Every switch port in the bundle, which is also referred to as a channel group, must be configured with the same speed and duplex settings. To configure a switch port to be a member of a particular channel group, you should issue the **channel-group** *number* **mode** {**on** | **active** | **passive** | {**auto** | **desirable**} [**non-silent**]} command. This command uses the *number* keyword to specify a particular channel group. The supported values for the *number* keyword vary depending on hardware platform and IOS revision.

The following table displays the channel-group configurations that will establish an EtherChannel:

SwitchB / SwitchA	**off**	**auto**	**desirable**	**passive**	**active**	**on**
off	NO	NO	NO	NO	NO	NO
auto	NO	NO	PAgP	NO	NO	NO

Day 23 - EtherChannel

Original Question 14/74 - continued

The **on** keyword configures the channel group to unconditionally create the channel with no LACP or PAgP negotiation. In the **on** mode, an EtherChannel exists only if a channel group that is in the on mode is connected to another channel group that is also in the **on** mode.

The **auto**, **desirable**, and **non-silent** keywords can be used only with PAgP. The **desirable** keyword configures the channel group to actively negotiate PAgP, and the **auto** keyword configures the channel group to listen for PAgP negotiation to be offered. Either or both sides of the link must be set to **desirable** to establish an EtherChannel over PAgP; setting both sides to auto will not establish an EtherChannel over PAgP. The optional **non-silent** keyword requires that a port receive PAgP packets before the port is added to the channel.

The **active** and **passive** keywords can be used only with LACP. The **active** keyword configures the channel group to actively negotiate LACP, and the **passive** keyword configures the channel group to listen for LACP negotiation to be offered. Either or both sides of the link must be set to active to establish an EtherChannel over LACP; setting both sides to **passive** will not establish an EtherChannel over LACP.

Reference:
Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Configuring PAgP

Day 23 - EtherChannel

Original Question 14/74 - continued

EtherChannel

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 1, Chapter 10: RSTP and EtherChannel Configuration, Configuring Layer 2 EtherChannel

Cisco: Catalyst 3750 Switch Cisco IOS Commands: channel-group

Cisco: Catalyst 3750 Switch Cisco IOS Commands: channel-protocol

Category:
Network Access

Day 24 - Dynamic Routing

Original Question 9/74

9/74 OVO

You administer a router that contains five routes to the same network: a static route, a RIPv2 route, an IGRP route, an OSPF route, and an internal EIGRP route. The default ADs are used. The link to the static route has just failed.

Which of the following routes will be used? *(Select the best answer.)*

- A. the OSPF route
- B. the EIGRP route
- C. the IGRP route
- D. the RIPv2 route
- E. both the RIPv2 route and the EIGRP route

Correct Answer(s): B

Explanation:

The Enhanced Interior Gateway Routing Protocol (EIGRP) route will be used when the link to the static route goes down. When multiple routes to a network exist and each route uses a different routing protocol, a router prefers the routing protocol with the lowest administrative distance (AD). The following list contains the most commonly used ADs:

Route Source	AD
Directly connected route	0
Static route	1
EIGRP summary route	5
eBGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
iBGP	200
Unknown	255

In this scenario, the static route has the lowest AD. Therefore, the static route is used instead of the other routes. When the static route fails, the EIGRP route is preferred, because internal EIGRP has an AD of 90.

If the EIGRP route were to fail, the Interior Gateway Routing Protocol (IGRP) route would be preferred, because IGRP has an AD of 100. If the IGRP route were also to fail, the Open Shortest Path First (OSPF) route would be preferred, because OSPF has an AD of 110. The Routing Information Protocol version 2 (RIPv2) route would not be used unless all of the other links were to fail, because RIPv2 has an AD of 120.

ADs for a routing protocol can be manually configured by issuing the **distance** command in router configuration mode. For example, to change the AD of Routing Information Protocol (RIP) from 120 to 80, you should issue the following commands:

```
RouterA(config)#router rip
RouterA(config-router)#distance 80
```

You can view the AD of the best route to a network by issuing the **show ip route** command. The AD is the first number inside the brackets in the output. For example, the following router output shows an OSPF route with an AD of 160:

```
Router#show ip route
<output omitted>

Gateway of last resort is 10.19.54.20 to network 10.140.0.0

O E2 172.150.0.0 [160/5] via 10.19.54.6, 0:01:00, Ethernet2
```

The number 5 in the brackets above is the OSPF metric, which is based on cost. OSPF calculates cost based on the bandwidth of an interface: the higher the bandwidth, the lower the cost. When two OSPF paths exist to the same destination, the router will choose the OSPF path with the lowest cost.

Reference:
Boson CCNA Curriculum 200-301, Module 6: IP Routing, Understanding AD

Cisco: Describe Administrative Distance

Category:

IP Connectivity

Day 24 - Dynamic Routing

Original Question 21/74

21/74 OVO

Which of the following is most likely to be configured with a higher AD value than an OSPF path that is learned by the same router? *(Select the best answer.)*

- ○ A. a directly connected route
- ○ B. a floating static route
- ○ C. a fully specified static route
- ○ D. a default route

Correct

Correct Answer(s): B

Explanation:

Of the available choices, a floating static route is more likely than the others to be configured with a higher administrative distance (AD) value than an Open Shortest Path First (OSPF) path that is learned by the same router. Static routes are paths that are manually configured by a router administrator. A static route defines a specific path for the router to use to send packets that are either destined for a specific network or destined for any network that is not otherwise specifically defined in the routing table.

Floating static routes are typically configured with an AD value that is numerically higher than a dynamic routing protocol, which ensures that the routing protocol path is always selected by the router unless the path becomes unavailable. For example, to create a floating static route that would be used as a backup link to an OSPF route, you might issue the following command:

ip route 192.168.171.0 255.255.255.0 192.168.1.5 111

Assuming that the default OSPF AD of 110 is used for the primary link, this floating static route would be used if the OSPF route to 192.168.171.0/24 were to become unavailable. The floating static route would then direct traffic to the next hop address 192.168.1.5. However, if the AD of the floating static route were set to a value of 109, it would always be preferred over the OSPF route.

A fully specified static route is not likely to be configured with a higher AD value than an OSPF path that is learned by the same router. A fully specified static route is a single manually configured path to a given network or host through a specific interface or next-hop Internet Protocol (IP) address. By default, a static route is considered less trustworthy than a route to a network that is directly connected to the router and more trustworthy than a route that has been learned dynamically. Because static routes are manually configured, the CPU is not tasked with calculating paths. Therefore, CPU overhead is kept low. One use for a static route might be to keep overhead low on a

stub router. Stub routers are routers for stub networks, which are networks that do not forward path information to other networks and do not receive path information from other networks.

A default route is not likely to be configured with a higher AD value than an OSPF path that is learned by the same router. A default route is used to send packets that are destined for a location that is not listed elsewhere in the routing table. If a router were required to store a specific route to every destination network on the Internet, routing tables would grow exponentially and router performance would deteriorate. Default routes help keep routing tables small. A default route can be a static route manually configured by an administrator, or it can be a dynamically learned route advertised by another router.

A directly connected route is not likely to be configured with a higher AD value than an OSPF path that is learned by the same router. A directly connected route automatically appears in the routing table for each interface that has an IP address and that is in an active state. If an interface is configured with an IP address in a particular network, a directly connected route to that network will not appear in the routing table until the interface becomes active. For example, an interface that is configured with an IP address but that is also in an administratively down state will not cause a directly connected route to automatically appear in the routing table. In addition, if the interface was previously in an active state, the existing directly connected route is automatically removed from the routing table when the interface enters an inactive state. By default, directly connected routes are considered to be the most trustworthy.

Reference:
Boson CCNA Curriculum 200-301, Module 6: IP Routing, Configuring a Floating Static Route

Cisco: Configuring Static Routing: Information About Static Routing

Cisco: Cisco IOS IP Addressing Services Command Reference: ip route

Category:
IP Connectivity

Day 24 - Dynamic Routing

Original Question 21/74 - continued

Day 25 - RIP & EIGRP

Original Question 35/74

35/74

You are trying to configure Router1 to perform unequal-cost load balancing over OSPF. However, only one OSPF route exists to the destination network.

Which of the following must you do to enable unequal-cost load balancing? *(Select the best answer.)*

- ○ A. Issue the **ip ospf cost 1** command on all interfaces.
- ○ B. Configure the variance to a value higher than 1.
- ○ C. Configure the routers to use EIGRP.
- ○ D. Adjust the OSPF process IDs so that they match throughout the network.
- ○ E. Adjust the hello timers and dead timers so that they match throughout the network.

Incorrect

Correct Answer(s): C

Explanation:

You should configure the routers to use Enhanced Interior Gateway Routing Protocol (EIGRP). EIGRP supports load balancing over equal-cost and unequal-cost paths. By contrast, Open Shortest Path First (OSPF) supports equal-cost load balancing but does not support unequal-cost load balancing. If multiple OSPF paths to a destination exist and each path has the same bandwidth, OSPF will load balance between the paths.

Issuing the **ip ospf cost 1** command on all interfaces will not enable Router1 to perform unequal-cost load balancing. You can manually configure the OSPF cost of a path through an interface by issuing the **ip ospf cost** *cost* command in interface configuration mode, where *cost* is the path cost that you want to assign. OSPF uses cost, which is based on bandwidth, as its metric. The higher the bandwidth, the lower the cost. OSPF selects the lowest-cost path, which is the path with the highest bandwidth, to a destination.

OSPF does not use variance; therefore, configuring variance to a value higher than 1 will not enable

Router1 to perform unequal-cost load balancing unless you also configure Router1 to use EIGRP. The **variance** command is used to determine whether EIGRP feasible successors can be used for unequal-cost load balancing.

Adjusting the OSPF process IDs so that they match throughout the network will not enable Router1 to perform unequal-cost load balancing. The OSPF process ID is locally significant to the router and can be any positive integer from 1 through 65535. You can specify the OSPF process ID by issuing the **router ospf** *process-id* command.

Adjusting the hello timers and dead timers so that they match throughout the network will not enable Router1 to perform unequal-cost load balancing. OSPF hello timers have to match and OSPF dead timers have to match in order for two routers to form an adjacency. However, all of the routers on the network are set to a hello interval of 10 seconds and a dead interval of 40 seconds; thus the OSPF timers are configured correctly. The hello timer is used to specify the amount of time between sending hello packets, and the dead timer is used to specify the amount of time to wait for hello packets before declaring a neighbor to be down. By default, the hello timer is set to 10 seconds on point-to-point and broadcast links and 30 seconds on nonbroadcast multiaccess (NBMA) links. The dead timer is set to four times the hello timer value by default. To manually configure the hello timer interval, you should issue the **ip ospf hello-interval** *seconds* command in interface configuration mode. To manually configure the dead timer interval, you should issue the **ip ospf dead-interval** *seconds* command in OSPF router configuration mode.

Reference:
Boson CCNA Curriculum 200-301, Module 6: IP Routing, Choosing Between OSPF and EIGRP

Cisco: Troubleshooting Load Balancing Over Parallel Links Using Cisco Express Forwarding: Before Cisco Express Forwarding - Collecting Multiple Paths or Routes

Cisco: How Does Unequal Cost Path Load Balancing (Variance) Work in IGRP and EIGRP?

Category:
IP Connectivity

Days 26-28 - OSPF

Original Question 4/74

4/74

Guidelines | Topology | Tasks

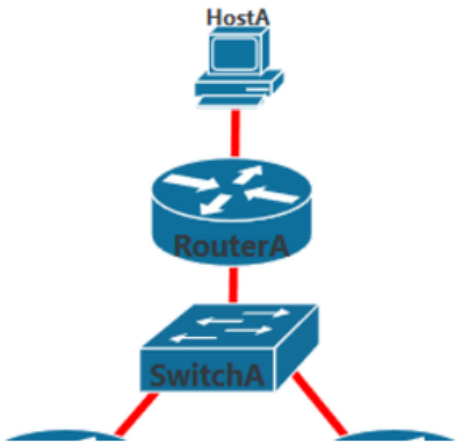
Guidelines

This is a lab item in which tasks are performed on Boson's NetSim virtual devices.

- Refer to the **Tasks** tab to view the tasks for this lab item.
- Refer to the **Topology** tab to access the device console(s) and perform the tasks.
- Console access is available for all required devices by clicking the device icon or by using the tab(s) above.
- All necessary preconfigurations have been applied.
- Do not change the enable password or the host name for any device.

ⓘ You **must** complete all lab tasks before you can move to the next question. Once you navigate away, the lab closes and cannot be resumed.

Guidelines | Topology | Tasks



Days 26-28 - OSPF

Original Question 4/74 - continued

Guidelines | Topology | Tasks

Tasks

Your company's network includes three Cisco routers: RouterA, RouterB, and RouterC. Each of these routers is connected to SwitchA over their respective FastEthernet 0/0 interfaces. IP addresses have been configured on all devices in the topology.

Perform the following configurations:

- Configure OSPF on each router by using OSPF process ID 1.
- Configure OSPF to advertise the specific subnetworks that are configured on each router interface.
- Configure all OSPF interfaces to operate in OSPF Area 0.
- Configure RouterA so that, in the future, it will almost always assume the role as the DR of the routers' shared segment, even if other routers are connected to the shared segment.
- Configure RouterC so that, in the future, it will never assume the role as the DR of the routers' shared segment.

Days 26-28 - OSPF

Original Question 16/74

16/74

You are attempting to configure OSPF between RouterA and RouterB on your network. However, the routers are unable to form an OSPF adjacency. You issue the **show ip ospf interface fastethernet 0/0** command on each router and receive the following output:

```
RouterA#show ip ospf interface fastethernet 0/0
FastEthernet 0/0 is up, line protocol is up
  Internet Address 10.1.1.13/30, Area 1
  Process ID 101, Router ID 192.168.15.237, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.15.237, Interface address 10.1.1.13
  No Backup Designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:07
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 2, maximum is 2
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

```
RouterB#show ip ospf interface fastethernet 0/0
FastEthernet 0/0 is up, line protocol is up
  Internet Address 10.1.1.14/30, Area 0
  Process ID 102, Router ID 192.168.15.238, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.15.237, Interface address 10.1.1.14
  No Backup Designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 2, maximum is 2
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

Why are the routers unable to form an adjacency? *(Select the best answer.)*

Why are the routers unable to form an adjacency? *(Select the best answer.)*

- ○ A. The dead timer value must match the hello timer value.
- ○ B. The IP addresses are configured with the wrong subnet mask.
- ○ C. The OSPF areas must match.
- ○ D. The cost must be higher than 1.
- ○ E. The OSPF process IDs must match.
- ○ F. The router IDs must not match.

Days 26-28 - OSPF

Original Question 18/74

18/74 OVO

You issue the **show running-config | include router** command on RouterA and receive the following output:

```
RouterA#show running-config | include router
router ospf 100
 router-id 5.5.5.5
```

You issue the **show ip interface brief** command on RouterA and receive the following output:

```
RouterA#show ip interface brief
Interface              IP-Address      OK? Method Status            Protocol
FastEthernet0/0        198.51.100.1    YES manual up                up
FastEthernet0/1        198.51.100.5    YES manual up                up
Loopback0              10.10.10.10     YES manual up                up
Loopback1              5.5.5.5         YES manual up                up
```

You issue the following commands on RouterA:

```
RouterA#configure terminal
RouterA(config)#router ospf 100
RouterA(config-router)#no router-id 5.5.5.5
```

Which of the following will become the OSPF router ID on RouterA if the OSPF process is restarted? *(Select the best answer.)*

- A. 5.5.5.5
- B. 198.51.100.1
- C. 198.51.100.5
- D. 10.10.10.10

Incorrect

Correct Answer(s): D

Explanation:

The Internet Protocol (IP) address of 10.10.10.10 will become RouterA's Open Shortest Path First (OSPF) router ID if the OSPF process is restarted. Issuing the **no router-id 5.5.5.5** command removes the manual router ID configuration for OSPF process 100 on RouterA. The IP address of 10.10.10.10 is the highest IP address assigned to a loopback interface on RouterA. If no manual router ID is configured on an OSPF router, the router will use the highest IP address assigned to a loopback interface as the router ID. If no loopback interfaces have been configured with IP addresses, the router will use the highest IP address assigned to an active physical interface.

The **router-id** *value* command can be used to manually configure an OSPF router ID in OSPF router configuration mode. An OSPF router will always select a manually configured router ID over any interface IP addresses. Issuing the **no router-id** *value* command causes the router to calculate the OSPF router ID based on IP addresses assigned to active interfaces on the router.

Neither the IP address of 198.51.100.1 nor the IP address of 198.51.100.5 will become the router ID, because RouterA's loopback interfaces have been configured with IP addresses. Had RouterA not been configured with loopback interfaces or had RouterA's loopback interfaces not been configured with IP addresses, RouterA would have selected the IP address of 198.51.100.5 as its router ID after you issued the **no router-id 5.5.5.5** command. The IP address of 198.51.100.5 is the highest IP address assigned to an active physical interface on RouterA.

The IP address of 5.5.5.5 will not become RouterA's OSPF router ID. Although RouterA was originally configured with a manual router ID of 5.5.5.5, the Loopback 1 interface's IP address of 5.5.5.5 is lower than the Loopback 0 interface's IP address of 10.10.10.10. If the 10.10.10.10 IP address had not been assigned to the Loopback 0 interface, RouterA would have kept the router ID of 5.5.5.5 after you issued the **no router-id 5.5.5.5** command because the IP address of 5.5.5.5 would have been the highest IP address assigned to a loopback interface.

Reference:

Boson CCNA Curriculum 200-301, Module 6: IP Routing, Understanding OSPF Router IDs

Boson CCNA Curriculum 200-301, Module 6: IP Routing, Understanding OSPF Router IDs

Days 26-28 - OSPF

Original Question 62/74

62/74

You issue the **show ip ospf neighbor** command on Router1 and see that Router2 is in the 2WAY/DROTHER state.

Which of the following statements is true regarding Router2? *(Select the best answer.)*

- ○ A. Router1 is the DR for the segment.
- ○ B. The MTU settings are mismatched between Router1 and Router2.
- ○ C. Router2 is the DR for the segment.
- ○ D. Router1 and Router2 are normal neighbor routers that are operating correctly.

Incorrect

Correct Answer(s): D

Explanation:

Router1 and Router2 are normal neighbor routers that are operating correctly. Neighbor routers that are neither the designated router (DR) nor the backup designated router (BDR) remain in the 2-Way state. The DR and BDR will proceed to the Full state.

Neither Router1 nor Router2 is the DR for the segment. If Router1 were the DR, the output of the **show ip ospf neighbor** command would show Router2 in the `FULL/DROTHER` state. If Router2 were the DR, the output of the **show ip ospf neighbor** command would show Router2 in the `FULL/DR` state.

The maximum transmission unit (MTU) settings are not mismatched between Router1 and Router2. If the MTU settings were mismatched, the routers would be stuck in the Exstart, Exchange, or Loading states.

When an OSPF neighbor router is powered on, it transitions through the following neighbor states:

- Down
- Init
- 2-Way
- Exstart
- Exchange
- Loading
- Full

An OSPF neighbor router begins in the Down state. A neighbor in the Down state has not yet sent a Hello packet.

When a Hello packet is received from the neighbor router but the Hello packet does not contain the receiving router's ID, the neighbor router is in the Init state. The receiving router replies to the

receiving router's ID, the neighbor router is in the Init state. The receiving router replies to the neighbor router with a Hello packet that contains the neighbor router's ID as an acknowledgment that the receiving router received the neighbor's Hello packet. If a router is stuck in the Init state, it has sent Hello packets but has not received any from the neighbor router.

The neighbor router replies with a Hello packet that contains the receiving router's ID. When this occurs, the neighbor router is in the 2-Way state. At the end of the 2-Way state, the DR and BDR are elected for broadcast and nonbroadcast multiaccess (NBMA) networks. On broadcast and NBMA networks, neighbor routers will proceed to the Full state only with the DR and BDR; other neighbor adjacencies will remain in the 2-Way state. If all routers on a segment remain in the 2-Way state, you should verify whether all routers on the segment are set to a priority of 0, which prevents any of them from becoming the DR or BDR.

After the DR and BDR are elected, neighbor routers form primary-subordinate relationships in order to establish the method for exchanging link-state information. Routers in this state are in the Exstart state. If a router is stuck in the Exstart state, you should verify whether there is a problem with mismatched maximum transmission unit (MTU) settings or duplicate router IDs.

Neighbor routers then exchange database descriptor (DBD) packets. These DBD packets contain link-state advertisement (LSA) headers that describe the contents of the link-state database (LSDB). Routers in this state are in the Exchange state. If a router is stuck in the Exchange state, you should verify whether there is a problem with mismatched MTU settings or duplicate router IDs.

Routers then send link-state request (LSR) packets to request the contents of the neighbor router's OSPF database. The neighbor router replies with link-state update (LSU) packets that contain the routing database information. Routers in this state are in the Loading state. If a router is stuck in the Loading state, you should verify whether there is a problem with mismatched MTU settings or corrupted LSR packets.

After the OSPF databases of neighbor routers are fully synchronized, the routers transition to the Full state, which is the normal OSPF router state for DRs and BDRs. A router will periodically send Hello packets to its neighbors to indicate that it is still functional. If a router does not receive a Hello packet from a neighbor within the dead timer interval, the neighbor router will transition back to the Down state.

Reference:
Boson CCNA Curriculum 200-301, Module 6: IP Routing, Understanding OSPF Adjacencies

Cisco: Understand OSPF Neighbor States

Category:
IP Connectivity

Day 29 - First Hop Redundancy Protocols

Original Question 47/74

47/74

Which HSRP router or routers will use the HSRP virtual IP address and will respond to ARP requests with the HSRP virtual MAC address? *(Select the best answer.)*

- A. only the standby router
- B. only the active router
- C. only the active router and the standby router
- D. all HSRP routers in a group

Incorrect

Correct Answer(s): B

Explanation:

Only the active router will use the Hot Standby Router Protocol (HSRP) virtual Internet Protocol (IP) address and will respond to Address Resolution Protocol (ARP) requests with the HSRP virtual Media Access Control (MAC) address. HSRP is a Cisco-proprietary protocol that enables two or more routers to act as a single virtual router. Multiple routers are assigned to an HSRP group, and the routers function as a single gateway. The HSRP virtual IP address can then be configured as the default gateway address for client devices.

Each HSRP group is identified by a group number from 0 through 255. The default HSRP group value is 0. Based on priority value, HSRP elects a single active router and a standby router for each group. To participate in the active and standby router election process, each HSRP router must be a member of the same group. The active router is the router with the highest priority; it forwards packets, responds to ARP requests with a virtual MAC address, and can be the only router that is explicitly configured with the virtual IP address. The standby router is the router with the second-highest priority. If multiple HSRP routers have the same priority, the router with the highest IP address is elected as the active router. The router with the second-highest IP address is elected as the standby router, which will assume the role of the active router if the active router fails. Other routers in the HSRP group are in the listen state.

By default, Hello packets are sent by the active router every three seconds. Only the standby router monitors the active router's Hello packets. If the standby router does not receive a Hello packet from the active router for the duration configured in the holdtime, the standby router will take over the role of the active router. By default, the holdtime is set to 10 seconds.

Day 29 - First Hop Redundancy Protocols

Original Question 47/74 - continued

To differentiate the virtual MAC addresses of the various groups, HSRP uses a special format for the virtual MAC address that uses the well-known virtual MAC address 0000.0c07.ac*xx*, where *xx* is the group number in hexadecimal format. For example, the virtual MAC address for HSRP group 11 is 0000.0c07.ac0b; 0b is the hexadecimal equivalent of 11 in decimal notation.

Reference:
Boson CCNA Curriculum 200-301, Module 6: IP Routing, Understanding HSRP

Cisco: Configuring HSRP and VRRP: Understanding HSRP

Category:
IP Connectivity

Day 30 - TCP & UDP

Original Question 56/74

56/74

Which of the following Application layer protocols uses TCP for reliable, connection-oriented data transfer? *(Select the best answer.)*

- ○ A. TFTP
- ○ B. SNMP
- ○ C. FTP
- ○ D. DHCP

Incorrect

Correct Answer(s): C

Explanation:

File Transfer Protocol (FTP) uses Transmission Control Protocol (TCP) for reliable, connection-oriented data transfer. TCP is a Transport layer protocol that uses sequencing and error-checking to ensure that transmitted data can be easily reordered if packets arrive out of sequence and can be retransmitted if any packets are lost. FTP, which is used to transfer files over a network, uses TCP ports 20 and 21. Cisco devices can reliably transfer IOS images by using FTP. FTP requires the transmission of authentication credentials, even if anonymous FTP is in use, but those credentials are transmitted in plain text. Other common TCP protocols are Hypertext Transfer Protocol (HTTP), which is used to transfer webpages over the Internet, Simple Mail Transfer Protocol (SMTP), which is used to send email messages, Post Office Protocol 3 (POP3), which is used to retrieve email messages, and Telnet,

which is used to manage network devices.

Dynamic Host Configuration Protocol (DHCP), Simple Network Management Protocol (SNMP), and Trivial FTP (TFTP) use User Datagram Protocol (UDP) and not TCP. UDP is a Transport layer protocol that is used for unreliable, connectionless datagram transfer. Because UDP does not use sequence numbers or establish synchronized connections, transmitted datagrams can appear out of sequence or can be dropped without notice.

DHCP is used to assign Internet Protocol (IP) addressing and configuration information to clients. SNMP is used to monitor and manage network devices. TFTP is used to transfer files without authentication over a network. For example, administrators can use TFTP to transfer Cisco IOS images from a server to a device to perform firmware upgrades. Other common Application layer protocols that use UDP include Network Time Protocol (NTP), which is used to coordinate time on a network, and Remote Authentication Dial-In User Service (RADIUS), which is used to authenticate users.

Reference:
Boson CCNA Curriculum 200-301, Module 7: IP Services, FTP

IANA: Service Name and Transport Protocol Port Number Registry

Category:
IP Services

Days 31-33 - IPv6

Original Question 34/74

34/74 OVO

Which of the following can you do to compress an IPv6 address as efficiently as possible? *(Select 2 choices.)*

- ☐ A. Remove only trailing zeros from each hextet.
- ☐ B. Remove leading and trailing zeros from each hextet.
- ☐ C. Replace one or more contiguous hextets consisting of zeros with a double colon.
- ☐ D. Replace every hextet consisting of zeros with double colons.
- ☐ E. Replace only one hextet of zeros with a double colon.
- ☐ F. Remove only leading zeros from each hextet.

Incorrect

Correct Answer(s): C, F

Explanation:

To compress an Internet Protocol version 6 (IPv6) address as efficiently as possible, you can replace one or more contiguous hextets consisting of zeros with a double colon and you can remove only leading zeros from each hextet. An IPv6 address is a 128-bit (16-byte) address that is typically written as eight groups of four hexadecimal characters. Each group of four characters is called a hextet and is separated by a colon. A double colon can be used at the beginning, middle, or end of an IPv6 address in place of one or more contiguous hextets consisting of all zeros. However, only one double colon can be used in an IPv6 address. Therefore, you cannot replace every hextet consisting of zeros with double colons. Typically, the largest contiguous set of zeros is replaced with a double colon. For example, the IPv6 address 2001:0000:0000:070D:0000:50A0:0001:0024 can be expressed as the following addresses:

- 2001:0000:0000:070D:0000:50A0:0001:0024
- 2001::070D:0000:50A0:0001:0024
- 2001:0:0:70D:0:50A0:1:24
- 2001::70D:0:50A0:1:24

You are not limited to replacing only one hextet consisting of zeros with a double colon. However, you can replace only a single set of contiguous hextets consisting of zeros with a double colon.

Trailing zeros cannot be removed from each hextet; only leading zeros can be removed from each hextet. If a hextet consisting of all zeros cannot be replaced with a double colon, you can reduce the four-character group of zeros to a single zero.

Reference:

Boson CCNA Curriculum 200-301, Module 2: Network Addressing and Transport, Abbreviating IPv6 Addresses

Cisco: IPv6 Basics (PDF)

Category:

Network Fundamentals

Days 31-33 - IPv6

Original Question 64/74

64/74

What command would you issue on RouterA so that traffic can be routed to RouterC? *(Select the best answer.)*

- A. **ipv6 route 2001:DB8:1::/64 2001:DB8:1::2**
- B. **ipv6 route 2001:DB8:2::/64 2001:DB8:1::2**
- C. **ipv6 route 2001:DB8:2::/64 2001:DB8:2::2**
- D. **ipv6 route 2001:DB8:1::/64 2001:DB8:2::2**

Incorrect

Correct Answer(s): B

Explanation:

Of the choices provided, you would issue the **ipv6 route 2001:DB8:2::/64 2001:DB8:1::2** command on RouterA so that traffic can be routed to RouterC. The basic syntax of the **ipv6 route** command is **ipv6 route** *prefix/length* {*interface* | *next-hop*} [*administrative-distance*], where *prefix/length* is the destination network, *interface* is the output interface on the local router, *next-hop* is the next-hop Internet Protocol version 6 (IPv6) address, and *administrative-distance* is an optional administrative distance (AD).

You should not issue the **ipv6 route 2001:DB8:2::/64 2001:DB8:2::2** command, because it incorrectly specifies the next-hop IPv6 address. The next-hop IPv6 address should be the inbound interface on RouterB, 2001:DB8:1::2.

You should not issue the **ipv6 route 2001:DB8:1::/64 2001:DB8:1::2** command, because it incorrectly specifies the destination network. The destination network should be the network between RouterB and RouterC, 2001:DB8:2::/64.

You should not issue the **ipv6 route 2001:DB8:1::/64 2001:DB8:2::2** command, because it incorrectly specifies both the destination network and the next-hop IPv6 address.

Reference:

Boson CCNA Curriculum 200-301, Module 6: IP Routing, IPv6 Static Routes

Cisco: Cisco IOS IPv6 Command Reference: ipv6 route

Cisco: IPv6 Routing: Static Routing: Information About IPv6 Routing: Static Routing

Category:
IP Connectivity

Days 31-33 - IPv6

Original Question 69/74

69/74

Which of the following IPv6 addresses is a link-local multicast address that is used to send a packet to all routers on a segment? *(Select the best answer.)*

- A. FF05::2
- B. FF02::2
- C. FF05::1
- D. FF02::1

Incorrect

Correct Answer(s): B

Explanation:

The Internet Protocol version 6 (IPv6) address FF02::2 is a link-local multicast address that is used to send a packet to all routers on a segment. All IPv6 router interfaces automatically join this multicast group. Link-local multicast addresses in IPv6 work similarly to how multicast addresses work in IP version 4 (IPv4). A multicast address is used to send packets to multiple devices that are configured with that multicast address. The following table shows common IPv4 multicast addresses and their respective IPv6 multicast addresses:

Multicast Address	IPv4	IPv6
All Hosts	224.0.0.1	FF02:0:0:0:0:0:0:1
All Routers	224.0.0.2	FF02:0:0:0:0:0:0:2
All OSPF Routers	224.0.0.5	FF02:0:0:0:0:0:0:5
All OSPF DRs	224.0.0.6	FF02:0:0:0:0:0:0:6
All RIP Routers (except RIPv1)	224.0.0.9	FF02:0:0:0:0:0:0:9
All EIGRP Routers	224.0.0.10	FF02:0:0:0:0:0:0:A

Similar to IPv4, IPv6 uses three types of addresses, including unicast and multicast addresses

Similar to IPv4, IPv6 uses three types of addresses, including unicast and multicast addresses. However, IPv6 does not use broadcast addresses. Instead, IPv6 uses anycast addresses, which are not available in IPv4. Functions that were performed by broadcast addresses in IPv4 are performed by multicast and anycast addresses in IPv6. Unicast addresses in IPv6 work similarly to how they work in IPv4. A unicast address is used to send packets to a single device. Multicast addresses in IPv6 also work similarly to how they work in IPv4. Anycast addresses are used to send packets to the closest device that is configured with the anycast address. Therefore, an anycast address can be described as a one-to-nearest address. The closest device is selected by the routing protocol that is used by the router. Because anycast addresses use the same address for multiple devices in a group, anycast addresses are ideal for load balancing.

The IPv6 prefix FF00::/8 is used for all multicast addresses. IPv6 addresses in the FF00::/8 range begin with the characters from FF00 through FFFF. However, certain address ranges are used to indicate the scope of the multicast address. The following IPv6 multicast scopes are defined:

- FF01::/16 – node-local
- FF02::/16 – link-local
- FF05::/16 – site-local
- FF08::/16 – organization-local
- FF0E::/16 – global

The IPv6 address FF02::1 is a link-local multicast address that is used to send a packet to all nodes on a segment. All IPv6 nodes automatically join this multicast group. Therefore, IPv6-enabled routers and hosts on a segment receive multicast packets sent to FF02::1.

The IPv6 address FF05::2 is a site-local multicast address that is used to send a packet to all routers in a site, not just on the local segment. The IPv6 address FF05::1 is a site-local multicast address that is used to send a packet to all nodes in a site.

Reference:
Boson CCNA Curriculum 200-301, Module 2: Network Addressing and Transport, Understanding IPv6 Address Types

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 1, Chapter 23: IPv6 Addressing and Subnetting, Unique Local Unicast Addresses

RIPE: IPv6 Address Types (PDF)

Category:
Network Fundamentals

Days 34-35 - Access Control Lists (ACLs)

Original Question 12/74

You issue the following commands on RouterA:

```
RouterA(config)#access-list 10 permit 10.1.0.0 0.0.255.255
RouterA(config)#interface fastethernet 0/1
RouterA(config-if)#ip access-group 10 out
```

HostA sends a packet that is destined for HostB.

When will the router check access list 10? *(Select the best answer.)*

- A. RouterA will check access list 10 before the packet enters interface FastEthernet 0/0.
- B. RouterA will check access list 10 after the packet enters interface FastEthernet 0/0 but before the packet is routed to interface FastEthernet 0/1.
- **C. RouterA will check access list 10 after the packet is routed to interface FastEthernet 0/1.**
- D. RouterA will check access list 10 after the packet enters interface FastEthernet 0/0 and again after the packet is routed to interface FastEthernet 0/1.
- E. RouterA will not check access list 10 for this packet.

Correct

Correct Answer(s): C

Explanation:

RouterA will check access list 10 after the packet is routed to interface FastEthernet 0/1. You have configured access list 10 as an outbound access list on interface FastEthernet 0/1. Therefore, access list 10 will be processed for outbound traffic on interface FastEthernet 0/1.

You can configure each interface on a router with one inbound access list and one outbound access list. Incoming traffic is checked against inbound access lists. To configure an inbound access list for an interface, you should issue the **ip access-group** *acl-identifier* **in** command in interface configuration mode. Outgoing traffic is checked against outbound access lists. To configure an outbound access list for an interface, you should issue the **ip access-group** *acl-identifier* **out** command in interface configuration mode.

Outbound access lists are processed after the router receives and routes a packet to the outbound interface. When a packet is routed to the outbound interface, the Cisco IOS checks each outbound access list statement in the order it appears in the access list. If the packet matches a statement that allows the packet, the router transmits the packet out the outbound interface. If the packet matches a statement that denies the packet, the router discards the packet. If the packet does not match any of the access list statements, the router drops the packet due to the *implicit deny* rule.

RouterA will not check access list 10 after the packet enters interface FastEthernet 0/0 but before the packet is routed to interface FastEthernet 0/1, because access list 10 has not been applied as an inbound access list on interface FastEthernet 0/0. Inbound access lists are processed after a router receives a packet but before the packet is routed to the outbound interface. When a packet is received by a router, the Cisco IOS checks each inbound access list statement in the order it appears in the access list. If the packet matches a statement that allows the packet, the router continues to process the packet. If the packet matches a statement that denies the packet, the router discards the packet. If the packet does not match any of the access list statements, the router drops the packet due to the implicit deny rule. The following commands would configure access list 10 as an inbound access list on interface FastEthernet 0/0:

```
RouterA(config)#interface fastethernet 0/0
RouterA(config-if)#ip access-group 10 in
```

RouterA would check access list 10 after the packet entered interface FastEthernet 0/0 and again after the packet was routed to interface FastEthernet 0/1 only if access list 10 were configured as an inbound access list on interface FastEthernet 0/0 and as an outbound access list on interface FastEthernet 0/1. RouterA will not check access list 10 before the packet enters interface FastEthernet 0/0, because a router cannot attempt to match a packet to an access list until the router receives the packet.

RouterA will check access list 10 for any outbound traffic on interface FastEthernet 0/1. If HostB were to send a packet to HostA, RouterA would not check access list 10, because the packet from HostB would be inbound on interface FastEthernet 0/1, not outbound.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Configuring IPv4 ACLs to Control Interface Access

Cisco: Access Control List Overview and Guidelines

Category:
Security Fundamentals

Days 34-35 - Access Control Lists (ACLs)

Original Question 45/74

45/74

Which of the following statements is true regarding the **ip access-group** command? *(Select the best answer.)*

- A. It can be used to apply an IPv4 ACL to a vty line.
- B. It can be used to apply an IPv6 ACL to the AUX line.
- C. It can be used to apply an ACL to an IPv4 interface.
- D. It can be used to apply an ACL to an IPv6 interface.

Incorrect

Correct Answer(s): C

Explanation:

The **ip access-group** command can be used to apply an access control list (ACL) to an Internet Protocol version 4 (IPv4) interface. You can configure each interface on a router with one inbound ACL and one outbound ACL. Incoming traffic is checked against inbound ACLs. To configure an inbound ACL for an interface, you should issue the **ip access-group** *list-number* **in** command in interface configuration mode. Outgoing traffic is checked against outbound ACLs. To configure an outbound ACL for an interface, you should issue the **ip access-group** *list-number* **out** command in interface configuration mode.

The **ipv6 traffic-filter** command can be used in interface configuration mode to apply an ACL to an IP version 6 (IPv6) network interface. The syntax for the **ipv6 traffic-filter** command is **ipv6 traffic-filter** *ipv6-acl* [**in** | **out**], where the **in** keyword configures the ACL to apply to inbound traffic and the **out** keyword configures the ACL to apply to outbound traffic.

The **access-class** command can be used to apply an IPv4 ACL to a virtual terminal (vty) line, the auxiliary (AUX) line, or the console line. The syntax for the **access-class** command is **access-class** *acl-name-or-number* [**in** | **out**], where the **in** keyword configures the ACL to apply to inbound traffic and the **out** keyword configures the ACL to apply to outbound traffic. Likewise, the **ipv6 access-class** command can be used to apply an IPv6 ACL to a vty line, the AUX line, or the console line. The syntax for the **ipv6 access-class** command is **ipv6 access-class** *ipv6-acl* [**in** | **out**].

Reference:

Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Configuring IPv4 ACLs to Control Interface Access

Cisco: IPv6 Access Control Lists: Configuration Examples for IPv6 Access Control Lists

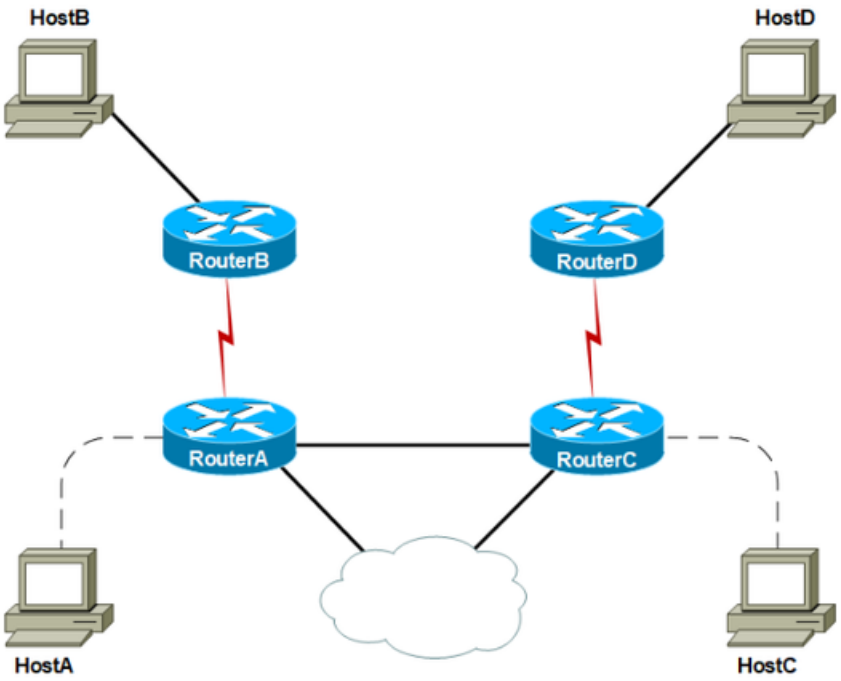
Category:
Security Fundamentals

Days 34-35 - Access Control Lists (ACLs)

Original Question 60/74

60/74

You administer the network shown in the following exhibit:



You issue the **show running-config | section line** command on RouterB and receive the following output:

```
RouterB#show running-config | section line
line con 0
line aux 0
line vty 0 4
 access-class 10 in
 login
 password cisco
```

Which of the following ACL types is applied to the vty lines on RouterB? *(Select the best answer.)*

- ○ A. dynamic
- ○ B. named
- ○ C. standard
- ○ D. extended

Incorrect

Correct Answer(s): C

Explanation:

A standard access control list (ACL) is applied to the virtual terminal (vty) lines on RouterB, as shown in the following output:

```
RouterB#show running-config | section line
line con 0
line aux 0
line vty 0 4
 access-class 10 in
 login
 password cisco
```

In the output above, the **access-class 10 in** command configures vty lines from 0 through 4 to apply ACL 10 in the inbound direction. To configure a standard ACL on a Cisco router, you should use the

following command syntax:

access-list *acl-number* {**deny** | **permit**} {**host** *ip-address* | *source source-wildcard* | **any**}

A Cisco router can identify the type of ACL by the *acl-number* value. An ACL is a standard access list when the *acl-number* value is from 1 through 99 or from 1300 through 1999.

No extended ACL is applied to the vty lines on RouterB. An access list is an extended access list when the *acl-number* value is from 100 through 199 or from 2000 through 2699. Extended ACLs can permit or deny packets based on source IP address, destination Internet Protocol (IP) address, protocol, and port. Because of how extended access lists filter traffic, Cisco recommends that you apply extended ACLs as close to the source of the packets being filtered as possible. Placing an extended ACL as close as possible to the source of the traffic prevents traffic from unnecessarily traversing the network, thereby consuming bandwidth and router resources.

No named ACL is applied to the vty lines on RouterB. A named ACL is an ACL that is identified by a name instead of an ACL number. Named ACLs can be either standard ACLs or extended ACLs depending on the keyword issued when the ACL is created.

To configure a standard access list by name instead of number, you should issue the **ip access-list standard** *name* command, where *name* is a unique identifier consisting of any alphanumeric character except the space character. After you issue the **ip access-list standard** *name* command, the device will be placed in standard ACL configuration mode, where you can issue one or more statements to configure the named ACL. To create these ACL statements, you should use the syntax [*sequence-number*] {**deny** | **permit**} {**host** *ip-address* | *source source-wildcard* | **any**}, where *sequence-number* is an optional sequence number that indicates the order in which the ACL statements will be evaluated. If you do not issue a sequence number, the statements will be processed in the order in which they are issued.

To configure an extended access list by name instead of number, you should issue the **ip access-list**

extended *name* command, where *name* is a unique identifier consisting of any alphanumeric character except the space character. After you issue the **ip access-list extended** *name* command, the router will be placed in extended ACL configuration mode, where you can issue one or more statements to configure the named ACL. To create these ACL statements, you should use the syntax [*sequence-number*] {**deny** | **permit**} *protocol source source-wildcard* [*operator port*] *destination destination-wildcard* [*operator port*], where *sequence-number* is an optional sequence number that indicates the order in which the ACL statements will be evaluated. If you do not issue a sequence number, the statements will be processed in the order in which they are issued.

No dynamic ACL is applied to the vty lines on RouterB. Dynamic ACLs offer additional security by forcing users to authenticate before permission to send packets over a network is granted. Dynamic ACL–enabled interfaces initially use an extended ACL to deny all traffic. Users who want to send packets beyond the protected interface are first required to establish a Telnet connection to the router and provide credentials to authenticate. If authentication is verified, an ACL will be automatically generated to allow the authenticated user access and the Telnet connection will be disconnected.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Configuring IPv4 ACLs to Control Remote Access

Cisco: Configure IP Access Lists: Standard ACLs

Category:
Security Fundamentals

Day 36 - CDP & LLDP

Original Question 38/74

38/74

Which of the following statements are true regarding LLDP? *(Select 2 choices.)*

- ☐ A. It cannot be disabled at the interface level.
- ☐ B. An LLDP interface can be configured to transmit LLDP messages but not receive them.
- ☐ C. An LLDP interface can be configured to receive LLDP messages but not transmit them.
- ☐ D. It is enabled by default on Cisco switches.
- ☐ E. It can convey VTP information.

Incorrect

Correct Answer(s): B, C

Explanation:

A Link Layer Discovery Protocol (LLDP) interface can be configured to receive LLDP messages but not transmit them, or it can be configured to transmit LLDP messages but not receive them. LLDP is an Open Systems Interconnections (OSI) Layer 2 open-standard discovery protocol that is used to facilitate interoperability between Cisco devices and non-Cisco devices. Attributes that can be learned from neighboring devices contain Type, Length, Value (TLV) information including port description, system description, and management address.

By default, all interfaces are configured to transmit and receive LLDP packets when LLDP is globally enabled. However, you can issue the **lldp transmit** and **lldp receive** commands from interface configuration mode to control the sending and receiving of LLDP packets for a specific interface. For example, the **no lldp transmit** command configures an interface not to send LLDP packets, whereas the **lldp receive** command configures an interface that was previously configured not to receive LLDP packets to resume listening for LLDP packets. You can issue the **show lldp interface** command to examine the LLDP status of a particular interface. For example, the following sample output indicates that the FastEthernet 0/4 interface is configured to receive LLDP updates but not to send them:

Day 36 - CDP & LLDP

Original Question 38/74 - continued

```
SwitchA#show lldp interface fastethernet 0/4

FastEthernet0/4:
    Tx: disabled
    Rx: enabled
    Tx state: INIT
    Rx state: WAIT PORT OPER
```

By default, a Cisco switch will send LLDP advertisements every 30 seconds when LLDP is enabled on an interface. These advertisements are used by neighboring devices to update the LLDP information learned about each neighbor. They are also used as keepalive messages to ensure that a discovered neighbor continues to be available on the network. You can issue the **lldp timer** *rate* command from global configuration mode to configure the frequency at which LLDP advertisements are sent by a switch. The default *rate* value is 30 seconds; however, the rate can be configured to any integer value from 5 through 65534 seconds. A Cisco switch will retain LLDP information for 120 seconds when LLDP is enabled on an interface. This time interval is known as the LLDP holdtime. You can issue the **lldp holdtime** *seconds* command from global configuration mode to configure the LLDP holdtime to any integer value from 0 through 65535 seconds. Whenever a new LLDP advertisement is received, the hold timer is reset and the LLDP information is considered current. When the hold timer expires for a particular neighbor, the LLDP information regarding that neighbor is considered stale and is discarded.

You can issue the **show lldp** command from privileged EXEC mode to display the current LLDP configuration. The following sample output shows the default settings for a Cisco 3560 series switch after LLDP has been enabled globally:

```
SwitchA#show lldp

Global LLDP Information:
    Status: ACTIVE
    LLDP advertisements are sent every 30 seconds
    LLDP hold time advertised is 120 seconds
    LLDP interface reinitialisation delay is 2 seconds
```

LLDP is disabled by default on Cisco switches. You can issue the **lldp run** command from global configuration mode to enable LLDP globally on a switch.

Cisco Discovery Protocol (CDP), not LLDP, can provide VLAN Trunking Protocol (VTP) information in advertisements. VTP is a protocol that is used to centrally manage virtual local area network (VLAN) changes and to propagate those changes across a network to all devices in the VTP domain. Like LLDP, CDP is an OSI Layer 2 protocol that is used to advertise and discover only directly connected Cisco devices on a local network. However, unlike LLDP, CDP is a Cisco-proprietary protocol rather than open-standard. CDP sends advertisements every 60 seconds by default. Although CDP is enabled by default on many Cisco devices, it can be disabled for security purposes.

Reference:
Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Disabling and Enabling LLDP

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 13: Device Management Protocols, Configuring and Verifying LLDP

Cisco: Configuring CDP: Default CDP Configuration

Day 36 - CDP & LLDP

Original Question 38/74 - continued

Cisco: Configuring LLDP, LLDP-MED, and Wired Location Service: Default LLDP Configuration

Category:
Network Access

Day 36 - CDP & LLDP

Original Question 40/74

40/74

You issue the **show cdp neighbors** command on RouterB and receive the following output:

```
RouterB#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater

Device ID  Local Intrfce  Holdtme Capability  Platform    Port ID
Boson-1    Gig 0/1        117        S I      WS-C2960-2  Gig 0/2
Boson-2    Fas 0/1        102       R S I     1841        Fas 0/2
```

Which of the following interfaces on RouterB is connected to another router? *(Select the best answer.)*

- A. Gi0/1
- B. Gi0/2
- C. Fa0/1
- D. Fa0/2

Incorrect

Correct Answer(s): C

Explanation:

The Fa0/1 interface on RouterB is connected to another router. In the output of the **show cdp neighbors** command, the `Local Intrfce` field indicates the interface on the local device. The device named Boson-2 is a router, as indicated by the `Capability` field, and it is a Cisco 1841 router, as indicated by the `Platform` field. Therefore, the Fa0/1 interface of RouterB is connected to Boson-2.

CDP is used to collect information about neighboring Cisco devices and is enabled by default. Because CDP operates at the Data Link layer, which is Layer 2 of the Open Systems Interconnection (OSI) model, CDP is not dependent on any particular Layer 3 protocol addressing, such as Internet Protocol (IP) addressing. Therefore, if CDP information is not being exchanged between devices, you should check for Physical layer and Data Link layer connectivity problems. CDP is enabled by default on Cisco devices. You can globally disable CDP by issuing the **no cdp run** command in global configuration mode. You can disable CDP on a per-interface basis by issuing the **no cdp enable** command in interface configuration mode.

The Fa0/2 interface on RouterB is not connected to another router. In the output of the **show cdp neighbors** command, the `Port ID` field indicates an interface on the neighboring device, not an interface on the local device. Therefore, the Fa0/1 interface of RouterB is connected to the Fa0/2 interface of Boson-2.

The Gi0/1 interface of RouterB is not connected to a router; it is connected to a switch. The device named Boson-1 is a switch, as indicated by the `Capability` field, and it is a Cisco 2960 switch, as indicated by the `Platform` field. The `Local Intrfce` field indicates that the Gi0/1 interface of RouterB is connected to Boson-1, and the `Port ID` field indicates that the Gi0/2 interface of Boson-1 is connected to RouterB.

Reference:

Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Displaying CDP and LLDP Neighbors

Reference:

Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Displaying CDP and

Day 36 - CDP & LLDP

Original Question 40/74 - continued

Boson CCNA Curriculum 200-301, Module 9: Switching and Network Access, Displaying CDP and LLDP Neighbors

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 13: Device Management Protocols, Examining Information Learned by CDP

Cisco: Configuring Cisco Discovery Protocol on Cisco Routers and Switches Running Cisco IOS

Category:
Network Access

Day 37 - NTP

Original Question 36/74

36/74

Which of the following commands should you issue to enable symmetric active mode on a Cisco router?
(Select the best answer.)

- ○ A. **ntp broadcast client**
- ○ B. **ntp master**
- ○ C. **ntp authenticate**
- ○ D. **ntp server**
- ○ E. **ntp peer**
- ○ F. **ntp access-group serve**

Incorrect

Correct Answer(s): E

Explanation:

You should issue the **ntp peer** command from global configuration mode to enable symmetric active mode on a Cisco router. Network Time Protocol (NTP) is used to synchronize the time on network devices. A device in symmetric active mode attempts to mutually synchronize with another NTP host. When symmetric active mode is enabled, the host might synchronize the peer or it might be synchronized by the peer. The syntax of the **ntp peer** command is **ntp peer** *ip-address*, where *ip-address* is the Internet Protocol (IP) address of the NTP host.

You should not issue the **ntp broadcast client** command to enable symmetric active mode on a Cisco router. Issuing the **ntp broadcast client** command from interface configuration mode configures a Cisco router to be an NTP broadcast client. An NTP broadcast client listens on the configured interface for NTP broadcasts from an NTP server, which the NTP client uses to adjust its time. The difference between a broadcast client and a static client is that a broadcast client can receive its time from any NTP server. By contrast, a static client receives its time from the NTP server specified in the **ntp server** command.

You should not issue the **ntp server** command to enable symmetric active mode on a Cisco router. Issuing the **ntp server** command from global configuration mode configures a Cisco router to be an NTP static client and be synchronized by an NTP server. The syntax of the **ntp server** command is **ntp**

server *ip-address*, where *ip-address* is the IP address of the NTP server that the client will use to receive its time.

You should not issue the **ntp authenticate** command to enable symmetric active mode on a Cisco router. Issuing the **ntp authenticate** command from global configuration mode enables NTP authentication on a Cisco router. Authentication can be used with NTP to provide source verification for NTP synchronization. To configure the router with the NTP Message Digest 5 (MD5) authentication key, you should issue the **ntp authentication-key** *key-number* **md5** *key* [*encryption-type*] command.

You should not issue the **ntp master** command to enable symmetric active mode on a Cisco router. Issuing the **ntp master** command from global configuration mode configures a Cisco router to be an NTP server; NTP clients are synchronized by NTP servers. The syntax of the **ntp master** command is **ntp master** [*stratum*], where *stratum* is an NTP stratum value from 1 through 15; if the stratum value is not specified, the NTP server uses the default stratum value of 8. A primary server connected to a reference clock is called a stratum 1 NTP server. A stratum 2 device typically receives its time from a stratum 1 device, a stratum 3 device typically receives its time from a stratum 2 device, and so on.

You should not issue the **ntp access-group serve** command to enable symmetric active mode on a Cisco router. Issuing the **ntp access-group serve** *acl* command configures a Cisco router to accept requests from NTP clients that match the specified access control list (ACL). Issuing the **ntp access-group peer** *acl* command configures a Cisco router to synchronize with NTP peers that match the specified ACL.

Reference:
Boson CCNA Curriculum 200-301, Module 7: IP Services, NTP Server Configuration

Cisco: Cisco IOS Basic System Management Command Reference: ntp peer

Category:
IP Services

Day 38 - DNS

Original Question 67/74

67/74

Which of the following protocols uses both TCP and UDP? *(Select the best answer.)*

- ○ A. Telnet
- ○ B. DNS
- ○ C. FTP
- ○ D. DHCP
- ○ E. TFTP

Incorrect

Correct Answer(s): B

Explanation:

Domain Name System (DNS) uses both Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) over port 53. DNS translates host names to Internet Protocol (IP) addresses.

Day 38 - DNS

Original Question 67/74 - continued

TCP is a Transport layer protocol that is used for reliable, connection-oriented transfer of data. Data sent by TCP is ordered and checked for errors, and any lost packets are retransmitted. File Transfer Protocol (FTP), which is used to transfer files over a network, uses TCP ports 20 and 21. Telnet, a terminal emulation protocol that can be used to remotely log on to a router, uses TCP port 23. Other Application layer protocols that use TCP include Simple Mail Transfer Protocol (SMTP), which uses TCP port 25; Hypertext Transfer Protocol (HTTP), which uses TCP port 80; and Post Office Protocol 3 (POP3), which uses TCP port 110. Neither FTP nor Telnet uses UDP to communicate.

UDP is a Transport layer protocol that is used for unreliable, connectionless datagram transfer. Transmitted datagrams can appear out of sequence or can be dropped without notice. Dynamic Host Configuration Protocol (DHCP), which assigns IP addressing information to clients, uses UDP ports 67 and 68. Trivial File Transfer Protocol (TFTP), which is used to transfer files over a network, uses UDP port 69. Other Application layer protocols that use UDP include Network Time Protocol (NTP), which uses UDP port 123; Simple Network Management Protocol (SNMP), which uses UDP ports 161 and 162; and Remote Authentication Dial-In Service (RADIUS), which uses UDP ports 1812 and 1813. Neither DHCP nor TFTP uses TCP to communicate.

Reference:
Boson CCNA Curriculum 200-301, Module 7: IP Services, DNS

IETF: RFC 7766: DNS Transport over TCP - Implementation Requirements: Section 5. Transport Protocol Selection

Category:
IP Services

Day 39 - DHCP

Original Question 3/74

3/74

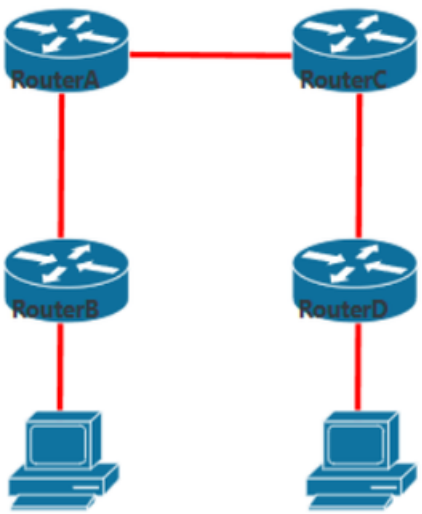
Guidelines | Topology | Tasks

Guidelines

This is a lab item in which tasks are performed on Boson's NetSim virtual devices.

- Refer to the **Tasks** tab to view the tasks for this lab item.
- Refer to the **Topology** tab to access the device console(s) and perform the tasks.
- Console access is available for all required devices by clicking the device icon or by using the tab(s) above.
- All necessary preconfigurations have been applied.
- Do not change the enable password or the host name for any device.

ⓘ You **must** complete all lab tasks before you can move to the next question. Once you navigate away, the lab closes and cannot be resumed.



Day 39 - DHCP

Original Question 3/74 - continued

HostB HostD

Tasks

You want to configure dynamic IP addressing on RouterB for its connected hosts. The first 10 addresses on the subnet should be reserved for static addresses; the remainder of the subnet should be available for dynamic IP address allocation.

Your supervisor has asked you to complete the following tasks:

- Configure a pool named **pool1** on RouterB to assign IP addresses for the hosts on the network connected to the FastEthernet 0/0 interface of RouterB.
- Ensure that the hosts use RouterB as the default gateway.
- Ensure that the hosts use the DNS server at 192.168.11.3.
- Ensure that the hosts use the domain name **boson.com**.
- Ensure that the hosts use the NetBIOS server at 192.168.11.4.

Configure only RouterB to complete these tasks. Host configuration will be performed by another administrator.

All router interfaces are operational, and the proper IP addresses have been configured.

All passwords are set to **cisco**.

Day 39 - DHCP

Original Question 42/74

42/74

You administer the network in the topology diagram.

RouterA is configured as a DHCP server. In an effort to centralize the administration of DHCP services, you have decided to configure RouterB as a DHCP relay agent rather than configure it as a DHCP server. You issue the **show running-config** command on RouterA and receive the following partial output:

```
RouterA#show running-config
<output omitted>
ip dhcp pool DHCP
   network 192.168.111.16 255.255.255.240
   default-router 192.168.111.30
   dns-server 192.168.111.67
<output omitted>
```

Which of the following commands should you issue on RouterB? *(Select the best answer.)*

- ○ A. **ip helper-address 192.168.100.18**
- ○ B. **network 192.168.111.32 255.255.255.248**
- ○ C. **service dhcp**
- ○ D. **ip helper-address 192.168.111.30**
- ○ E. **ip default-network 192.168.111.30**

Incorrect

Correct Answer(s): D

Explanation:
Of the choices provided, you should issue the **ip helper-address 192.168.111.30** command on RouterB's Fa0/0 interface. The **ip helper-address** command configures Dynamic Host Configuration Protocol (DHCP) relay, which enables an interface to forward DHCP broadcasts across a network to the IP address of a DHCP server. The DHCP server will respond to the request and begin the DHCP discovery process. The syntax of the **ip helper-address** command is **ip helper-address** *ip-address*, where *ip-address* is the IP address of the DHCP server. Therefore, you should not issue the **ip helper-address 192.168.100.18** command.

By default, the **ip helper-address** command configures an interface to forward broadcasts to the following User Datagram Protocol (UDP) ports:

- UDP port 37 – Time Protocol
- UDP port 49 – Terminal Access Controller Access-Control System (TACACS)
- UDP port 53 – Domain Name System (DNS)
- UDP port 67 – Bootstrap Protocol (BOOTP) and DHCP Server
- UDP port 68 – BOOTP and DHCP Client
- UDP port 69 – Trivial File Transfer Protocol (TFTP)
- UDP port 137 – Network Basic Input/Output System (NetBIOS) Name Service
- UDP port 138 – NetBIOS Datagram

You should not issue the **network 192.168.111.32 255.255.255.248** command on RouterB, because you are not creating a DHCP server on RouterB. However, you would create a DHCP scope on the DHCP server, RouterA, that contains the IP addresses that should be distributed by RouterA. The syntax of the **network** command is **network** *address* [*mask* | */prefix*], where *address* is the network address, *mask* is the subnet mask, and *prefix* is the prefix length in Classless Inter-Domain Routing (CIDR) notation.

You should not issue the **service dhcp** command on RouterB, because you are not configuring RouterB as a DHCP server in this scenario. The **service dhcp** command enables the DHCP service on a Cisco router if the service is disabled. However, the DHCP service and DHCP relay agent services are typically enabled by default.

You should not issue the **ip default-network 192.168.111.30** command on RouterB, because you are not configuring RouterB's gateway of last resort. The **ip default-network** *ip-address* command can be used to configure a gateway of last resort on a router when IP routing is enabled. If IP routing is not enabled on a router, you can instead issue the **ip default-gateway** *ip-address* command to configure a gateway of last resort.

Reference:
Boson CCNA Curriculum 200-301, Module 7: IP Services, DHCP Helper Address Configuration

Day 39 - DHCP

Original Question 42/74 - continued

Cisco: Cisco IOS IP Application Services Command Reference: ip helper-address

Category:
IP Services

Day 40 - SNMP

Original Question 11/74

11/74

Which of the following protocols uses a MIB to store information? *(Select the best answer.)*

- A. DNS
- B. CDP
- **C. SNMP**
- D. DHCP

Incorrect

Correct Answer(s): C

Explanation:

Simple Network Management Protocol (SNMP) uses a Management Information Base (MIB) to store information. SNMP is a protocol that a Network Management System (NMS) can use to communicate with a managed device. The MIB is a database that contains information about a managed device. When the device is queried by an NMS, the device can reference its MIB and return the required value to the NMS. The MIB stores information about a managed device in a standard format specified by Abstract Syntax Notation One (ASN.1). Because the information is stored in a machine-independent format, the information is guaranteed to be accessible by any SNMP-compliant management station.

Dynamic Host Configuration Protocol (DHCP) servers store lease information in a database file, not in a MIB. DHCP is a protocol that can be used to dynamically assign Internet Protocol (IP) addresses to network devices. In addition, DHCP can be used to provide the client with other configuration parameters, such as boot file locations and local time server addresses.

Domain Name System (DNS) servers typically store information in text files or in a database. DNS is a protocol that is used to resolve names to IP addresses. DNS servers provide authoritative responses for local addresses and cache data from nonlocal addresses. Traditionally, DNS data was stored in text files; however, current implementations can store the data in common database formats.

Cisco Discovery Protocol (CDP) stores information in a local table, not in a MIB. CDP is a Cisco-proprietary protocol that is used to advertise and discover Cisco devices on a local network. When CDP is enabled, a device periodically broadcasts information about its capabilities. Other devices hear the broadcasts and store the information in a local table.

Reference:

Boson CCNA Curriculum 200-301, Module 7: IP Services, SNMP

Cisco: Configuring SNMP Support: Components of SNMP

Cisco: Configuring Simple Network Management Protocol: SNMP Overview

Day 41 - Syslog

Original Question 46/74

46/74

You are troubleshooting Router1 and are attempting to view messages at the informational level and below. You want to display log messages to the console. However, you want to minimize the number of log messages as much as possible.

Which of the following commands should you issue? *(Select the best answer.)*

- A. **logging console 5**
- B. **logging console 3**
- C. **logging console 6**
- D. **logging console 4**

Incorrect

Correct Answer(s): C

Explanation:

You should issue the **logging console 6** command. You can filter log messages on the console by severity level by issuing the **logging console** *severity-level* command. Alternatively, you can filter log messages to a Syslog server by issuing the **logging trap** *severity-level* command. When the **logging console** or **logging trap** command is issued with a *severity-level* parameter, messages with the specified severity level and all lower-numbered severity levels will be displayed or sent, respectively.

Cisco debug messages and log messages are divided into the following severity levels:

- 0 – emergencies
- 1 – alerts
- 2 – critical
- 3 – errors
- 4 – warnings
- 5 – notifications
- 6 – informational
- 7 – debugging

Messages are formatted in the Berkeley Software Distribution (BSD) Syslog format, which is a percent sign (%) followed by a facility code, a severity code, and a mnemonic code. The three codes are separated by dashes. Consider the following message:

```
*Dec  1 14:20:07.059: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to down
```

In this Syslog message, the facility code is `LINEPROTO`, the severity code is `5`, which is equivalent to notifications, and the mnemonic code is `UPDOWN`. The dash-separated code is followed by a colon and the human-readable text of the log message.

The **logging console 6** command displays all messages at the informational level and below. Informational messages are routine messages but still might indicate a problem. Although you could issue the **logging console 7** command to display these messages, this command would cause Router1 to also display messages at the debugging level.

You should not issue the **logging console 5** command. The **logging console 5** command displays all messages at the notifications level and below. Route flaps, neighbor adjacencies, and interface protocol up/down transitions are displayed as notifications.

You should not issue the **logging console 4** command or the **logging console 3** command. Level 4 is used to display messages at the warnings level and below, level 3 is used to display messages at the errors level and below, and level 2 is used to display messages at the critical level and below. Critical, error, and warning messages indicate something that might impact the device. For example, interface up/down state changes are displayed as errors at level 3.

Emergencies, which are displayed at level 0, and alerts, which are displayed at level 1, indicate a severe hardware or software problem with the device. These messages need to be addressed immediately.

Reference:
Boson CCNA Curriculum 200-301, Module 7: IP Services, Log Severity Levels

Cisco: System Message Logging: Table 3 Message Logging Level Keywords

Cisco: Cisco IOS Embedded Syslog Manager Command Reference: logging console

Category:
IP Services

Day 42 - SSH

Original Question 2/74

2/74

Guidelines | Topology | Tasks

Guidelines

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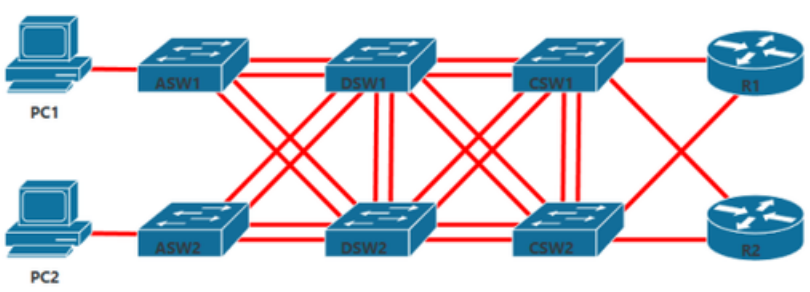
Guidelines | Topology | Tasks



Day 42 - SSH

Original Question 2/74 - continued

Guidelines | Topology | Tasks

Tasks

You are responsible for configuring switches CSW1 and CSW2. You have been asked to achieve the following goals:

- Configure the console port on CSW1 to disconnect the current session after three minutes of inactivity.
- Configure the console port on the CSW2 to never disconnect an idle session.
- Configure the first five vty ports on CSW1 to accept only the SSH protocol for inbound connections.
- Configure the first five vty ports on CSW2 to accept no protocols for inbound connections.
- Enable the SSH server on CSW1. Use the domain **boson.com** and the default modulus value when generating RSA keys for the SSH server.
- Ensure that CSW1 will accept only SSHv2 connections.

Day 42 - SSH

Original Question 43/74

43/74

Which of the following commands will automatically enable SSH on a router? *(Select the best answer.)*

- ○ A. **enable secret**
- ○ B. **crypto key zeroize rsa**
- ○ C. **no transport input telnet**
- ○ D. **transport input ssh**
- ○ E. **crypto key generate rsa**

Incorrect

Correct Answer(s): E

Explanation:

The **crypto key generate rsa** command will automatically enable Secure Shell (SSH) on a router. This command creates a set of RSA security keys that can be used for SSH sessions. RSA is an asymmetric encryption algorithm that can be used to create a public/private key pair. SSH is a cryptographic protocol that provides a secure connection between two devices. A router uses the security keys generated by RSA to secure SSH connections. Information sent by using SSH is encrypted and, thus, is not viewable by using packet sniffing applications.

SSH is often used as a secure replacement for Telnet to manage network devices. In order for SSH to be enabled on a Cisco device, the device must be running a K9 IOS image, which provides cryptographic functionality. To enable SSH for virtual terminal (vty) lines on a Cisco router, you should complete the following steps:

1. Configure the router with a host name other than Router by issuing the **hostname** command.
2. Configure the router with a domain name by issuing the **ip domain-name** command.

3. Generate an RSA key pair for the router by issuing the **crypto key generate rsa** command.
4. Configure the vty lines to use SSH by issuing the **transport input ssh** command from line configuration mode.

The **transport input ssh** command does not enable SSH on the router. The **transport input ssh** command only configures the vty lines to use SSH if SSH has already been configured.

The **crypto key zeroize rsa** command removes RSA keys from a router. You may want to remove RSA keys in order to generate new keys. However, removing RSA keys does not automatically enable SSH on a router.

The **enable secret** command can be used to help prevent unauthorized access to privileged EXEC mode. Using the **enable secret** command is more secure than using the **enable password** command because the **enable secret** command configures the enable password to be stored as a Message Digest 5 (MD5) hash, whereas the **enable password** command configures the enable password to be stored as plain text. However, the **enable secret** command does not automatically enable SSH on a router.

The **no transport input telnet** command can be used to prevent Telnet access to a router. Because Telnet information is sent unencrypted as plain text, Telnet is not as secure as SSH. Therefore, to ensure that remote management connections to the router are encrypted, you can issue the **no transport input telnet** command after enabling SSH. Although the **no transport input telnet** command disables Telnet access, this command does not automatically enable SSH.

Reference:
Boson CCNA Curriculum 200-301, Module 7: IP Services, SSH Configuration

Cisco: Configure SSH on Routers: Set Up a Cisco IOS Router as SSH Client

Day 42 - SSH

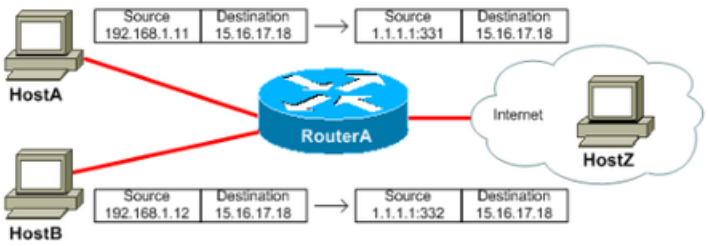
Original Question 43/74 - continued

Category:
IP Services

Days 44-45 - NAT

Original Question 8/74

8/74 Nat overlaping



Which of the following terms most accurately defines the type of NAT that is used on the network above?
(Select the best answer.)

- A. NAT overloading
- B. dynamic NAT
- C. NAT overlapping
- D. static NAT

Incorrect

Correct Answer(s): A

Explanation:

The network in this scenario uses Network Address Translation (NAT) overloading. NAT overloading uses ports to translate inside local addresses to one or more inside global addresses. The NAT router uses port numbers to keep track of which packets belong to each host. NAT overloading is also called Port Address Translation (PAT).

NAT translates between public and private Internet Protocol (IP) addresses to enable hosts on a privately addressed network to access the Internet. Public addresses are routable on the Internet, and private addresses are routable only on internal networks. Request for Comments (RFC) 1918 defines several IP address ranges that are reserved for private, internal use:

- 10.0.0.0 through 10.255.255.255
- 172.16.0.0 through 172.31.255.255
- 192.168.0.0 through 192.168.255.255

Because NAT performs address translation between private and public addresses, NAT effectively hides the address scheme used by the internal network, which can increase security. NAT also reduces the number of public IP addresses that a company needs to allow its devices to access Internet resources, thereby conserving IP version 4 (IPv4) address space.

An inside local address is typically an RFC 1918–compliant IP address that represents an internal host to the internal network. An inside global address is used to represent an internal host to an external

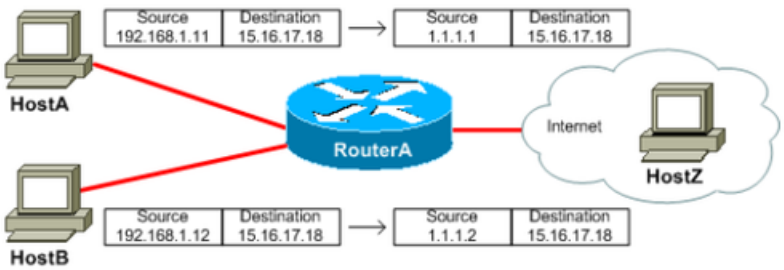
network.

Static NAT translates a single inside local address to a single inside global address or a single outside local address to a single outside global address. You can configure a static inside local–to–inside global IP address translation by issuing the **ip nat inside source static** *inside-local inside-global* command. To configure a static outside local–to–outside global IP address translation, you should issue the **ip nat outside source static** *outside-global outside-local* command.

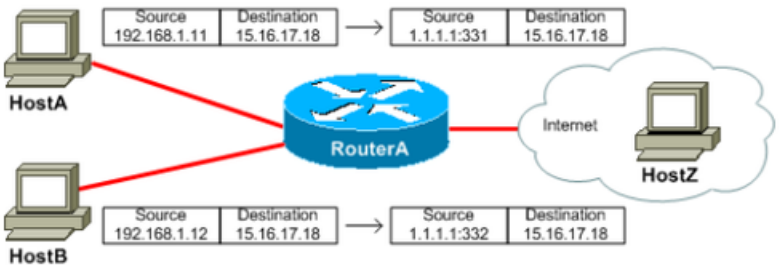
Dynamic NAT translates local addresses to global addresses that are allocated from a pool. To create a NAT pool, you should issue the **ip nat pool** *nat-pool start-ip end-ip* {**netmask** *mask* | **prefix-length** *prefix*} command. To enable translation of inside local addresses, you should issue the **ip nat inside source list** *access-list* **pool** *nat-pool* [**overload**] command.

When a NAT router receives an Internet-bound packet from a local host, the NAT router performs the following tasks:

- It checks the static NAT mappings to verify whether an inside global address mapping exists for the local host.
- If no static mapping exists, it dynamically maps the inside local address to an unused inside global address, if one is available, from the NAT pool.
- It changes the inside local address in the packet header to the inside global address and forwards the packet to its destination:



When all the inside global addresses in the NAT pool are mapped, no other inside local hosts will be able to communicate on the Internet. This is why NAT overloading is useful. When NAT overloading is configured, an inside local address, along with a port number, is mapped to an inside global address. The NAT router uses port numbers to keep track of which packets belong to each host:



You can issue the **ip nat inside source list** *access-list* **interface** *outside-interface* **overload** command to configure NAT overload with a single inside global address, or you can issue the **ip nat inside source list** *access-list* **pool** *nat-pool* **overload** command to configure NAT overload with a NAT pool.

You should use NAT overlapping when the addresses on an internal network conflict with the addresses on another network. The internal addresses must be translated to unique addresses on the external network, and addresses on the external network must be translated to unique addresses on the internal network; the translation can be performed through either static or dynamic NAT.

Reference:
Boson CCNA Curriculum 200-301, Module 7: IP Services, NAT/PAT Translation Methods

Cisco: Configure Network Address Translation: Examples: 1. Allow Internal Users to Access the Internet

Days 46-47 - QoS & Voice VLANs

Original Question 23/74

23/74 OVO

Which of the following commands should you issue in interface configuration mode to configure a switch so that voice traffic and data traffic are separated but do not require a unique voice VLAN to be created? *(Select the best answer.)*

- A. **switchport voice vlan none**
- B. **switchport voice vlan 10**
- C. **switchport voice vlan dot1p**
- D. **switchport voice vlan untagged**

Incorrect

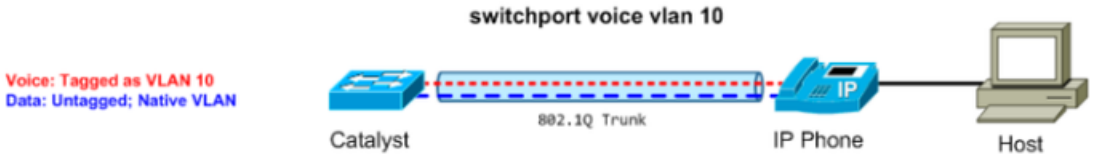
Correct Answer(s): C

Explanation:

You should issue the **switchport voice vlan dot1p** command in interface configuration mode to configure the switch so that voice traffic and data traffic are separated but do not require a unique voice virtual local area network (VLAN) to be created. The **switchport voice vlan** command has four options: *vlan-id*, **dot1p**, **untagged**, and **none**. Each of the four options determines how the voice traffic will be encapsulated and whether the voice traffic will be carried over a unique voice VLAN or over the native VLAN. The *vlan-id*, **dot1p**, and **untagged** modes all use the special-case 802.1Q trunk that is automatically negotiated by Dynamic Trunking Protocol (DTP) and Cisco Discovery Protocol (CDP) if a trunk is needed. The **dot1p** mode sends voice traffic with a default 802.1p priority of 5 and uses VLAN 0, which requires a VLAN ID, but does not require a unique voice VLAN to be created, as shown in the exhibit below:

Issuing the **switchport voice vlan none** command, which reflects the default configuration of the switch port, enables the Internet Protocol (IP) phone to use its configuration to send untagged voice traffic and untagged data traffic over the access VLAN. With the **none** mode, voice traffic does not use 802.1p priority tagging or Class of Service (CoS) and voice traffic is transmitted with data traffic, as shown in the exhibit below:

The **switchport voice vlan 10** command sends voice traffic over VLAN 10, which is a unique VLAN. Valid VLAN IDs range from 1 through 4094. The *vlan-id* mode is the most versatile because a unique voice VLAN is created to carry voice traffic, whereas data traffic is carried over the native VLAN. Voice packets on the voice VLAN include the CoS bits in the 802.1p encapsulation field. In this scenario, VLAN 10 was specified as the unique VLAN to carry the voice packets, as shown in the exhibit below:



The **switchport voice vlan untagged** interface configuration command configures the IP phone to send both untagged voice traffic and untagged data traffic over the native VLAN. When the **untagged** keyword is used, both voice traffic and data traffic are transmitted through a special-case 802.1Q trunk, as shown in the exhibit below:

Reference:
Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Configuring the Voice VLAN

Cisco: Configuring Voice VLAN: Configuring IP Phone Voice Traffic

Category:
Network Access

Days 46-47 - QoS & Voice VLANs

Original Question 53/74

53/74

Which of the following QoS mechanisms involves queue creation, traffic classification, and queue scheduling? *(Select the best answer.)*

- ○ A. congestion management
- ○ B. traffic policing
- ○ C. traffic shaping
- ○ D. congestion avoidance

Incorrect

Correct Answer(s): A

Explanation:

Congestion management is a Quality of Service (QoS) mechanism that involves three main components:

- Queue creation
- Traffic classification
- Queue scheduling

All congestion management mechanisms involve the creation of at least one queue. For example, on a Cisco Adaptive Security Appliance (ASA), you must create a high-priority queue on an interface before you can use priority queuing to manage congestion. When an interface receives more traffic than it can send, the interface places the excess traffic in a queue. Some congestion management mechanisms allow you to tune the size of the interface hardware queue, also known as the Transmit Ring (TX Ring), and the size of the software queue. Some congestion management mechanisms also classify traffic by priority. Each traffic class is assigned a particular queue, and the packets that match each class are placed into the appropriate queue when congestion occurs. How an interface handles the packets in each queue is referred to as queue scheduling.

Congestion avoidance is a method of monitoring network traffic to detect and possibly prevent congestion. The simplest method of congestion avoidance is the tail drop mechanism, which drops all packets on an interface once a queue has reached its capacity. By contrast, the weighted random early detection (WRED) mechanism avoids congestion by dropping low-priority packets before a queue reaches its capacity. Congestion avoidance does not involve queue creation or queue scheduling.

Traffic policing is a method of limiting the rate of traffic that passes through an interface. With traffic policing, packet flows that exceed the configured thresholds are typically dropped. Alternatively, traffic can be re-marked with a lower priority before being transmitted. On an ASA, traffic policing can be applied to an interface in either the inbound or outbound direction. Although traffic policing involves the classification of traffic, traffic policing does not involve queue creation or queue scheduling.

scheduling.

Similarly, traffic shaping is the process of limiting the rate of traffic that passes through an interface. However, with traffic shaping, flows that exceed the configured thresholds are typically buffered and not dropped. Because traffic is buffered to maintain a desired packet rate, erratic patterns are smoothed into a uniform flow. On an ASA, traffic shaping can only be applied to an interface in the outbound direction. In addition, traffic shaping can only be applied to the class-default class map, which matches all traffic. Although traffic shaping involves the classification of traffic, traffic shaping does not involve queue creation or queue scheduling.

Reference:
Boson CCNA Curriculum 200-301, Module 7: IP Services, Congestion Management

Cisco: Congestion Management Overview: Congestion Management Overview

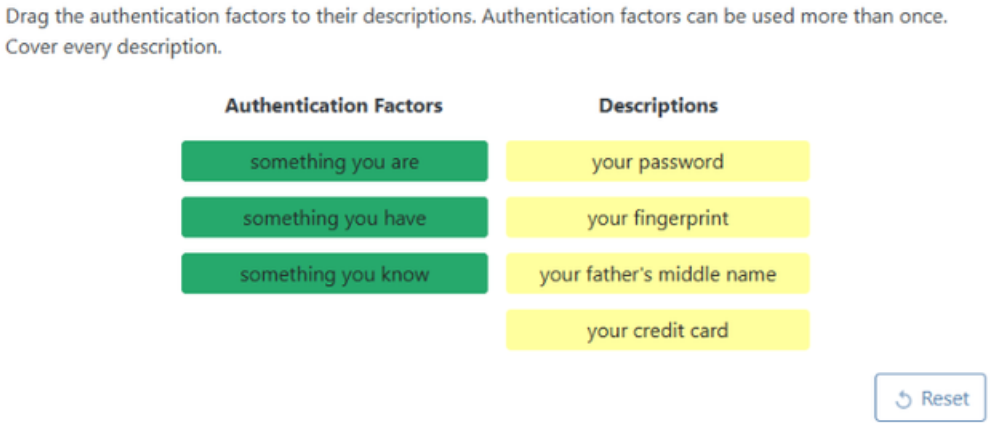
Cisco: Configuring QoS: Information About QoS

Category:
IP Services

Day 48 - Security Fundamentals

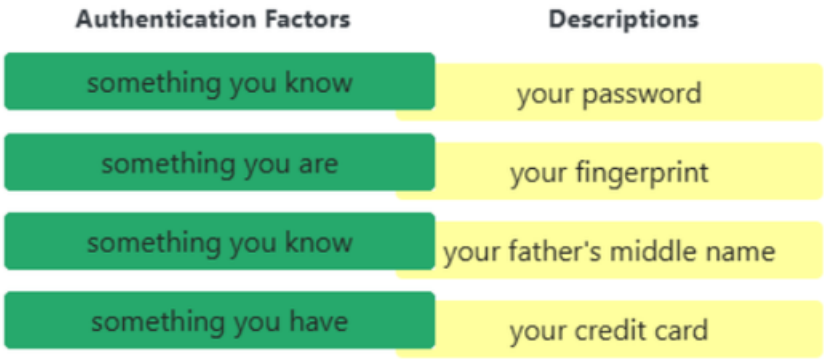
Original Question 5/74

5/74



Incorrect

Answer:



Explanation:

Your fingerprints are an example of authentication by something you are. Your password and your father's middle name are examples of authentication by something you know. Your credit card is an example of authentication by something you have. There are three typical methods of authentication for gaining access to a secure environment: something you know, something you have, and something you are. A fourth

possible method is authentication by someplace you are, which means that you can be granted access to a secure system by virtue of your workstation's location on a network, such as an Internet Protocol (IP) address or your physical location in the world.

Authentication by one type of factor, such as something you know, is known as single-factor authentication. Authentication by more than one type of factor, such as something you know and something you are, is known as multifactor authentication. However, requiring more than one of any single factor, such as two knowledge factors, is not considered multifactor authentication.

Your password or your father's middle name is an example of authentication by something you know. For example, a bank's website might choose to ask you to provide both a password and the answer to a security question such as your father's middle name. Although the bank's website prompts you for two forms of verification, both of those prompts are for information that you store in your memory. Authentication by something you know is also known as Type 1 authentication. Authentication by something you know is considered the weakest form of authentication because such authentication can often be easily guessed or broken by brute force.

Your credit card is an example of authentication by something you have. Authentication by something you have is the process of verifying your identity by using a device or document that you carry with you, such as a fob, a driver's license, a smart card, or a mobile phone with an authenticator application. For example, a police officer who stops you can verify your identity by comparing an image on your driver's license to your physical appearance. Authentication by something you have is also known as Type 2 authentication. Authentication by something you have is considered a stronger form of authentication than authentication by something you know because it requires the user to carry some sort of authenticating electronic access control (EAC) token.

Your fingerprints are an example of authentication by something you are. Authentication by something you are is the process of verifying your identity by using something that is unique about you and that cannot be easily changed, such as your iris, your retina, or your fingerprints. For example, your company could connect your workstation to a fingerprint scanner instead of requiring you to unlock your workstation with keystrokes and a password. Authentication by something you are is also known as Type 3 authentication. Authentication by something you are is considered the strongest form of authentication because of the unique biometrics of individuals.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Authentication Factors

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 9: Security Architectures, Password Alternatives

Category:
Security Fundamentals

Day 48 - Security Fundamentals

Original Question 6/74

6/74

You have installed a TACACS+ server on the network that you administer. You want to configure AAA on a router so that the router uses the TACACS+ server for authentication when a user attempts to log in to the router. However, if the server is unavailable, you want to ensure that the router uses its local user database.

Which of the following commands should you issue? *(Select the best answer.)*

- ○ A. **aaa authentication login default local**
- ○ B. **aaa authentication login user**
- ◉ C. **aaa authentication login user group tacacs+**
- ○ D. **aaa authentication login default group tacacs+ local**

Incorrect
Correct Answer(s): D

Explanation:
You should issue the **aaa authentication login default group tacacs+ local** command to ensure that the router uses its local user database if the Terminal Access Controller Access Control System Plus (TACACS+) server is unavailable. Authentication, Authorization, and Accounting (AAA) is used to control access to a router or switch. When

implementing AAA, you can configure a router to authenticate users against a local database, against a Remote Authentication Dial-In User Service (RADIUS) server, or against a TACACS+ server. In order to use AAA authentication with an authentication server, you must ensure that an authentication server is available on the network. To prepare for the possibility that the authentication server might become unavailable, you can configure a router so that it will use another method, such as the local user database or the enable password, in the absence of the authentication server. To accomplish this, you should issue the **aaa authentication login default group tacacs+ local** command or the **aaa authentication login default group tacacs+ enable** command. The **aaa authentication login** command configures AAA authentication on a router or a switch to limit login access to the router. The first parameter,

default, specifies that the command is the default authentication method used on the router. The **group tacacs+** keywords specify that the TACACS+ server should be used. The final keyword, which should be **local** in this scenario, specifies that if the TACACS+ server is unavailable, the local user database on the router should be used for authentication.

Issuing the **aaa authentication login default local** command would configure AAA authentication to use the local database for authentication purposes. Issuing this command would not configure the router to use the TACACS+ server for authentication as specified in the scenario.

The **aaa authentication login user** command is an incomplete command. Issuing this command would create an authentication list named **user**, but it would not define any authentication methods for the list. You must specify at least one authentication method when defining an authentication list.

Issuing the aaa authentication login user group tacacs+ command would create an authentication list named user that uses the TACACS+ server to authenticate users. However, because local is not specified after group

tacacs+, the local user database will not be used if the TACACS+ server is unavailable.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Configuring TACACS+

Cisco: Cisco IOS Security Command Reference: aaa authentication login

Category:
Security Fundamentals

Day 48 - Security Fundamentals

Original Question 19/74

19/74

Which of the following best describes accounting? *(Select the best answer.)*

- A. the process of verifying a user's identity
- B. the process of establishing a user's accounts upon hire
- C. the process of recording the use of resources
- D. the process of verifying the level of access configured for a user

Incorrect

Correct Answer(s): C

Explanation:

Accounting is the process of recording the use of resources. The following list defines the three phases of the Authentication, Authorization, and Accounting (AAA) process:

- Authentication – the process of verifying a user's identity
- Authorization – the process of verifying the level of access configured for a user
- Accounting – the process of recording the use of resources

AAA systems manage user activity. AAA systems are typically more sophisticated than simple password authentication systems, such as a local password database. Two common AAA systems are Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access-Control System Plus (TACACS+).

RADIUS is a standard AAA protocol created by the Internet Engineering Task Force (IETF). Compared to TACACS+, RADIUS has several limitations. For example, RADIUS encrypts only the password in Access-Request packets; it does not encrypt the entire contents of the packet like TACACS+ does.

TACACS+ is a Cisco-proprietary protocol used during AAA operations. TACACS+ provides more granular and flexible control over user access privileges. For example, the AAA operations are separated by TACACS+, whereas RADIUS combines the authentication and authorization services into a single function. Because TACACS+ separates these functions, administrators have more control over access to configuration commands. In addition, TACACS+ encrypts the entire contents of packets, thus providing additional security.

Reference:

Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, AAA

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 9: Security Architectures,

Day 48 - Security Fundamentals

Original Question 19/74 - continued

Controlling and Monitoring User Access

Security Fundamentals

Day 48 - Security Fundamentals

Original Question 49/74

49/74

You issue the following command set on a Cisco router:

enable secret B0$0n!
line vty 0 4
password C1$c0!
login
service password-encryption

Which of the following password types is the **C1$c0!** password stored as in the running configuration? *(Select the best answer.)*

- A. Type 5
- B. Type 0
- C. Type 8
- D. Type 7
- E. Type 9

Incorrect

Correct Answer(s): D

Explanation:

The **C1$c0!** password will be stored as a Type 7 password because the **service password-encryption** command has been issued in this scenario. When you issue the **service password-encryption** command on a Cisco device, all current and future Type 0, or clear-text, passwords are encrypted as Type 7 passwords in the device's running configuration. Type 7 passwords are encrypted by using the Cisco-proprietary Vigenère encryption algorithm. This type of encryption prevents anyone who views the device's running configuration from easily reading the passwords that are configured there. However, Type 7 encryption is weak. Type 7 passwords can be easily decrypted by using tools that are freely available online. Where possible, you should use a stronger encryption method to encrypt passwords in the running configuration.

The **C1$co!** password will not be stored as a Type 0 password in this scenario. Type 0 passwords are passwords that are stored without encryption in the running configuration. Some examples of Cisco commands that, by default, store passwords as Type 0 passwords are the **enable password** *password*

command, the line configuration mode **password** *password* command, and the global configuration mode **username** *user-name* **password** *password* command.

The **C1$co!** password will not be stored as a Type 5 password in this scenario. However, the **B0$0n!** password will be stored as a Type 5 password because it is configured by using the **enable secret** command. The **enable secret** command by default stores a password in Message Digest 5 (MD5)-encrypted format in the device's running configuration. In some Cisco IOS versions prior to 15.3(3), the **enable secret** command by default stores an encrypted password in the device's running configuration by using a Secure Hash Algorithm (SHA) 256-bit hash. As of Cisco IOS 15.3(3), Type 4 passwords have been deprecated because of a security flaw in their implementation. Because a password that is configured by using the **enable secret** command is already encrypted as a Type 5 password in the running configuration, that password will not be re-encrypted as a Type 7 password by the **service password-encryption** command. Similar to the **enable secret** command, the **username** *user-name* **secret** *password* command encrypts the password associated with the given user as a Type 5 password in the running configuration. It is important to note that if both the **enable password** *password* command and the **enable secret** *password* command are configured on the same device, the device will prefer the Type 5 password when a user is prompted for the enable mode password.

There is nothing in this scenario to indicate that the **C1$co!** password will be stored as a Type 8 password. Type 8 passwords are encrypted by using a different implementation of SHA-256 than the Type 4 implementation of SHA-256. Type 8 passwords are encrypted by using a Password-Based Key Derivation Function 2 (PBKDF2) with a SHA-256 secret. Type 8 passwords are not reversible and are considered to be stronger than Type 4, Type 5, and Type 7 passwords. You can configure a Cisco device to use Type 8 passwords by default when you issue the **enable secret** *password* command. To configure Type 8 passwords as the default secret type, issue the **enable algorithm-type sha256** command in Cisco IOS versions later than 15.3(3)M3.

There is nothing in this scenario to indicate that the **C1$co!** password will be stored as a Type 9 password. Type 9 passwords are encrypted in the running configuration by using the scrypt encryption algorithm. Like Type 8 passwords, Type 9 passwords are not reversible. You can configure a Cisco device to use Type 9 passwords by default when you issue the **enable secret** *password* command. To configure Type 9 passwords as the default secret type, issue the **enable algorithm-type scrypt** command in Cisco IOS versions later than 15.3(3)M3. It is important to note that if you plan to downgrade a device to a Cisco IOS version earlier than 15.3(3)M3 and you have configured either Type 8 or Type 9 encryption, you should first issue the **enable algorithm-type md5** command in IOS 15.3(3)M3 and reconfigure the **enable secret** *password* command. Otherwise, you will be locked out of the Cisco device after downgrading.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Creating a Secure Password for Privileged EXEC Mode Access

Cisco: Networking Knowledge Base: Understanding the differences between the Cisco password \

Day 48 - Security Fundamentals

Original Question 49/74 - continued

secret Types

Category:
Security Fundamentals

Day 48 - Security Fundamentals

Original Question 59/74

59/74

Which of the following are most likely to be considered forms of authentication? *(Select 2 choices.)*

- ☐ A. verifying a user's password
- ☐ B. assigning a role to a verified user
- ☐ C. verifying a user's fingerprint pattern
- ☐ D. logging a verified user's file access
- ☐ E. allowing a user to access a specific file

Incorrect

Correct Answer(s): A, C

Explanation:

Verifying a user's fingerprint pattern and verifying a user's password are both likely to be considered forms of authentication. Authentication, Authorization, and Accounting (AAA) systems manage user activity. Authentication is the process of verifying a user's identity. Authentication by itself does not grant access to a given resource. AAA systems are typically more sophisticated than simple password authentication systems, such as a local password database. Two common AAA systems are Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access-Control System Plus (TACACS+).

RADIUS is a standard AAA protocol created by the Internet Engineering Task Force (IETF). Compared to TACACS+, RADIUS has several limitations. For example, RADIUS encrypts only the password in Access-Request packets; it does not encrypt the entire contents of the packet like TACACS+ does.

TACACS+ is a Cisco-proprietary protocol used during AAA operations. TACACS+ provides more granular and flexible control over user access privileges. For example, the AAA operations are separated by TACACS+, whereas RADIUS combines the authentication and authorization services into a single function. Because TACACS+ separates these functions, administrators have more control over access to configuration commands. In addition, TACACS+ encrypts the entire contents of packets, thus providing additional security.

The following list defines the three phases of the AAA process:

- Authentication – the process of verifying a user's identity
- Authorization – the process of verifying the level of access configured for a user
- Accounting – the process of recording the use of resources

Allowing a user to access a specific file is a form of authorization, not authentication. Similarly, assigning a role to a verified user is a form of authorization. Allowing access to a specific file can involve assigning specific user or group permissions directly to the file, matching a rule, such as an access control list (ACL) that associates a specific user with a file, or assigning a user to a specific role that has permission to access the file.

Logging a verified user's file access is a form of accounting, not authentication. Accounting is a feature of AAA systems that enables administrators to track resource usage across a network. If a security incident occurs, accounting can aid the effort to track the incident back to its source.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, AAA

Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Authentication Factors

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 9: Security Architectures, Controlling and Monitoring User Access

Category:
Security Fundamentals

Day 48 - Security Fundamentals

Original Question 63/74

63/74

You issue the following commands on a Cisco router named RouterA:

enable password !b0s0nu$3r!
enable secret b0$0n4dm!n
line console 0
password b0$0n4dm1n
line vty 0 15
login
password b0s0nu$3r
service password-encryption

Another user has been asked to examine the running configuration on RouterA but not make any configuration changes. The user connects to RouterA by using Telnet.

Which of the following will the user require in order to perform this task? *(Select the best answer.)*

- A. the enable password and the console password
- B. the console password alone
- C. the enable password and the vty line password
- D. the enable secret password and the vty line password
- E. the console password and the vty line password
- F. the vty line password alone
- G. the enable secret password and the console password

Day 49 - Port Security

Original Question 58/74

58/74

In which of the following Layer 2 attacks does an attacker generate thousands of forged frames every minute with the intention of overwhelming the switch's MAC address table? *(Select the best answer.)*

- A. ARP poisoning
- B. VLAN hopping
- C. MAC spoofing
- D. MAC flooding
- E. DHCP spoofing

Incorrect

Correct Answer(s): D

Explanation:

In a Media Access Control (MAC) flooding attack, an attacker generates thousands of forged frames every minute with the intention of overwhelming the switch's MAC address table. A switch builds the MAC address table by analyzing the MAC addresses of ingress, or inbound, traffic on each port. However, once the MAC table is full, the switch can no longer make intelligent forwarding decisions and all traffic is flooded out each port. This allows the attacker to view all data sent through the switch because all traffic will be sent out each port. Implementing port security can help mitigate MAC flooding attacks by limiting the number of MAC addresses that can be learned on each interface to a maximum of 128. A MAC flooding attack is also known as a Content Addressable Memory (CAM) table overflow attack.

In a MAC spoofing attack, an attacker uses the MAC address of another known host on the network in order to bypass port security measures. MAC spoofing can also be used to impersonate another host on the network. Implementing port security with sticky secure MAC addresses can help mitigate MAC spoofing attacks.

In an Address Resolution Protocol (ARP) poisoning attack, which is also known as an ARP spoofing attack, the attacker sends a gratuitous ARP (GARP) message to a host. The GARP message associates the attacker's MAC address with the Internet Protocol (IP) address of a valid host on the network. Subsequently, traffic sent to the valid host address will go through the attacker's computer rather than directly to the intended recipient. Implementing Dynamic ARP Inspection (DAI) can help mitigate ARP poisoning attacks.

In a virtual local area network (VLAN) hopping attack, an attacker attempts to inject packets into other VLANs by accessing the VLAN trunk and double-tagging 802.1Q frames. A successful VLAN hopping attack enables an attacker to send traffic to other VLANs without the use of a router. You can prevent VLAN hopping by disabling Dynamic Trunking Protocol (DTP) on trunk ports, by changing the native VLAN, and by configuring user-facing ports as access ports.

In a Dynamic Host Configuration Protocol (DHCP) spoofing attack, an attacker installs a rogue DHCP server on the network in an attempt to intercept DHCP requests. The rogue DHCP server can then respond to the DHCP requests with its own IP address as the default gateway address; hence all traffic is routed through the rogue DHCP server. You should enable DHCP snooping to help prevent DHCP spoofing attacks.

Reference:

Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Restricting Ports by Client MAC Address

Cisco: Layer 2 Security Features on Cisco Catalyst Layer 3 Fixed Configuration Switches Configuration Example: Background Information

Huntress: What is MAC Flooding

Category:

Security Fundamentals

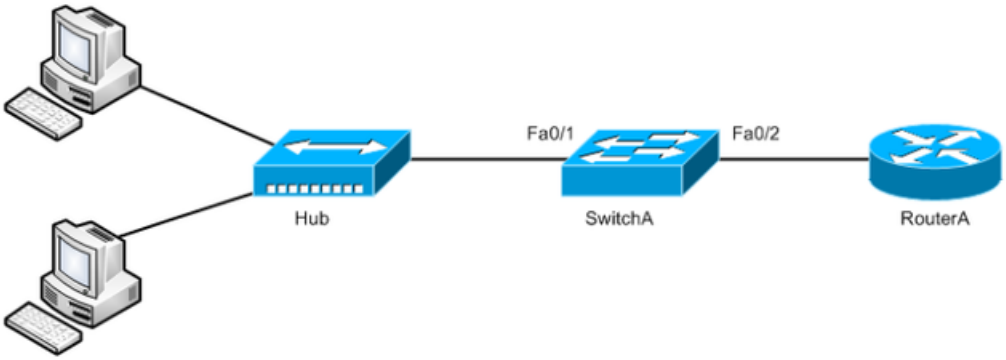
Day 49 - Port Security

Original Question 58/74 - continued

Day 49 - Port Security

Original Question 68/74

68/74



You administer the network shown in the exhibit. You issue the following commands on SwitchA:

```
SwitchA#configure terminal
SwitchA(config)#interface fastethernet 0/1
SwitchA(config-if)#switchport port-security
SwitchA(config-if)#switchport port-security violation protect
```

Which of the following statements are true regarding this configuration? *(Select 2 choices.)*

- ☐ A. Static MAC addresses cannot be configured on the interface.
- ☐ B. The interface will dynamically learn a MAC address.
- ☐ C. Any previously learned MAC addresses will be removed.
- ☐ D. The interface will remain up but will drop all packets generated by the hosts connected to the hub.

- [ ] E. One of the hosts will not be able to access the network.

Incorrect

Correct Answer(s): B, E

Explanation:

After you issue this command sequence on SwitchA, one of the hosts will not be able to access the network and the interface will dynamically learn a Media Access Control (MAC) address. You can configure port security to limit traffic on a switch port so that only traffic sent from one or more statically configured or dynamically learned source MAC addresses is allowed into a switch port. If port security has not been previously configured for an interface, issuing the **switchport port-security** command will result in port security becoming enabled with all of the following default settings:

- Sticky address learning is disabled.
- A maximum of one MAC address will be allowed on the port.
- The Shutdown violation mode is enabled.
- The port security aging time is configured to zero.
- The port security static aging is disabled.
- The port security aging type is configured to absolute.

It is not necessary to re-enable port security if you modify port security settings, such as the maximum number of MAC addresses allowed on the port, after having previously issued the **switchport port-security** command without parameters.

The **switchport port-security violation protect** command configures an interface to discard traffic from unauthorized hosts. The **switchport port-security violation** command can be issued with either the **restrict** keyword, the **protect** keyword, or the **shutdown** keyword. The **restrict** keyword configures the interface to remain up if more than the maximum number of addresses is learned, but traffic from the violating devices is dropped and a log entry is generated. Similarly, the **protect** keyword configures the interface to remain up if more than the maximum number of addresses is

learned and to drop traffic from violating devices; however, unlike the **restrict** keyword, the **protect** keyword does not create a log of the violations. The **shutdown** keyword configures the port to shut down if more than the maximum allowed MAC addresses are learned on the interface.

Because the FastEthernet 0/1 interface is configured to permit the default maximum number of MAC addresses and has a violation mode of **protect**, the interface will permit only one of the two hosts in this scenario to access the network. When the second host attempts to access the network, the FastEthernet interface will discard all traffic from the second host and will continue to function normally for the first host.

You can issue the **switchport port-security maximum** command to specify the number of authorized MAC addresses for a particular interface. For example, the **switchport port-security maximum 2** command limits the number of secure MAC addresses stored on the interface to two.

You can configure static MAC addresses on an interface that is configured for port security. In addition, you can allow the interface to dynamically learn the MAC addresses of connected devices. Port security supports both methods of learning MAC addresses.

You can issue the **clear port-security dynamic** [**address** *mac-address* | **interface** *type mod/num*] command to remove a MAC address from an interface. The command sequence shown in this scenario will not remove a MAC address from interface FastEthernet 1/1.

The command sequence shown in this scenario will not cause the FastEthernet 0/1 interface to remain up but will drop all packets originating from the hosts connected to the hub. The interface will remain up and will accept packets from only one of the hosts connected to it. However, if the violation mode of the interface were changed to **shutdown**, the interface would be shut down when it detected the MAC address of the second host and then neither host would be able access the network.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Restricting Ports by Client MAC Address

Day 49 - Port Security

Original Question 68/74 - continued

Cisco: Configuring Port-Based Traffic Control: Configuring Port Security

Category:
Security Fundamentals

Day 51 - Dynamic ARP Inspection

Original Question 33/74

33/74

You issue the following commands on Switch1:

```
Switch1(config)#ip arp inspection vlan 11,14,18
Switch1(config)#interface fastethernet 0/1
Switch1(config-if)#switchport access vlan 14
Switch1(config-if)#switchport mode access
Switch1(config-if)#ip arp inspection trust
Switch1(config-if)#interface range fastethernet 0/2 - 4
Switch1(config-if-range)#switchport access vlan 14
Switch1(config-if-range)#switchport mode access
```

Which of the following statements are true? *(Select 2 choices.)*

- ☐ A. All ports in VLAN 14 are trusted ports.
- ☐ B. The FastEthernet 0/1 port in VLAN 14 is a trusted port.
- ☐ C. Ports in every VLAN except VLAN 14 are trusted ports.
- ☐ D. Every port in VLANs 11, 14, and 18 is an untrusted port.
- ☐ E. Every port except the FastEthernet 0/1 port in VLAN 14 is an untrusted port.

Incorrect

Correct Answer(s): B, E

Explanation:

In this scenario, the FastEthernet 0/1 port in virtual local area network (VLAN) 14 is a trusted port. In addition, every port except the FastEthernet 0/1 port in VLAN 14 is an untrusted port. The **ip arp inspection vlan 11,14,18** command configures Dynamic ARP Inspection (DAI) globally for VLANs 11, 14, and 18. This means that every port in those VLANs is automatically configured with DAI. When DAI is configured on a switch, ports are labeled as either trusted or untrusted. An Address Resolution Protocol (ARP) packet that is received on a trusted port is not inspected, whereas an ARP packet received on an untrusted port is inspected. If an ARP reply received on an untrusted port does not match a trusted entry on the switch, the packet will be dropped.

By default, a port is configured as an untrusted port when DAI is enabled on that port. Therefore, configuring VLANs 11, 14, and 18 with DAI ensures that any port operating in those VLANs is automatically an untrusted port. When DAI is configured for an entire VLAN, you can override the default configuration for a given port by issuing the **ip arp inspection trust** command in interface configuration mode.

Day 51 - Dynamic ARP Inspection

Original Question 33/74 - continued

In this scenario, a range of four interfaces has been configured to operate on VLAN 14. These interfaces include the FastEthernet 0/1 interface, the FastEthernet 0/2 interface, the FastEthernet 0/3 interface, and the FastEthernet 0/4 interface. However, only the FastEthernet 0/1 interface has been configured with the **ip arp inspection trust** command. Therefore, only the FastEthernet 0/1 interface is a trusted port.

DAI can be enabled on a single VLAN or on multiple VLANs. To enable DAI, you should use the **ip arp inspection vlan** global configuration command. The syntax of the **ip arp inspection vlan** command is **ip arp inspection vlan** {*vlan-ID* | *vlan-range*}. A range of VLANs can be entered by using a comma-separated list and/or a dash-separated pair of VLAN numbers indicating the range of VLANs to include. For example, each of the following commands enables DAI on VLANs 2 through 4: **ip arp inspection vlan 2,3,4** and **ip arp inspection vlan 2-4**. Configuring DAI on each VLAN ensures that traffic sent from each host is inspected. In addition, each port is by default an untrusted port.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, DAI

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 12: DHCP Snooping and ARP Inspection, Dynamic ARP Inspection Configuration

Cisco: Configuring Dynamic ARP Inspection: Interface Trust States and Network Security

Category:
Security Fundamentals

Day 51 - Dynamic ARP Inspection

Original Question 66/74

66/74

You issue the **show vlan brief** command on Switch1 and receive the following partial output:

```
Switch1#show vlan brief

VLAN Name                         Status    Ports
---- ---------------------------- --------- -------------------------------
1    default                      active    Gi0/1, Gi0/2
11   VLAN0011                     active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                            Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                            Fa0/9, Fa0/10
12   VLAN0012                     active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                            Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                            Fa0/19
14   VLAN0014                     active    Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                            Fa0/24
<output omitted>
```

You issue the following commands on Switch1:

```
Switch1#configure terminal
Switch1(config)#ip arp inspection vlan 11-12,14
Switch1(config)#interface range gigabitethernet 0/1 - 2
Switch1(config-if-range)#ip arp inspection trust
Switch1(config-if-range)#switchport access vlan 11
Switch1(config-if-range)#switchport mode access
```

Which of the following statements is true? *(Select the best answer.)*

- (●) A. Only GigabitEthernet 0/1 and GigabitEthernet 0/2 ports are untrusted ports.
- ( ) B. Only GigabitEthernet 0/1 and GigabitEthernet 0/2 are trusted ports.
- ( ) C. Only VLAN 11 ports are trusted ports.
- ( ) D. All ports on the switch are untrusted ports.
- ( ) E. All ports on the switch are trusted ports.

Day 51 - Dynamic ARP Inspection

Original Question 66/74 - continued

Incorrect

Correct Answer(s): B

Explanation:

Only GigabitEthernet 0/1 and GigabitEthernet 0/2 are trusted ports in this scenario because the **ip arp inspection trust** command has been issued on those interfaces. An interface-level Dynamic ARP Inspection (DAI) configuration overrides a global virtual local area network (VLAN) DAI configuration, even if the port that is configured with the interface-level command is assigned to a VLAN after the interface-level command has been issued.

When DAI is configured on a switch, ports are labeled as either trusted or untrusted. An Address Resolution Protocol (ARP) packet that is received on a trusted port is not inspected, whereas an ARP packet received on an untrusted port is inspected. If an ARP reply received on an untrusted port does not match a trusted entry on the switch, the packet will be dropped.

By default, a port is configured as an untrusted port when DAI is enabled on that port. Therefore, configuring VLANs 11, 12, and 14 with DAI in this scenario ensures that any port operating in those VLANs is automatically an untrusted port. However, when DAI is configured for an entire VLAN, you can override the default configuration for a given port by issuing the **ip arp inspection trust** command in interface configuration mode. In this scenario, the **ip arp inspection vlan 11-12,14** command configures DAI globally on VLANs 11, 12, and 14. The **ip arp inspection trust** command overrides the global configuration on the GigabitEthernet 0/1 and GigabitEthernet 0/2 ports.

DAI can be enabled on a single VLAN or on multiple VLANs. To enable DAI, you should use the **ip arp inspection vlan** global configuration command. The syntax of the **ip arp inspection vlan** command is **ip arp inspection vlan** {*vlan-ID* | *vlan-range*}. A range of VLANs can be entered by using a comma-separated list and/or a dash-separated pair of VLAN numbers indicating the range of VLANs to include. For example, each of the following commands enables DAI on VLANs 2 through 4: **ip arp inspection vlan 2,3,4** and **ip arp inspection vlan 2-4**. Configuring DAI on each VLAN ensures that traffic sent from each host is inspected. In addition, each port is by default an untrusted port.

Reference:

Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, DAI

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 12: DHCP Snooping and ARP Inspection, Dynamic ARP Inspection Configuration

Category:

Security Fundamentals

Day 52 - LAN Architectures

Original Question 51/74

51/74

Which of the following devices cannot be connected to leaf nodes in the Cisco ACI architecture? *(Select the best answer.)*

- A. EPGs
- B. application servers
- C. leaf nodes
- D. spine nodes
- E. APICs

Incorrect

Correct Answer(s): C

Explanation:

In the Cisco Application Centric Infrastructure (ACI), leaf nodes cannot connect to each other. Cisco ACI is a data center technology that uses switches, categorized as spine and leaf nodes, to dynamically implement network application policies in response to application-level requirements. Network application policies are defined on a Cisco Application Policy Infrastructure Controller (APIC) and are implemented by the spine and leaf nodes.

The spine and leaf nodes create a scalable network fabric that is optimized for east-west data transfer, which in a data center is typically traffic between an application server and its supporting data services, such as database or file servers. Each spine node requires a connection to each leaf node; however, spine nodes do not interconnect nor do leaf nodes interconnect. Despite its lack of fully meshed connections between spine nodes or between leaf nodes, this physical topology enables nonlocal traffic to pass from any ingress leaf interface to any egress leaf interface through a single, dynamically selected spine node. By contrast, local traffic is passed directly from an ingress interface on a leaf node to the appropriate egress interface on the same leaf node.

Because a spine node has a connection to every leaf node, the scalability of the fabric is limited by the number of ports on the spine node, not by the number of ports on the leaf node. For example, if additional access ports are needed, a new leaf node can be added to the infrastructure as long as there is a sufficient number of ports remaining on the existing spine nodes to support the new leaf node. In addition, redundant connections between a spine and leaf pair are unnecessary because the nature of the topology ensures that each leaf has multiple connections to the network fabric. Therefore, each spine node requires only a single connection to each leaf node.

Redundancy is also provided by the presence of multiple APICs, which are typically deployed as a cluster of three controllers. APICs are not directly involved in forwarding traffic and are therefore not required to connect to every spine or leaf node. Instead, the APIC cluster is connected to one or more leaf nodes in much the same manner that other endpoint groups (EPGs), such as application servers, are connected. Because APICs are not directly involved in forwarding traffic, the failure of an APIC does not affect the ability of the fabric to forward traffic.

Reference:

Boson CCNA Curriculum 200-301, Module 1: Network Fundamentals, Spine-Leaf Topology

Day 52 - LAN Architectures

Original Question 51/74 - continued

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 21: Introduction to Controller-Based Networking, ACI Physical Design: Spine and Leaf

Cisco: Cisco Application Centric Infrastructure: Cisco Application Centric Infrastructure Fabric Overview

Category:
Network Fundamentals

Day 53 - WAN Architectures & VPNs

Original Question 13/74

Which of the following statements are true about IPSec configured in tunnel mode? *(Select 2 choices.)*

- ☐ A. It is often used in networks with private IP addresses.
- ☐ B. It encrypts both the IP header and the payload.
- ☐ C. It encrypts the IP header but does not encrypt the payload.
- ☐ D. It encrypts the payload but does not encrypt the IP header.
- ☐ E. It is less secure than IPSec configured in transport mode.

Incorrect

Correct Answer(s): A, B

Explanation:

Internet Protocol Security (IPSec) configured in tunnel mode it is often used in networks with private Internet Protocol (IP) addresses and it encrypts both the IP header and the payload. IPSec is a suite of protocols that can be used to encrypt Generic Routing Encapsulation (GRE) tunnel traffic, such as over a virtual private network (VPN). IPSec supports two modes: transport and tunnel. In tunnel mode, IPSec encrypts the entire IP packet. IPSec is often used in networks with private IP addresses because it hides the private IP address within the encrypted packet.

By contrast, IPSec in transport mode does not encrypt the IP header. In transport mode, only the IP packet's payload is encrypted by IPSec, which means that the IP packet's header remains intact. Therefore, IPSec configured in tunnel mode is more secure than IPSec configured in transport mode.

Reference:

Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, IPSec Modes

Cisco: Introduction to Cisco IPSec Technology: Transport Mode and Tunnel Mode

Category:

Security Fundamentals

Day 53 - WAN Architectures & VPNs

Original Question 15/74

15/74 OVO

Which of the following connections provides the greatest total bandwidth? *(Select the best answer.)*

- A. T3
- B. E3
- C. T1
- D. E1

Incorrect

Correct Answer(s): A

Explanation:

Of the choices provided, a T3 connection has the greatest total bandwidth. A T3 connection has a total bandwidth of 44.736 megabits per second (Mbps). Primarily used in the United States and Japan, T3 connections transmit data across 672 DS0 channels 8,000 times per second. T3 connections are usually too cost prohibitive for use except by Internet and telephone service providers.

A T1 connection has a total bandwidth of 1.544 Mbps. To achieve the 1.544-Mbps rate, T1 connections encapsulate data in DS1 signaling frames. DS1 signaling frames include 24 DS0 channels and one framing bit. Each of the 24 DS0 channels transfers 8 bits of data at a time. The inclusion of the framing bit raises the total amount of data in a single DS1 frame to 193 bits. T1 connections transmit 8,000 DS1 frames per second, which brings the total bandwidth to 1,544,000 bits per second (bps), or 1.544 Mbps. T1 connections typically transfer data across two pairs of shielded copper wires, known as a T1 line, which extend from the service provider to the customer. One pair of wires within the T1 line is used to send data, whereas the other pair is used to receive data. T1 is primarily used in the United States and Japan.

An E1 connection has a total bandwidth of 2.048 Mbps. Primarily used in Europe, E1 connections operate similarly to T1 connections. However, instead of using 24 eight-bit DS0 channels, E1 connections use 32 eight-bit timeslots. E1 connections transmit 8,000 timeslot segments per second, which equals 2,048,000 bps, or 2.048 Mbps. Like T1s, E1 connections transfer data across two pairs of shielded copper wires.

An E3 connection has a total bandwidth of 34.368 Mbps. Primarily used in Europe, E3 connections consist of 16 E1 lines. E3 connections are usually too cost prohibitive for use except by Internet and telephone service providers.

Day 53 - WAN Architectures & VPNs

Original Question 15/74 - continued

Reference:

Boson CCNA Curriculum 200-301, Module 1: Network Fundamentals, WAN Topologies

How To Network: Wide Area Networks – WAN

Category:

Network Fundamentals

Day 54 - Virtualization & Cloud

Original Question 55/74

55/74

Which of the following statements is correct regarding containers and VMs? *(Select the best answer.)*

- ○ A. VMs require a hypervisor, whereas containers do not.
- ○ B. VMs can have dedicated resources, whereas containers cannot.
- ○ C. VMs are isolated from one another, whereas containers are not.
- ○ D. VMs share bins and libraries, whereas containers do not.

Incorrect

Correct Answer(s): A

Explanation:

Virtual machines (VMs) require a hypervisor, whereas containers do not. A VM is an isolated environment running a separate operating system (OS) while sharing a computer's hardware resources. Depending on a computer's hardware capabilities, multiple VMs can be installed on a single computer, which can help provide more efficient utilization of hardware resources.

A hypervisor is computer software, firmware, or hardware that runs VMs. A Type-1 hypervisor runs directly on the server hardware and does not require a guest OS. A Type-2 hypervisor runs on a guest OS.

Containers are simply a group of processes that run on a server. From inside the container, it looks like a normal VM. However, a container is much more efficient and lightweight than a VM, enabling you to install more container instances on a server than if you had installed only VMs.

Both VMs and containers are isolated from one another. One VM on a server cannot directly affect another VM on that server, and one container on a server cannot directly affect another container on that server. However, containers can share an OS's kernel, bins, and libraries. Therefore, if the host OS is attacked, it can affect all of the containers in that host OS. When VMs are used, a hypervisor protects the kernel by intercepting and relaying system calls from the VMs. As a result, security is

more of a concern with containers than it is with VMs.

Both VMs and containers can have dedicated resources, ensuring that there is enough memory or processor resources available for the VM or container. You can also set hard or soft quotas for memory and processor utilization so that a VM or container does not use too many resources.

Reference:
Boson CCNA Curriculum 200-301, Module 4: Virtualization Fundamentals, Device Virtualization

Cisco: Cisco Container Networking Overview and Roadmap (PDF)

Docker: Use containers to Build, Share and Run your applications

freeCodeCamp: A Beginner-Friendly Introduction to Containers, VMs and Docker

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 20: Cloud Architecture, Server Virtualization

Category:
Network Fundamentals

Day 55 - Wireless Fundamentals

Original Question 26/74

26/74

Which of the following is considered best practice when expanding an existing 802.11 wireless network? *(Select the best answer.)*

- A. configuring each AP with a unique SSID and a unique, nonoverlapping channel
- B. configuring each AP with the same SSID and a unique, overlapping channel
- C. configuring each AP with the same SSID and a unique, nonoverlapping channel
- D. configuring each AP with a unique SSID and the same channel

Correct Answer(s): C

Explanation:

Of the choices provided, configuring each wireless access point (AP) with the same Service Set Identifier (SSID) and a unique, nonoverlapping channel is considered best practice when expanding an existing Institute of Electrical and Electronics Engineers (IEEE) 802.11 wireless network. APs are devices that are used to establish communication between wireless devices and a wired network through the use of radio waves. APs are often placed in a centralized location within a network environment and are typically connected to a network through a wired connection. An SSID is a unique name used to identify a wireless network. When multiple APs are used to establish network connections to the same network, each AP must be configured with the same SSID.

APs transmit data by using a single channel at a time. A channel is a small portion of the spread-spectrum used for transmission. APs operating on the same channel and within close physical proximity to other APs may experience some interference. To avoid potential interference while continuing to offer the same physical range of coverage, technicians can set the channel used by the individual APs to a channel farther away on the spread-spectrum than the channel being used by the other AP in the same service radius. For example, if you have two APs within physical range of each other, you could set one to operate on channel 1 and the other to operate on channel 11. By default, APs that use the 2.4 gigahertz (GHz) spectrum, which include those on 802.11b, 802.11g, 802.11n, 802.11ax, and 802.11be networks, typically use the nonoverlapping channels 1, 6, or 11 to decrease the potential for interference issues by keeping the channels in use evenly spread apart.

Reference:

Boson CCNA Curriculum 200-301, Module 3: Wireless Networking Fundamentals, Wireless Bands and Channels

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 1: Fundamentals of Wireless

Day 55 - Wireless Fundamentals

Original Question 26/74 - continued

Networks, Wireless Bands and Channels

Cisco: Deployment

Category:
Network Fundamentals

Day 55 - Wireless Fundamentals

Original Question 48/74

48/74

Which of the following fields in an 802.11 MAC frame is used to indicate whether the frame is a management frame? *(Select the best answer.)*

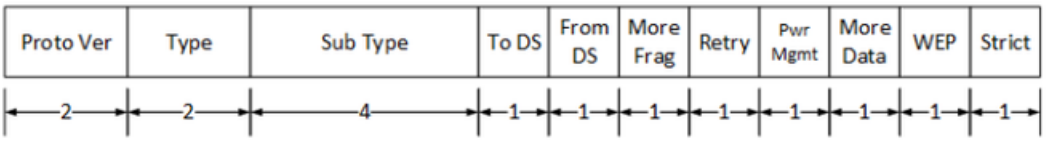
- A. FCS
- B. DUR
- C. FC
- D. SEQ

Incorrect

Correct Answer(s): C

Explanation:

The Frame Control (FC) field of an Institute of Electrical and Electronics Engineers (IEEE) 802.11 Media Access Control (MAC) frame is used to indicate whether the frame is a management frame. The FC field is the first field in an 802.11 frame, and its 2 bytes of data are subdivided into 11 related fields of information, such as wireless protocol, frame type, and frame subtype, as shown in the following diagram:



The Protocol Version field is a 2-bit field that identifies the 802.11 protocol used by the frame. The 2-bit Type field identifies what type of 802.11 frame follows. There are three general types of 802.11 frames: control, management, and data. The 4-bit Sub Type field further identifies what subset of frame can be expected. For example, if the Type field identifies the frame as a management frame, the Sub Type field might identify the frame as a beacon frame, which is a type of management frame.

The To DS and From DS fields are used to identify if the frame is destined to or transmitted from the distribution system (DS). For example, if the To DS field was set and the From DS field was unset, this would indicate that the frame was travelling from the wireless network to a computer that is connected to the wired network, or DS. If the To DS field is unset and the From DS field is set, then the frame is travelling to the wireless network from the wired network. If neither field is set, then the frame is not intended to leave the wireless network.

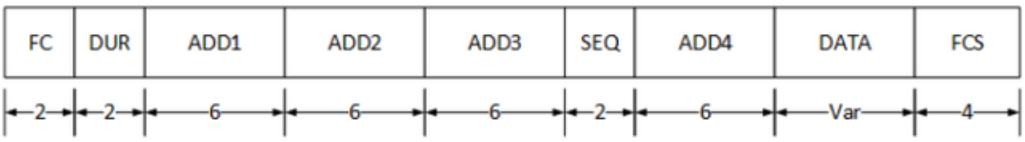
frame is not intended to leave the wireless network.

The More Fragments field indicates whether this frame is fragmented and if additional fragments follow. The Retry field indicates whether the frame has been retransmitted; this field is only applicable to management and data frames.

The Power Management field identifies the power mode of the client, which is commonly referred to as the station (STA). The STA can be in power save mode or active mode, and the Power Management field of a frame is used to inform the access point (AP) of the current state of the STA. The More Data field indicates that the AP has additional data buffered for the STA. This field is commonly used with broadcast and multicast frames.

The WEP field indicates whether the frame data is encrypted. Finally, the Strict field indicates whether data frames can be processed in an order different from which they were received.

The remaining fields in an 802.11 frame are as follows:



The Duration (DUR) field is a 2-byte field that is used mainly by control frames to indicate transmission timers. However, this field is also used by the Power Save (PS) Poll control frame to indicate the association identity (AID) of a client.

The address fields, Address 1 (ADD1), Address 2 (ADD2), Address 3 (ADD3), and Address 4 (ADD4), are 6-byte fields used to convey MAC address and Basic Service Set Identifier (BSSID) information. What information resides in which address field is entirely dependent on the type of frame. However, ADD1, ADD2, and ADD3 typically contain a source MAC address, destination MAC address, and BSSID with the order being dependent on whether the frame is entering the DS, leaving the DS, or passing directly between ad-hoc wireless devices. The ADD4 field is only present for frames passing between devices in the DS, such as from one AP to another AP.

The Sequence (SEQ) field is a 2-byte field that is subdivided to store two related pieces of information: the fragment number and sequence number of each frame.

The DATA portion of a frame varies in size and contains the frame's payload. For data frames, the payload is user data. However, for other frames, such as management frames, this portion of the frame might contain information such as supported data rates and cipher suites.

Finally, the Frame Check Sequence (FCS) field contains a 4-byte cyclic redundancy check (CRC) value calculated from all the 802.11 header fields, including the data portion of the frame. This value is used by the receiving station to determine whether the frame was corrupted during transit.

Reference:
Boson CCNA Curriculum 200-301, Module 3: Wireless Networking Fundamentals, 802.11 MAC Frames

SANS Institute: IEEE 802.11 Pocket Reference Guide (PDF)

Microsoft: How 802.11 Wireless Works: 802.11 MAC Frame

Category:
Network Fundamentals

Day 56 - Wireless Architectures

Original Question 10/74

10/74

Which of the following connects a wireless client to a wired network without requiring a separate wireless controller? *(Select the best answer.)*

- A. lightweight AP deployment
- B. embedded AP deployment
- C. autonomous AP deployment
- D. cloud-based AP deployment

Incorrect

Correct Answer(s): C

Explanation:

Of the available choices, an autonomous access point (AP) deployment connects a wireless client to a wired network without requiring a separate wireless controller. An AP is a device that connects a wireless client to a wired network. An autonomous AP contains network interfaces for both wireless and wired networks; it is typically deployed as part of an autonomous AP architecture in which APs are connected directly to the access layer of the three-tier hierarchical network model.

A lightweight AP deployment connects a wireless client to a wired network but requires a separate wireless controller. Wireless clients connect to lightweight APs, which are capable of performing real-time wireless network functions but rely on a Cisco wireless LAN controller (WLC) for management functions. The connection between a lightweight AP and a WLC is created by using two tunnels established by the Control and Provisioning of Wireless Access Points (CAPWAP) tunneling protocol. Information sent between lightweight APs and the WLC is encapsulated in Internet Protocol (IP) packets. This process enables a lightweight AP and WLC to manage connectivity to the same wireless local area network (WLAN) yet be separated by both physical and logical means. This type of deployment is also known as a split-MAC architecture because the lightweight AP handles the frames while the WLC handles the management functions.

A cloud-based AP deployment connects a wireless client to a wired network but requires a separate wireless controller. For example, a Cisco Meraki AP provides wireless access by connecting to a

wireless controller. For example, a Cisco Meraki AP provides wireless access by connecting to a centralized management system known as the Cisco Meraki Cloud. APs deployed at the access layer of the three-tier hierarchical network model contact the cloud in order to automatically configure themselves. APs are managed through a cloud-based dashboard.

An embedded AP deployment connects a wireless client to a wired network but requires a separate wireless controller. The primary difference between this deployment and others is that the WLC is embedded within a stack of switching hardware instead of existing as a separate entity. APs can connect to the WLC by connecting to switches that are directly hosting the WLC or switch ports that are operating on the same virtual local area network (VLAN) as the WLC.

Reference:
Boson CCNA Curriculum 200-301, Module 3: Wireless Networking Fundamentals, Associating with an AP

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 2: Analyzing Cisco Wireless Architectures, Autonomous AP Architecture

Category:
Network Access

Day 56 - Wireless Architectures

Original Question 32/74

32/74

Which of the following WLC interfaces is the only available interface when the WLC is booting? *(Select the best answer.)*

- ○ A. the service port interface
- ○ B. a dynamic interface
- ○ C. the management interface
- ○ D. the virtual interface
- ○ E. the AP-manager interface

Incorrect

Correct Answer(s): A

Explanation:

The service port interface of a wireless LAN controller (WLC) is the only available interface when the WLC is booting. A WLC can contain up to four static interfaces: the management interface, the AP-manager interface, the virtual interface, and the service port interface. The service port interface is used for maintenance purposes on a WLC. This interface is a physical interface on the WLC that can be used to recover the WLC in the event that the WLC fails. The service port can be used as a last-resort measure to access the WLC graphical user interface (GUI).

Alternatively, you can use the console port to access the WLC command-line interface (CLI). The console port is an out-of-band asynchronous interface and is typically accessed with a serial cable or a universal serial bus (USB) to RJ-45 adapter.

The virtual interface supports mobility management by providing a specific Internet Protocol (IP)

The virtual interface supports mobility management by providing a specific Internet Protocol (IP) address that is the same across multiple controllers when wireless clients roam among the controllers. This enables seamless roaming among the controllers. The virtual interface supports other uses as well. For example, the virtual interface can also be used in situations where web authorization has been enabled for clients; the user is redirected to the IP address of the virtual interface when the user opens a web browser. In addition, if Dynamic Host Configuration Protocol (DHCP) relay has been enabled on the controller, the virtual interface can be used as the DHCP server address on wireless clients.

The AP-manager interface contains the IP address that is used as the source IP address by which the lightweight access points (APs) communicate with the WLC. Because the AP-manager interface communicates with the lightweight APs on the wireless network, the IP address assigned to the AP-manager interface should be unique on the network. After the interface has been configured, the WLC uses the AP-manager interface to listen for Layer 3 Lightweight Access Point Protocol (LWAPP) communications.

The management interface is used for in-band management information. This interface is used for all Layer 2 LWAPP communications between the controller and the lightweight APs. In addition, the management interface is used to communicate with other WLCs on the wireless network.

In addition to the four static interfaces, a WLC can contain up to 512 dynamic interfaces. Dynamic interfaces are user-defined and are typically used for wireless client data. The dynamic interfaces function similarly to virtual local area networks (VLANs). For example, you can create a dynamic interface to segment traffic on the WLC.

Reference:
Boson CCNA Curriculum 200-301, Module 3: Wireless Networking Fundamentals, Cisco WLC Interfaces

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 4: Building a Wireless LAN, WLC Physical Ports

Cisco: Overview of Ports and Interfaces: Interfaces (PDF)

Category:
Network Access

Day 57 - Wireless Security

Original Question 7/74

7/74

Which of the following technologies is implemented in WPA3-Enterprise to improve authentication encryption? *(Select the best answer.)*

- A. SAE
- B. PMF
- C. ECDSA
- **D. AES-GCMP**

Correct

Correct Answer(s): D

Explanation:

Of the available choices, Advanced Encryption Standard (AES)-Galois/Counter Mode Protocol (GCMP) is the technology that is implemented in Wi-Fi Protected Access 3 (WPA3)-Enterprise to improve authentication encryption. WPA3 was introduced by the Wi-Fi Alliance in 2018 to address weaknesses in WPA2. WPA3-Enterprise uses 256-bit AES-GCMP.

WPA3-Enterprise uses Elliptic Curve Digital Signature Algorithm (ECDSA) with a 384-bit curve in addition to Elliptic Curve Diffie-Hellman (ECDH) for key establishment and authentication. This is not the same process as the authentication encryption that is provided by AES-GCMP.

Simultaneous Authentication of Equals (SAE) is the technology that is implemented in WPA3-Personal to replace the use of pre-shared keys (PSKs) in WPA2-Personal. SAE enables clients and access points (APs) to mutually authenticate

(APs) to mutually authenticate.

WPA3-Personal introduces support for Protected Management Frames (PMF). However, PMF does not replace PSKs. PMF secures management frame communication between an AP and a client.

Reference:
Boson CCNA Curriculum 200-301, Module 3: Wireless Networking Fundamentals, WPA3

Wi-Fi Alliance: Security

Category:
Security Fundamentals

Day 57 - Wireless Security

Original Question 30/74

30/74

You are configuring security on a new WLAN by using the WLC GUI.

Which of the following security settings are you most likely to configure by using the **Layer 2 Security** drop-down list box on the **Layer 2** tab? *(Select the best answer.)*

- A. **Web Authentication**
- B. **WPA+WPA2**
- C. **Web Passthrough**
- D. **VPN Pass-Through**

Incorrect

Correct Answer(s): B

Explanation:

Of the available choices, you are most likely to configure the **WPA+WPA2** setting by using the **Layer 2 Security** drop-down list box on the **Layer 2** tab of the Cisco wireless LAN controller (WLC) graphical user interface (GUI). There are two types of wireless local area networks (WLANs) that you can configure by using the WLC GUI: a WLAN and a Guest LAN. When you configure a new WLAN by using the WLC GUI, you can configure security settings by clicking the new WLAN's **Security** tab. By default, the **Layer 2** tab is selected when you click the **Security** tab. However, it is not possible to configure Layer 2 security on a Guest LAN.

On the **Layer 2** tab of the **Security** tab, you can select one of the following Layer 2 wireless security features from the **Layer 2 Security** drop-down list box:

- **None**, which disables Layer 2 security and allows open authentication to the WLAN
- **WPA+WPA2**, which enables Layer 2 security by using Wi-Fi Protected Access (WPA) or the more secure WPA2
- **802.1X**, which enables Layer 2 security by using Extensible Authentication Protocol (EAP) authentication combined with a dynamic Wired Equivalent Privacy (WEP) key
- **Static WEP**, which enables Layer 2 security by using a static shared WEP key
- **Static WEP + 802.1X**, which enables Layer 2 security by using either a static shared WEP key or EAP authentication
- **CKIP**, which enables Layer 2 security by using the Cisco Key Integrity Protocol (CKIP)
- **None + EAP Passthrough**, which enables Layer 2 security by using open authentication combined with remote EAP authentication

There are two different sets of Layer 3 security features that you can configure on a Cisco WLC: one set for a WLAN and one set for a Guest LAN. Depending on which type of WLAN you create and which Layer 2 security options you have selected, you can select one of the following Layer 3 wireless security features from the **Layer 3 Security** drop-down list box on the **Layer 3** tab of the **Security** tab in the WLC GUI:

- **None**, which disables Layer 3 security no matter which Layer 2 security option is configured and regardless of whether you are configuring a WLAN or a Guest LAN
- **IPSec**, which enables Layer 3 security for WLANs by using Internet Protocol Security (IPSec)
- **VPN Pass-Through**, which enables Layer 3 security for WLANs by allowing a client to establish a connection with a specific virtual private network (VPN) server
- **Web Authentication**, which enables Layer 3 security for Guest LANs by prompting for a user name and password when a client connects to the network
- **Web Passthrough**, which enables direct access to the network for Guest LANs without prompting for a user name and password

Not every Layer 3 security mechanism is compatible with every Layer 2 security mechanism. It is therefore important to first configure Layer 2 security options before you attempt to configure Layer 3 security options.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Configuring Cisco WLAN Layer 2 Security

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 4: Building a Wireless LAN, Configuring WLAN Security

Cisco: WLC Layer 2 Layer 3 Security Compatibility Matrix: Wireless LAN Controller Layer 2 – Layer 3 Security Compatibility Matrix

Cisco: WLANs Tab

Category:
Network Access

Day 57 - Wireless Security

Original Question 41/74

41/74

You are configuring Layer 2 security on a WLAN by using the WLC GUI. You select **WPA+WPA2** from the **Layer 2 Security** drop-down list box. You want to configure the WPA2 key in hexadecimal format.

Which of the following WPA2 key management methods should you select from the **Auth Key Mgmt** drop-down list box? *(Select the best answer.)*

- ○ A. **PSK**
- ○ B. **CCKM**
- ○ C. **802.1X**
- ○ D. **802.1X+CCKM**

Incorrect

Correct Answer(s): A

Explanation:

You should select the **PSK** Wi-Fi Protected Access 2 (WPA2) key management method from the **Auth Key Mgmt** drop-down list box if you want to configure the WPA2 key in hexadecimal format. The **PSK** method configures WPA or WPA2 to use the Pre-Shared Key (PSK) key management method. This method requires that an administrator configure each wireless client that will connect to the network with the key that is configured on the Cisco wireless LAN controller (WLC). The **PSK** option supports key entry as either an ASCII passphrase from 8 through 63 characters in length or a key of 64 hexadecimal values. Combining WPA or WPA2 with a PSK key management method is often known as WPA-PSK, or WPA Personal. WPA2 uses Advanced Encryption Standard (AES) and Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP) to provide message integrity checks (MICs) and encryption.

You should not select the **802.1X** key management method in this scenario. The Institute of Electrical and Electronics Engineers (IEEE) 802.1X standard defines a method of port-based network access control. On Cisco wireless local area networks (WLANs), the 802.1X key management method is the default method for both WPA and WPA2. It typically requires a Remote Authentication Dial-In User Service (RADIUS) server and uses various Extensible Authentication Protocol (EAP) implementations to authenticate users. Combining WPA or WPA2 with an 802.1X key management method is often known as WPA-8021X mode, or WPA Enterprise.

You should not select the **CCKM** key management method in this scenario. This option enables the Cisco Centralized Key Management (CCKM) key management method. CCKM is a Cisco-proprietary fast-rekeying method that enables a wireless client to roam from one access point (AP) to another without requiring intervention from the WLC. CCKM is typically used to reduce delay when wireless clients transition between APs so that delay-sensitive services, such as Voice over Internet Protocol (VoIP), operate smoothly.

You should not select the **802.1X+CCKM** key management method in this scenario. This option enables 802.1X clients to use the CCKM key management method to roam between APs without performing the complete authentication process again. Normally, 802.1X clients mutually authenticate to a new AP. This process likewise involves reauthenticating with the RADIUS server. The 802.1X+CCKM key management method removes the need to reauthenticate with the RADIUS server, thus reducing the amount of time it takes for an 802.1X client to roam between APs.

Reference:
Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Configuring Cisco WLAN Layer 2 Security

Cisco: WLAN Security: Configuring WPA1+WPA2 (GUI)

Category:
Security Fundamentals

Day 59 - Intro to Network Automation

Original Question 52/74

52/74

Which of the following is not an advantage of controller-based networks over traditional networks? *(Select the best answer.)*

- A. segregating software development processes from IT teams
- B. using northbound APIs to enable automation
- C. using southbound APIs to configure network devices
- D. configuring the entire network instead of each device

Incorrect

Correct Answer(s): A

Explanation:

Controller-based networks do not segregate software development processes from IT teams. By contrast, controller-based networks typically facilitate a DevOps approach to networking. Based on Agile production, delivery, management, operation, and maintenance, DevOps includes the added goal of fostering collaboration between Agile software development teams and IT operations teams throughout an application's life cycle. This process involves merging the work of quality assurance (QA) teams, operations teams, and development teams into a single major development process. Therefore, DevOps can help increase the speed and reliability of software development, testing, and deployment.

Controller-based networks allow the configuration of the entire network instead of per-device configuration. This configuration is accomplished by using southbound Application Programming Interfaces (APIs). Being able to configure the entire network is an advantage over traditional network configuration, which typically requires an administrator to connect to and configure each device individually. There are four specific southbound API standards that controllers use to communicate: OpenFlow, NETCONF, OnePK, and OpFlex.

Controller-based networks use northbound APIs to enable automation. This feature is an advantage over traditional networks, which require either an administrator or a programming functionality to intervene or to develop scripts to automate network tasks. A northbound API, which is also sometimes called a northbound interface, enables a controller to exchange information with applications. The applications send requests or instructions to the controller. The controller then uses that information to modify and manage network flow.

Reference:

Boson CCNA Curriculum 200-301, Module 9: Automation and Programming, Summary

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 21: Introduction to Controller-Based Networking, Comparing Traditional Versus Controller-Based Networks

Category:

Automation and Programmability

Day 59 - Intro to Network Automation

Original Question 71/74

71/74

Which of the following is a benefit of network automation? *(Select the best answer.)*

- A. Data models are human-interpreted from the output of **show** commands.
- B. Data models are formed from **show** command output that is processed by automation scripts.
- C. Data models are formalized and defined by a centralized controller.
- D. Data models are enhanced by APIs to provide only the most specific information.

Incorrect

Correct Answer(s): C

Explanation:

One benefit of network automation is that data models are formalized and defined by a centralized controller. In addition, network automation aids reliable and consistent deployment of device configurations and software updates throughout an enterprise. Network automation includes the Software-Defined Networking (SDN) architecture. An SDN architecture is one in which management software is used to centralize device intelligence.

SDNs use northbound Application Programming Interfaces (APIs) to send network instructions from software applications to the central controller. Southbound protocols, which connect to a network's physical devices, are typically linked to the SDN controller by using a service abstraction layer (SAL). The SAL is a database, or registry, of the services provided by the southbound APIs. The APIs are bound to the registry so that the SAL can service an application's request.

Data models are enhanced by APIs to provide more robust information, not only the most specific information, about a network. Because of the centralized controller, data can be collected throughout a network. The controller can then provide that data to APIs in ways that enable extrapolation and interpretation that is more difficult to achieve when data is manually collected.

Data models are not human-interpreted from the output of **show** commands when network automation is implemented. The issuing of Cisco IOS **show** commands to verify configurations and to troubleshoot networks is an action traditionally performed by an administrator. The administrator must then interpret the output of the **show** command to verify a configuration or to uncover problems with the configuration. Often, the output of a given **show** command must be compared to

the output of other **show** commands or to a configuration standard in order for the administrator to obtain complete information.

Data models are not formed from **show** command output that is processed by automation scripts. Automation scripts are typically human-created programs that parse the output of **show** commands in order to obtain specific information about a specific configuration. Although they might ease administrative burden for common and repetitive tasks, automation scripts are less robust than network automation.

Reference:
Boson CCNA Curriculum 200-301, Module 9: Automation and Programming, Automation and Programming Overview

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 21: Introduction to Controller-Based Networking, How Automation Impacts Network Management

Cisco: What Is Network Automation?

Cisco: Software Defined Networking

Category:
Automation and Programmability

Day 60 - JSON, XML & YAML

Original Question 29/74

29/74 OVO

A REST API query returns the following output:

```
{
   "id": 12345,
   "fname": "John",
   "lname": "Doe",
   "group": {
      "role": "Receivables",
      "read-only": [
         "Accounting Folder",
         "Sales Folder"
      ]
   }
}
```

Which of the following statements are true? *(Select 2 choices.)*

- ☐ A. The value of the **role** key is an object.
- ☐ B. The value of the **lname** key is not a text value.
- ☑ **C. The value of the fname key is a text value.**
- ☐ D. The value of the **read-only** key is a text value.
- ☐ E. The value of the **id** key is an array.
- ☑ **F. The value of the group key is an object.**

Correct

Correct Answer(s): C, F

Explanation:

Of the available choices, it is true that the value of the **fname** key in the Representational State Transfer (REST) Application Programming Interface (API) output above is a text value. In addition, it is true that the value of the **group** key in the output is an object. REST is an API architecture that uses Hypertext Transfer Protocol (HTTP) or HTTP Secure (HTTPS) to enable external resources to access and make use of programmatic methods that are exposed by the API. The REST API that is being queried in this scenario has returned output in JavaScript Object Notation (JSON) format. JSON is a data modeling language that is commonly used by REST APIs. The JSON data modeling language returns data in the form of an object that contains key and value pairs.

A single JSON object can contain multiple key and value pairs. Each key and value pair inside a JSON object is separated from the others by a comma (**,**). Furthermore, each pair's key is separated from its value by a colon (**:**). The element in quotation marks on the left side of each colon is the key. The element on the right side of each colon is the value, which might or might not be enclosed in quotation marks. There are several data value types that can be returned in JSON output: text, numeric, array, object, Boolean, and null.

The value of the **lname** key, the value of the **fname** key, and the value of the **role** key are all text values in the JSON object in this scenario. JSON keys and JSON text values are always enclosed in quotation marks. Text values typically contain alphanumeric or Unicode characters.

The value of the **id** key is numeric in the JSON object in this scenario. Numeric data is defined by digits that are not enclosed in quotation marks. If a string of digits is enclosed in quotation marks, it will be treated as text even if it contains only numeric characters. The numeric value of the id key in this scenario is **12345**.

The value of the **group** key in this scenario is another JSON object. A JSON object can be identified by the curly brackets that mark the beginning and the end of the object. A left curly bracket ({) marks the beginning of a JSON object. A right curly bracket (}) marks the end of a JSON object. JSON objects can be nested as values inside other JSON objects. In addition, JSON objects can be specified as values inside other JSON objects. In this scenario, the value of the **group** key in the primary JSON object is shown in the following JSON object:

```
{
   "role": "Receivables",
   "read-only": [
      "Accounting Folder",
      "Sales Folder"
   ]
}
```

As previously mentioned, a JSON object is a group of key and value pairs. The JSON object above contains two key and value pairs. The **role** key contains a text value of **Receivables**. The **read-only** key, on the other hand, contains an array.

The value of the **read-only** key in this scenario is an array that contains two text values: **Accounting Folder** and **Sales Folder**. Unlike JSON objects, arrays contain values only. They do not contain key and value pairs. Arrays are typically contained within square brackets. A left square bracket ([) indicates the beginning of the array. A right square bracket (]) indicates the end of the array. Each value inside an array is separated from the others by a comma (,). Although the array in this scenario contains two text values enclosed in quotation marks, an array can contain any of the other JSON data types. For example, an array can include JSON objects or other arrays as values. In addition, a given array need not contain only a single data type.

There are no Boolean or null values in the JSON output in this scenario. A given JSON key can be assigned a Boolean value of either **true** or **false**. Because these are Boolean values, not text values, they are typically not placed inside quotation marks. Null values technically have no value at all, although it is not exactly the same as a numeric value of 0. A given JSON key can be assigned a null value by specifying **null** without quotation marks.

Reference:
Boson CCNA Curriculum 200-301, Module 9: Automation and Programming, JSON

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 23: Understanding REST and JSON, Recognizing the Components of JSON

Category:
Automation and Programmability

Day 60 - JSON, XML & YAML

Original Question 29/74 - continued

Day 61 - REST APIs

Original Question 20/74

20/74 OVO

Which of the following statements is true about the HTTP POST method? *(Select the best answer.)*

- ○ A. It can be used to apply partial modifications to an object.
- ○ B. It can be used to remove an object.
- ○ C. It can be used to retrieve data from an object.
- ○ D. It can be used to implement a webhook.

Correct

Correct Answer(s): D

Explanation:

The Hypertext Transfer Protocol (HTTP) POST method can be used to implement a webhook; the HTTP POST method can also be used to create a new object or resource. A webhook, also known as a reverse Application Programming Interface (API), is a user-defined HTTP callback. Webhooks are event-driven rather than request-driven: instead of an application continuously requesting data, the application can wait for a particular event to occur before it reacts. Webhook notifications can be used on many Cisco platforms, including Meraki and Webex Teams.

The following HTTP methods are most commonly used:

- GET – retrieves data from an object
- PUT – adds information to an object
- POST – creates an object
- DELETE – removes an object
- PATCH – applies partial modifications to an object

Reference:

Boson CCNA Curriculum 200-301, Module 9: Automation and Programming, HTTP Verbs

MDN: POST request method

Prismatic: Anatomy of a Webhook HTTP Request

REST API Tutorial: HTTP Methods

Category:

Automation and Programmability

Day 61 - REST APIs

Original Question 61/74

61/74

Which of the following HTTP methods corresponds to the CRUD read operation? *(Select the best answer.)*

- ○ A. PUT
- ○ B. GET
- ○ C. DELETE
- ○ D. POST
- ○ E. PATCH

Incorrect

Correct Answer(s): B

Explanation:

The Hypertext Transfer Protocol (HTTP) GET method corresponds to the CRUD read operation. The acronym CRUD refers to the four actions that are performed by applications: create, read, update, and delete. Create allows the formation of new variables or data structures. Read allows a client to view data. Update allows a client to change data. Delete allows a client to remove data. The following table shows the relationship between HTTP methods and CRUD operations:

HTTP Methods	CRUD Operations
GET	Read
PATCH	Update
PUT	Update
POST	Create
DELETE	Delete

The HTTP POST method does not correspond to the CRUD read operation; it corresponds to the CRUD create operation. The HTTP PATCH method does not correspond to the CRUD read operation; it corresponds to the CRUD update operation. The HTTP PUT method does not correspond to the CRUD read operation; it corresponds to the CRUD update operation. The HTTP DELETE method does not correspond to the CRUD read operation; it corresponds to the CRUD delete operation.

Reference:

Boson CCNA Curriculum 200-301, Module 9: Automation and Programming, HTTP Verbs

Day 61 - REST APIs

Original Question 61/74 - continued

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 23: Understanding REST and JSON, Software CRUD Actions and HTTP Verbs

Microsoft: CRUD (Create, Read, Update, Delete)

Category:
Automation and Programmability

Day 62 - Software-Defined Networking

Original Question 22/74

Drag each Cisco SDA term to its description. Use all terms. Each term can be used only once.

Terms	Descriptions
northbound API	creates VXLAN tunnels between SDA switches
southbound API	is a collection of devices that comprises the IP network that connects to each fabric node
overlay	enables an SDN controller to communicate with applications in the application plane
underlay	is the entirety of the overlay network and the underlay network
fabric	enables an SDN controller to communicate with devices in the data plane

Reset

Incorrect

Answer:

overlay	creates VXLAN tunnels between SDA switches
underlay	is a collection of devices that comprises the IP network that connects to each fabric node
northbound API	enables an SDN controller to communicate with applications in the application plane
fabric	is the entirety of the overlay network and the underlay network
southbound API	enables an SDN controller to communicate with devices in the data plane

Explanation:
A Software-Defined Networking (SDN) controller uses two different sets of Application Programming Interfaces (APIs): one set to communicate with applications and another set to communicate with devices in the data plane. Northbound APIs enable an SDN controller to communicate with applications in the application plane. Applications use northbound APIs to send requests or instructions to the SDN controller, which uses that information to modify and manage network flow.

Southbound APIs enable an SDN controller to communicate with devices in the data plane. In both Software-Defined Access (SDA) and SDN deployments, the controller communicates with devices by using a southbound API. Communication with applications and user interfaces is accomplished by using a northbound API.

Cisco SDA is a Cisco-developed SDN for building local area networks (LANs) by using policies and automation. The Cisco Digital Network Architecture (DNA) controller, which is similar to an SDN controller, is the central component of a Cisco SDA network. Cisco DNA is a software-centric network architecture that uses a combination of APIs and a graphical user interface (GUI) to simplify network operations.

The overlay network creates Virtual Extensible LAN (VXLAN) tunnels between Cisco SDA switches. The tunnels send and receive traffic between fabric endpoints. The underlay network, on the other hand, is a more traditional network configuration of switches. It is a collection of devices, interfaces, and media that comprises the Internet Protocol (IP) network that connects each fabric node. The underlay network is part of a dynamic discovery process that is involved in creating the overlay network's VXLAN tunnels. When an endpoint in a Cisco SDA network sends traffic to another endpoint, the traffic flows from the endpoint through the overlay network's VXLAN tunnels.

The fabric is the entirety of the overlay network and the underlay network in a Cisco SDA network.

Reference:
Boson CCNA Curriculum 200-301, Module 9: Automation and Programming, SDN and Cisco SDA

Boson CCNA Curriculum 200-301, Module 9: Automation and Programming, Northbound APIs

Boson CCNA Curriculum 200-301, Module 9: Automation and Programming, Southbound APIs

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 21: Introduction to Controller-Based Networking, The Southbound Interface

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 21: Introduction to Controller-Based Networking, The Northbound Interface

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 22: Cisco Software-Defined Access (Cisco SD-Access), Cisco SD-Access Fabric, Underlay, and Overlay

Category:
Automation and Programmability

Day 62 - Software-Defined Networking

Original Question 65/74

65/74

Which of the following is centralized on an SDN network? *(Select the best answer.)*

- A. the data plane
- B. the application plane
- C. the control plane
- D. the management plane

Incorrect

Correct Answer(s): C

Explanation:

The control plane is centralized on a Software-Defined Networking (SDN) network, which is a type of controller-based network. The control plane is responsible for network decision making in both a controller-based network and a traditional network. However, the control plane in a traditional network is typically distributed among many devices. The Open Shortest Path First (OSPF) routing protocol running on a series of routers on a traditional network is one example of a traditional control plane. OSPF makes routing decisions for packets that require routing among Layer 3 devices. In a controller-based network, the decision-making logic for packet-handling policies is either moved to a central controller or monitored by a central controller.

The data plane is not centralized on an SDN network. Layer 2 switches, Layer 3 switches, and end devices typically operate in the data plane. In a controller-based network, the controller communicates with the data plane by using a southbound Application Programming Interface (API), such as NETCONF, OpenFlow, OpFlex, or OnePK. Network tasks that are typically performed in the data plane include the encapsulation and decapsulation of packets, the adding or removing of trunk headers, the matching of Media Access Control (MAC) addresses to a MAC address table, the matching of Internet Protocol (IP) addresses to paths in a routing table, the encryption of data, Network Address Translation (NAT), and filtering by using either access control lists (ACLs) or port security.

The management plane is not centralized on an SDN network. Network management protocols, such as Telnet, Secure Shell (SSH), Simple Network Management Protocol (SNMP), and Syslog typically operate in the management plane on both a traditional network and a controller-based network. All of these protocols enable an administrator to connect to and manage a network device.

The application plane is not centralized on an SDN network. The application plane is the component of a controller-based network in which applications that are written to allow interaction with the

centralized controller reside. These applications are typically designed to improve network management efficiency through network automation. A controller communicates with applications in the application plane by using a northbound API such as Representational State Transfer (REST) or Open Services Gateway initiative (OSGi).

Reference:

Boson CCNA Curriculum 200-301, Module 9: Automation and Programming, Device Management on Controller-Based Networks

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 21: Introduction to Controller-Based Networking, The Data, Control, and Management Planes

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 21: Introduction to Controller-Based Networking, Controllers and Software Defined Architecture

Category:

Automation and Programmability